# Deployment of Online Ticketing and Event Management Applications and Infrastructure

A DevOps Implementation for Scalable Movie Ticketing Platform

Domain: E-Commerce and Entertainment

## Project Overview:

BookMyShow is a popular online movie ticketing platform that allows users to browse movies, book tickets, and manage their bookings. The company is transitioning from a monolithic architecture to a microservices-based architecture to improve scalability, deployment
efficiency, and infrastructure management.

To achieve this, BookMyShow has decided to adopt DevOps practices with CI/CD pipelines, containerization, and automated infrastructure provisioning. The project will use AWS as the primary cloud provider for hosting servers, databases (if any), and deploying applications.

## Business Goals:

- Deliver frequent and reliable updates to production.
- Accelerate software delivery speed while maintaining high quality.
- Reduce manual intervention in deployments and testing.
- Improve scalability and monitoring of the application.

## Current Challenges:

- Complex Builds: Managing dependencies and incremental builds is difficult.
- Manual Testing: Testing multiple components manually is time-consuming.
- Infrastructure Setup: Manual server provisioning and configuration delays deployments.

## Monitoring Challenges:

- Lack of real-time monitoring leads to delayed issue detection.
- CI/CD Pipeline Implementation

The following tools will be used to automate the deployment process:

### Tools and Technologies

- Git – Version control for tracking code changes.
- Jenkins – CI/CD pipeline automation (build, test, deploy).
- Docker – Containerization of the application.
- Kubernetes (K8s) – Orchestration for scalable deployments.
- Ansible – Configuration management for infrastructure automation.
- AWS – Cloud infrastructure (EC2, EKS for Kubernetes).
- Prometheus and Grafana – Monitoring and alerting.

## Pipeline Workflow:

- GitHub Webhook Trigger
- When a developer pushes code to the master branch, Jenkins automatically triggers the pipeline.
- Build & Test Phase
- Jenkins checks out the code, creates docker images, pushes docker images, creates containers, deploys to EKS cluster
- TestNG generates an HTML test report.
- Dockerization: The application is packaged into a Docker image and pushed to AWS ECR (Elastic Container Registry).
- Kubernetes Deployment (Dev / Stage Environment)
- Ansible deploys the Docker container to a Kubernetes cluster (AWS EKS).
- Monitoring & Validation
- Prometheus collects metrics (CPU, memory, API latency).
- Grafana visualizes performance dashboards.
- If the deployment is stable, it proceeds to production.

## Production Deployment:

- After validation in staging, the same Docker image is deployed to the production Kubernetes cluster.

# DevOps Project: Book My Show App Deployment

Open below ports for the Security Group attached to the EC-2 Machine

| Type | Protocol | Port range |
|---|---|---|
| SMTP | TCP | 25 |

(Used for sending emails between mail servers)

| Type | Protocol | Port range |
|---|---|---|
| Custom TCP | TCP | 3000-10000 |

(Used by various applications, such as Node.js (3000), Grafana (3000), Jenkins (8080), and custom web applications.

| Type | Protocol | Port range |
|---|---|---|
| HTTP | TCP | 80 |

Allows unencrypted web traffic. Used by web servers (e.g., Apache, Nginx) to serve websites over HTTP.

| Type | Protocol | Port range |
|---|---|---|
| HTTPS | TCP | 443 |

Allows secure web traffic using SSL/TLS.

| Type | Protocol | Port range |
|---|---|---|
| SSH | TCP | 22 |

Secure Shell (SSH) for remote server access.

| Type | Protocol | Port range |
|---|---|---|
| Custom TCP | TCP | 6443 |

Kubernetes API server port. Used for communication between kubectl, worker nodes, and the Kubernetes control plane.

| Type | Protocol | Port range |
|---|---|---|
| SMTPS | TCP | 465 |

Secure Mail Transfer Protocol over SSL/TLS. Used for sending emails securely via SMTP with encryption.

| Type | Protocol | Port range |
|---|---|---|
| Custom TCP | TCP | 30000-32767 |

Kubernetes NodePort service range.

## Step 1: Basic Setup

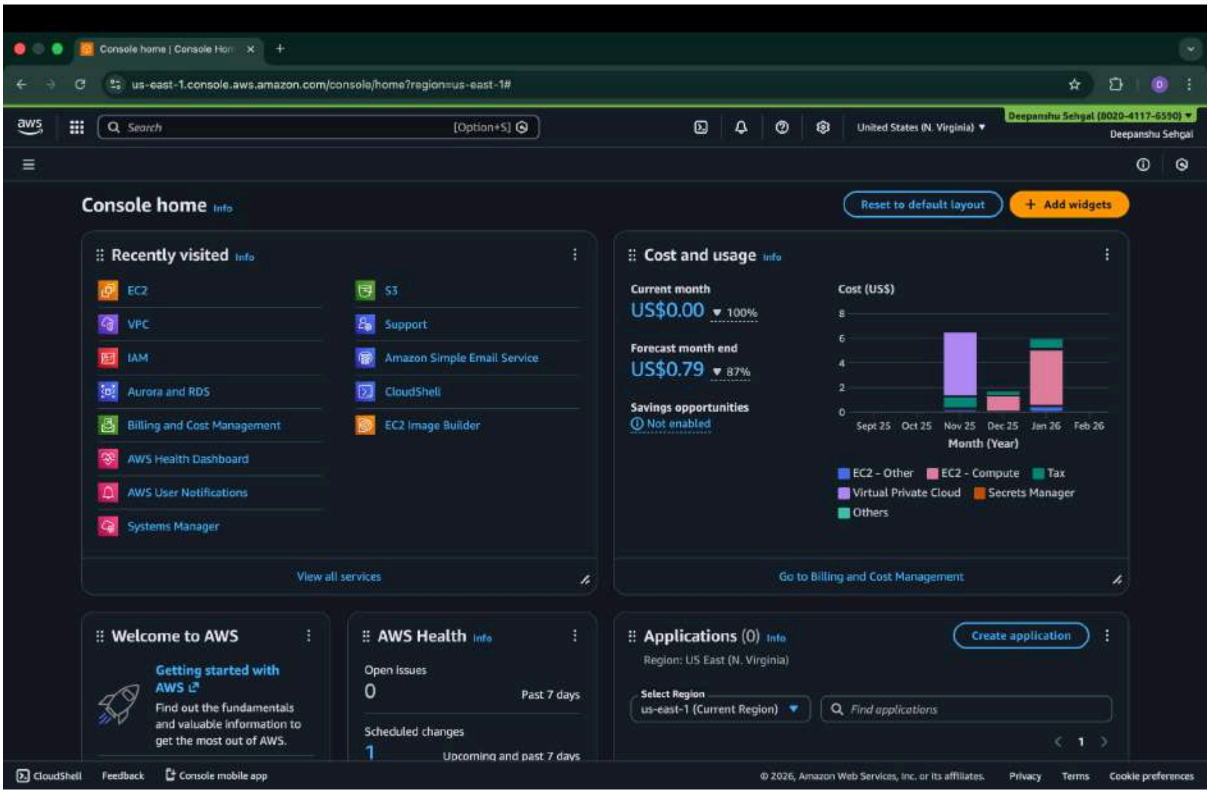
1.1 Launch EC2 Instance

Launch 1 Ubuntu VM:

- **OS:** Ubuntu 24.04
- **Instance Type:** t2.large
- **Storage:** 28 GB
- **Name:** BMS-Server

Security Group Ports

| Type | Protocol | Port | Purpose |
|---|---|---|---|
| SSH | TCP | 22 | Remote access |
| HTTP | TCP | 80 | Web traffic |
| HTTPS | TCP | 443 | Secure web traffic |
| SMTP | TCP | 25 | Mail transfer |
| SMTPS | TCP | 465 | Secure mail |
| Custom TCP | TCP | 3000–10000 | Node, Jenkins, Grafana etc. |
| Custom TCP | TCP | 6443 | Kubernetes API |
| Custom TCP | TCP | 30000–32767 | K8s NodePort |


Console home
Reset to default layout
Add widgets
Recently visited
EC2
VPC
IAM
Aurora and RDS
Billing and Cost Management
AWS Health Dashboard
AWS User Notifications
Systems Manager
S3
Support
Amazon Simple Email Service
CloudShell
EC2 Image Builder
View all services
Cost and usage
Current month
US$0.00
Forecast month end
US$0.79
Savings opportunities
Not enabled
Cost (US$)
Month (Year)
EC2 - Other
EC2 - Compute
Tax
Virtual Private Cloud
Secrets Manager
Others
Go to Billing and Cost Management
Welcome to AWS
Getting started with AWS
Find out the fundamentals and valuable information to get the most out of AWS.
AWS Health
Open issues
Past 7 days
Scheduled changes
Upcoming and past 7 days
Applications (0)
Create application
Region: US East (N. Virginia)
Select Region
us-east-1 (Current Region)
Find applications
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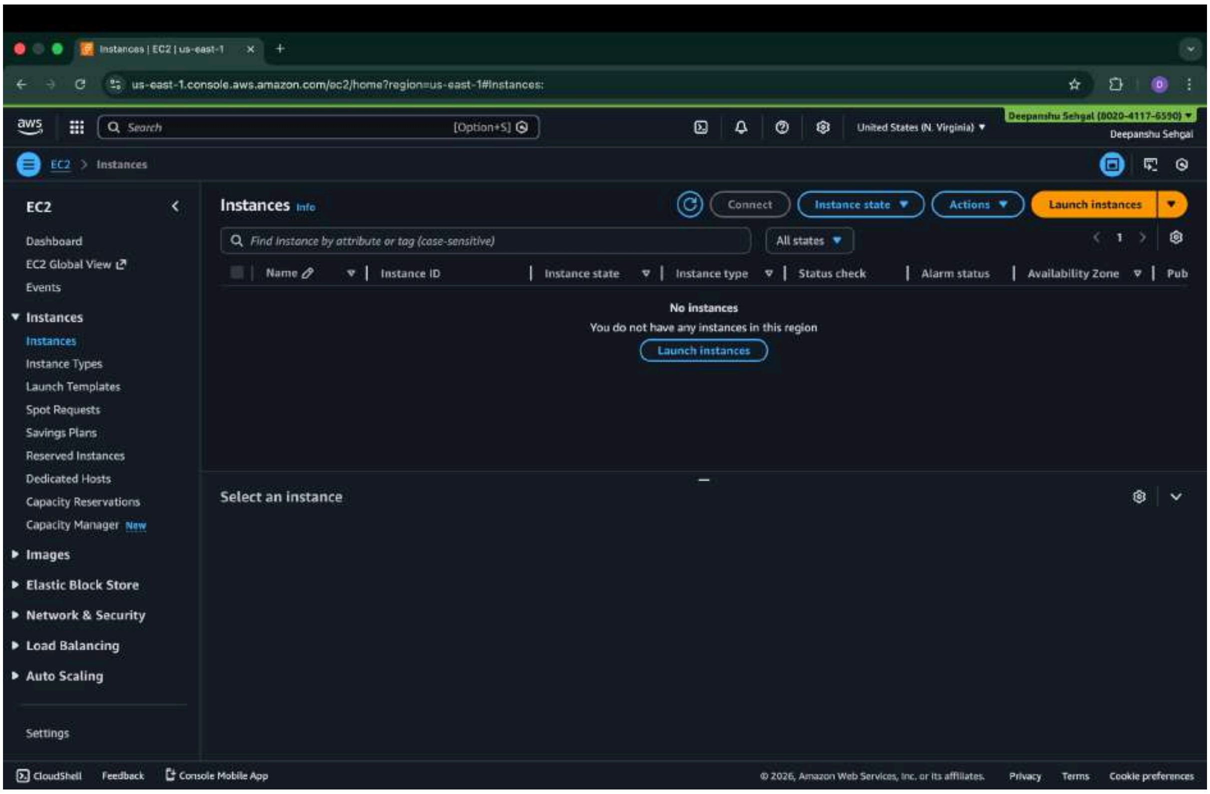
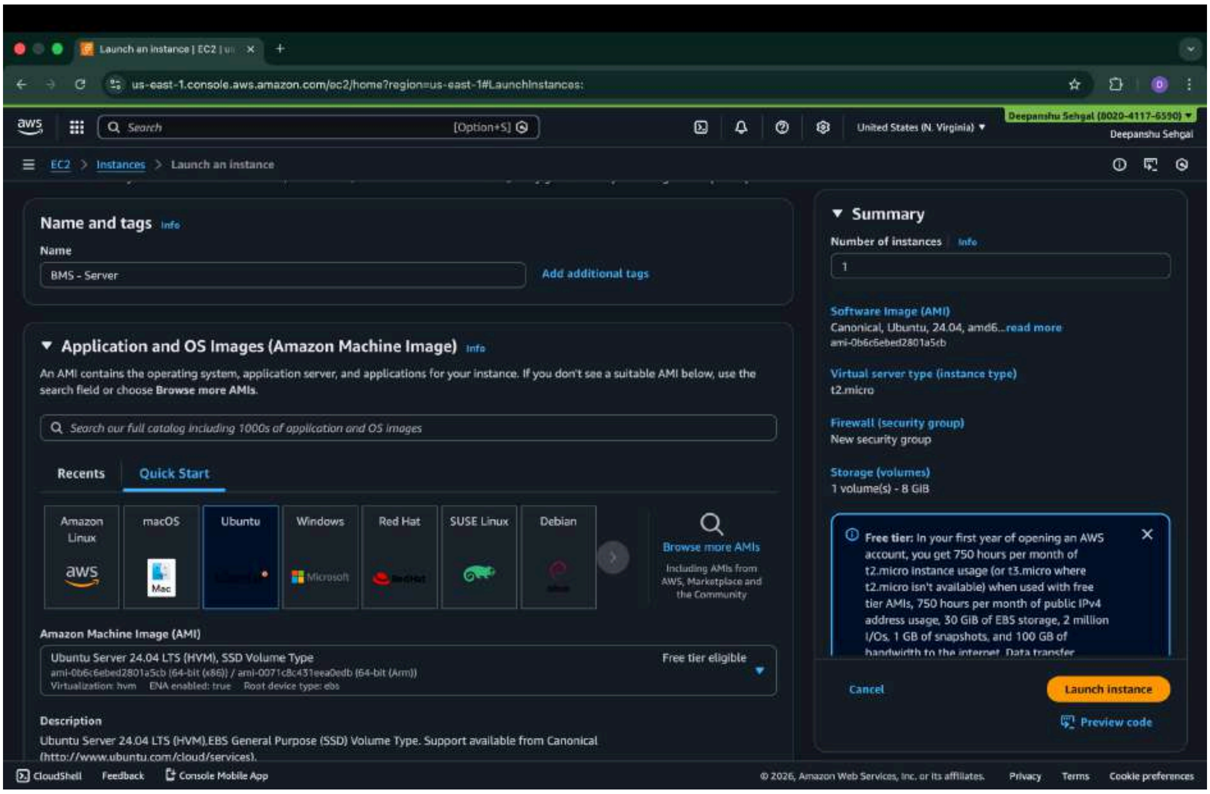

us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
EC2
Instances
Dashboard
EC2 Global View
Events
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager
Images
Elastic Block Store
Network & Security
Load Balancing
Auto Scaling
Settings
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
All states
Name
Instance ID
Instance type
Status check
Alarm status
Availability Zone
No instances
You do not have any instances in this region
Select an instance
© 2026, Amazon Web Services, Inc. or its affiliates.


Launch an instance | EC2 | us
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:
Search
[Option+S]
United States (N. Virginia)
Deepanshu Sehgal
EC2 > Instances > Launch an instance
Name and tags Info
Name
BMS - Server
Add additional tags
Application and OS Images (Amazon Machine Image) Info
An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose Browse more AMIs.
Search our full catalog including 1000s of application and OS images
Recents
Quick Start
Amazon Linux
macOS
Ubuntu
Windows
Red Hat
SUSE Linux
Debian
Browse more AMIs
Including AMIs from AWS, Marketplace and the Community
Amazon Machine Image (AMI)
Ubuntu Server 24.04 LTS (HVM), SSD Volume Type
Free tier eligible
Description
Ubuntu Server 24.04 LTS (HVM),EBS General Purpose (SSD) Volume Type. Support available from Canonical (http://www.ubuntu.com/cloud/services).
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
Virtual server type (instance type)
t2.micro
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer
Cancel
Launch instance
Preview code
CloudShell
Feedback
Console Mobile App
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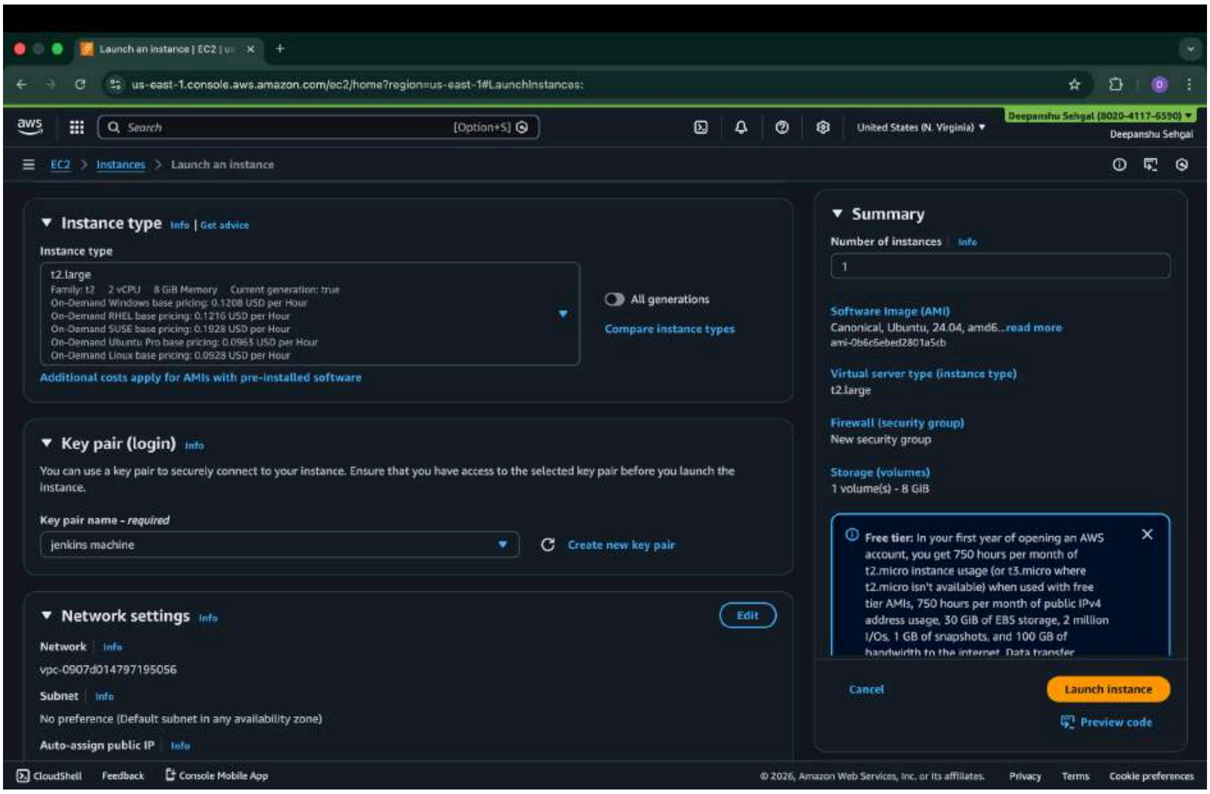
Launch an instance | EC2 | us
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:
Search
[Option+S]
United States (N. Virginia)
Deepanshu Sehgal
EC2 > Instances > Launch an instance
Instance type Info | Get advice
Instance type
t2.large
Family: t2 2 vCPU 8 GiB Memory Current generation: true
On-Demand Windows base pricing: 0.1208 USD per Hour
On-Demand RHEL base pricing: 0.1216 USD per Hour
On-Demand SUSE base pricing: 0.1928 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0963 USD per Hour
On-Demand Linux base pricing: 0.0928 USD per Hour
All generations
Compare instance types
Additional costs apply for AMIs with pre-installed software
Key pair (login) Info
You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.
Key pair name - required
jenkins machine
Create new key pair
Network settings Info
Edit
Network Info
vpc-0907d014797195056
Subnet Info
No preference (Default subnet in any availability zone)
Auto-assign public IP Info
Summary
Number of instances Info
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
Virtual server type (instance type)
t2.large
Firewall (security group)
New security group
Storage (volumes)
1 volume(s) - 8 GiB
Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GB of snapshots, and 100 GB of bandwidth to the internet. Data transfer
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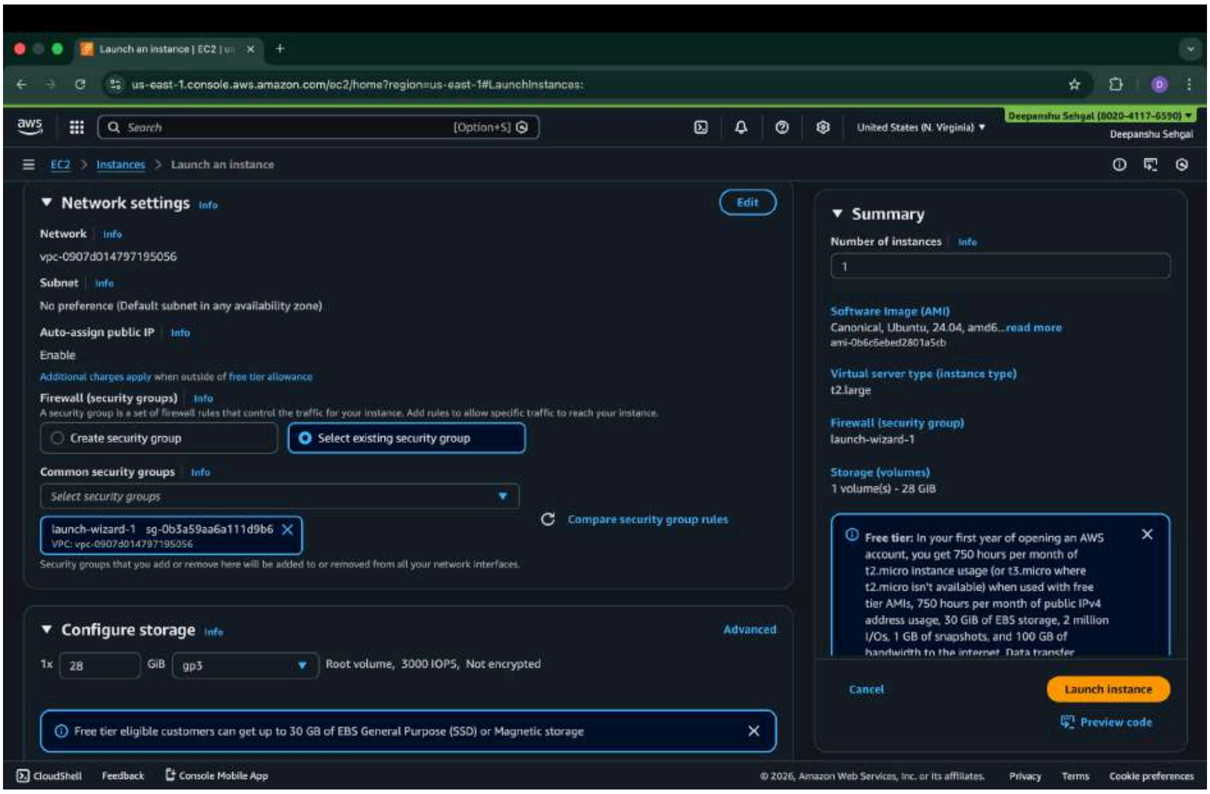
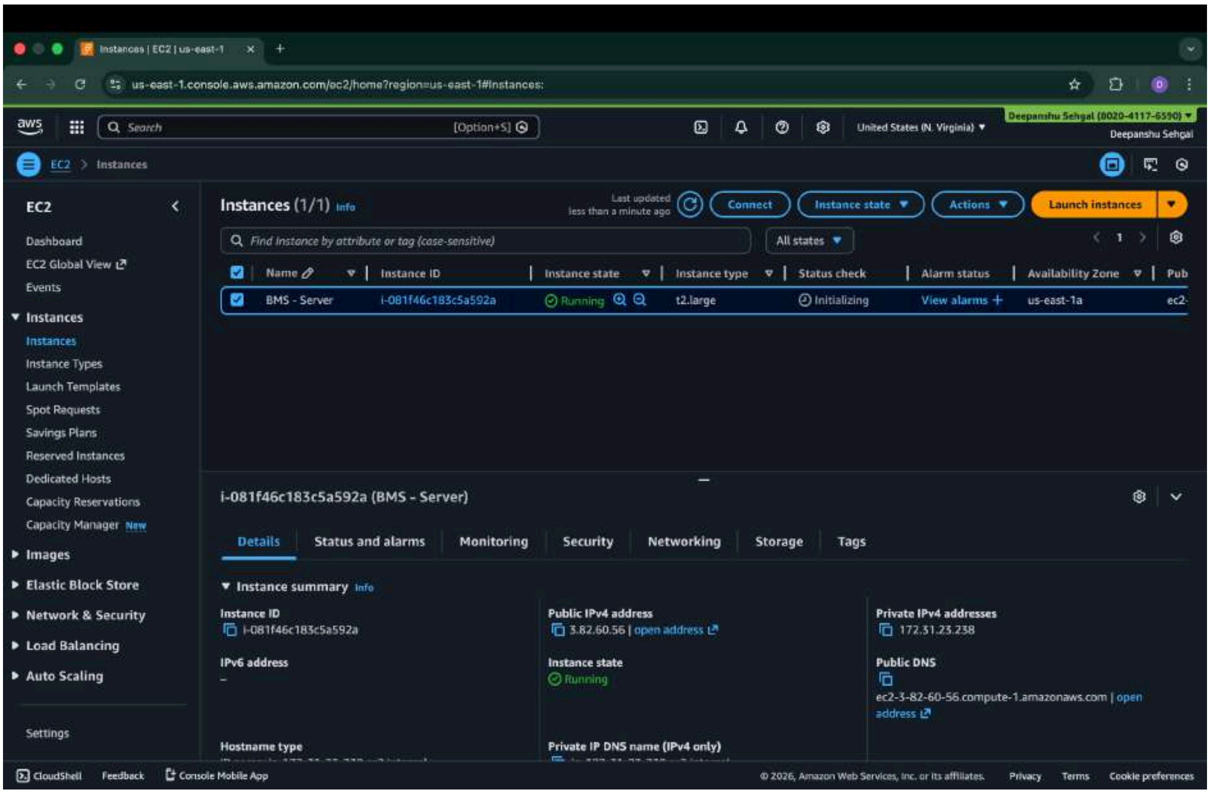
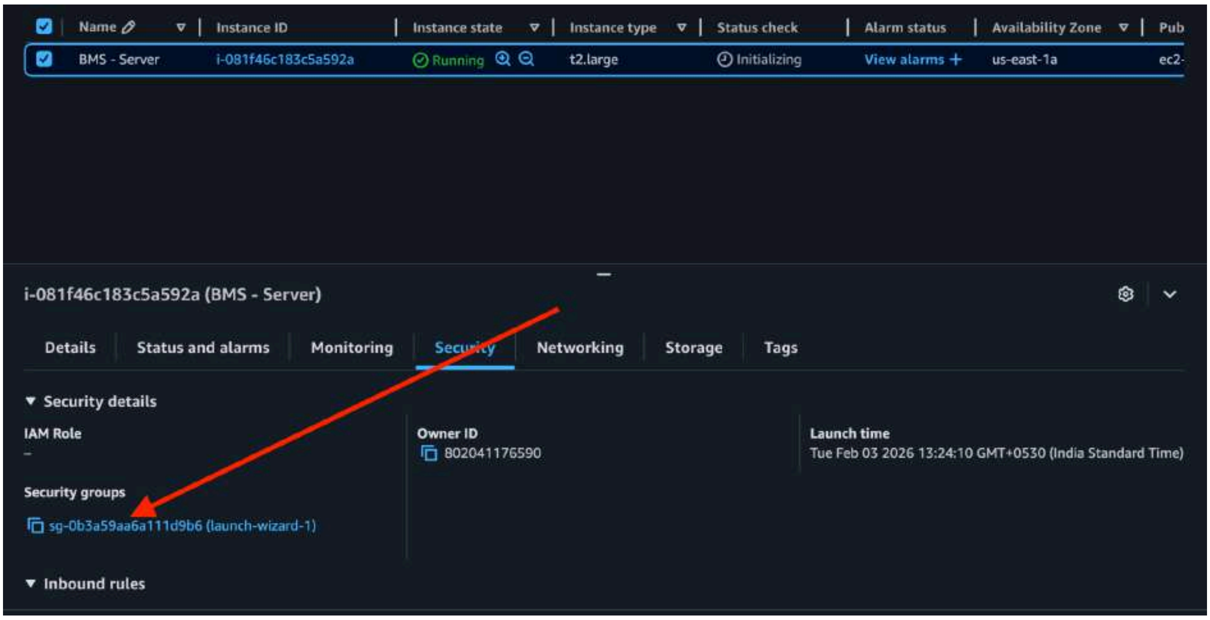
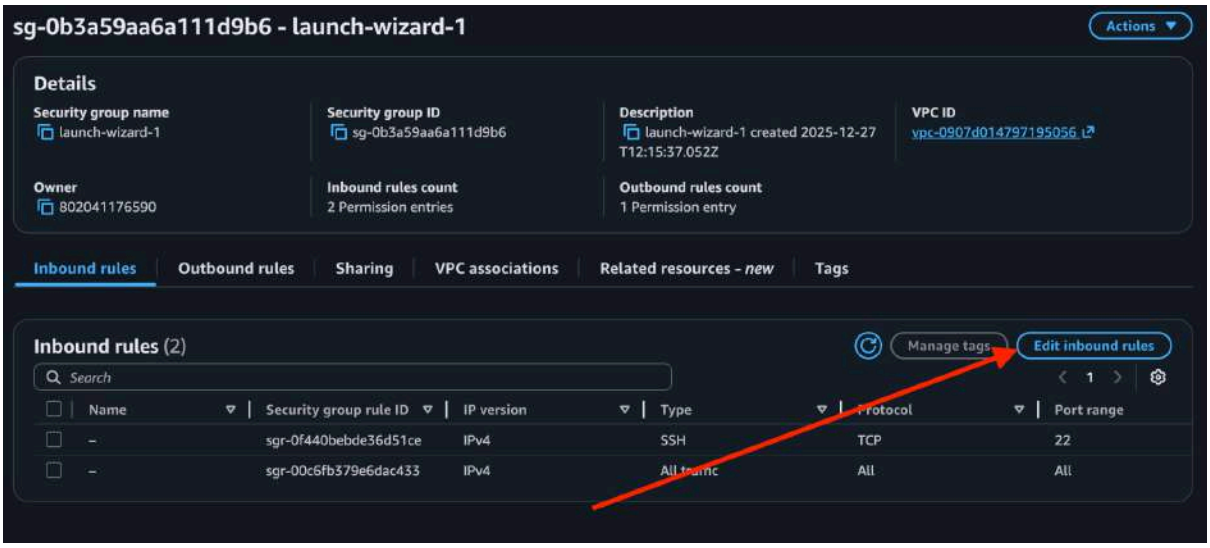
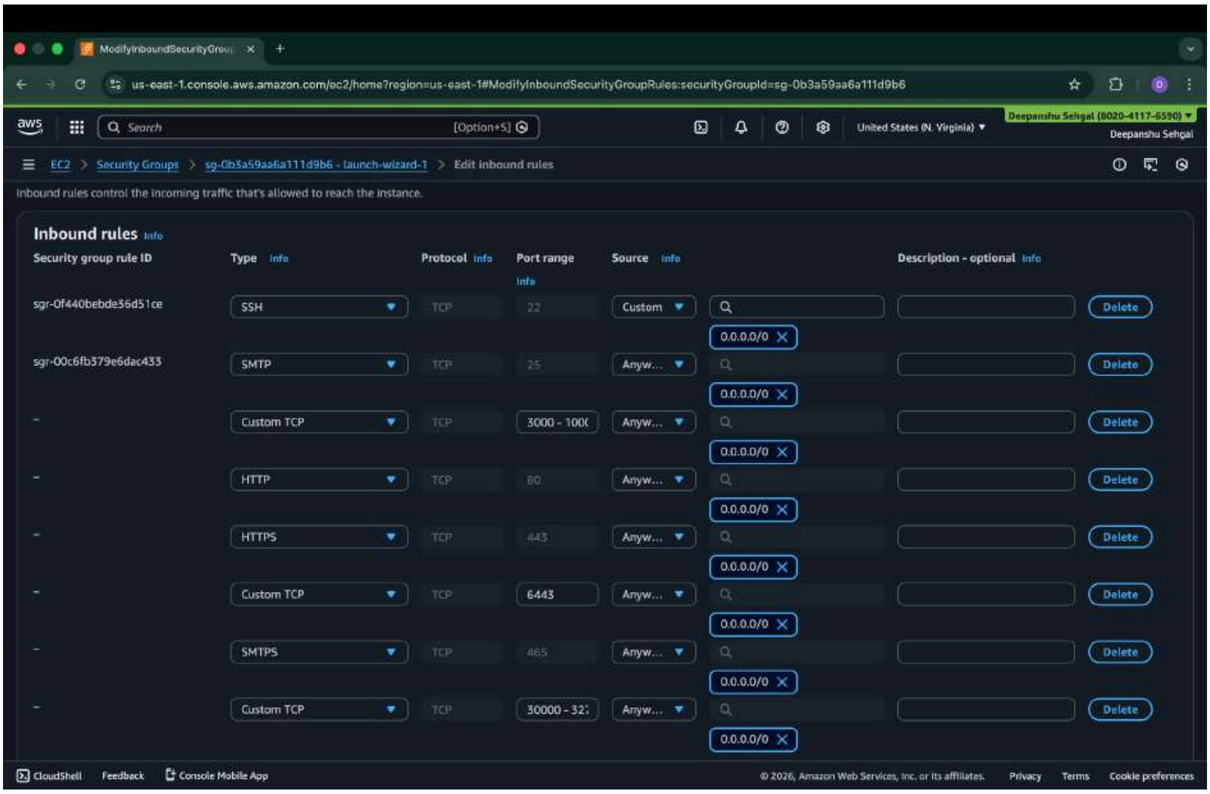

Launch an instance | EC2
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#LaunchInstances:
EC2 > Instances > Launch an instance
Network settings
Network
vpc-0907d014797195056
Subnet
No preference (Default subnet in any availability zone)
Auto-assign public IP
Enable
Firewall (security groups)
Create security group
Select existing security group
Common security groups
launch-wizard-1 sg-0b3a59aa6a111d9b6
Compare security group rules
Configure storage
1x 28 GiB gp3 Root volume, 3000 IOPS, Not encrypted
Free tier eligible customers can get up to 30 GB of EBS General Purpose (SSD) or Magnetic storage
Summary
Number of instances
1
Software Image (AMI)
Canonical, Ubuntu, 24.04, amd6...read more
Virtual server type (instance type)
t2.large
Firewall (security group)
launch-wizard-1
Storage (volumes)
1 volume(s) - 28 GiB
Cancel
Launch instance
Preview code


Instances | EC2 | us-east-1
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#Instances:
EC2 > Instances
Instances (1/1)
Connect
Instance state
Actions
Launch instances
BMS - Server
i-081f46c183c5a592a
Running
t2.large
Initializing
View alarms
us-east-1a
i-081f46c183c5a592a (BMS - Server)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary
Instance ID
i-081f46c183c5a592a
Public IPv4 address
3.82.60.56 | open address
Private IPv4 addresses
172.31.23.238
Instance state
Running
Public DNS
ec2-3-82-60-56.compute-1.amazonaws.com | open address


Name | Instance ID | Instance state | Instance type | Status check | Alarm status | Availability Zone | Pub
BMS - Server | i-081f46c183c5a592a | Running | t2.large | Initializing | View alarms + | us-east-1a | ec2-
i-081f46c183c5a592a (BMS - Server)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Security details
IAM Role
–
Owner ID
802041176590
Launch time
Tue Feb 03 2026 13:24:10 GMT+0530 (India Standard Time)
Security groups
sg-0b3a59aa6a111d9b6 (launch-wizard-1)
Inbound rules


sg-0b3a59aa6a111d9b6 - launch-wizard-1
Actions
Details
Security group name
launch-wizard-1
Security group ID
sg-0b3a59aa6a111d9b6
Description
launch-wizard-1 created 2025-12-27 T12:15:37.052Z
VPC ID
vpc-0907d014797195056
Owner
802041176590
Inbound rules count
2 Permission entries
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (2)
Manage tags
Edit inbound rules
Search
Name | Security group rule ID | IP version | Type | Protocol | Port range
– | sgr-0f440bebde36d51ce | IPv4 | SSH | TCP | 22
– | sgr-00c6fb379e6dac433 | IPv4 | All traffic | All | All


ModifyInboundSecurityGroup
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#ModifyInboundSecurityGroupRules:securityGroupId=sg-0b3a59aa6a111d9b6
Search
[Option+S]
United States (N. Virginia)
Deepanshu Sehgal (8020-4117-6590)
Deepanshu Sehgal
EC2 > Security Groups > sg-0b3a59aa6a111d9b6 - launch-wizard-1 > Edit inbound rules
Inbound rules control the incoming traffic that's allowed to reach the instance.
Inbound rules Info
Security group rule ID
Type Info
Protocol Info
Port range Info
Source Info
Description - optional Info
sgr-0f440bebde36d51ce
SSH
TCP
22
Custom
0.0.0.0/0
Delete
sgr-00c6fb379e6dac433
SMTP
TCP
25
Anyw...
0.0.0.0/0
Delete
Custom TCP
3000 - 100(
HTTP
80
HTTPS
443
Custom TCP
6443
SMTPS
465
Custom TCP
30000 - 32
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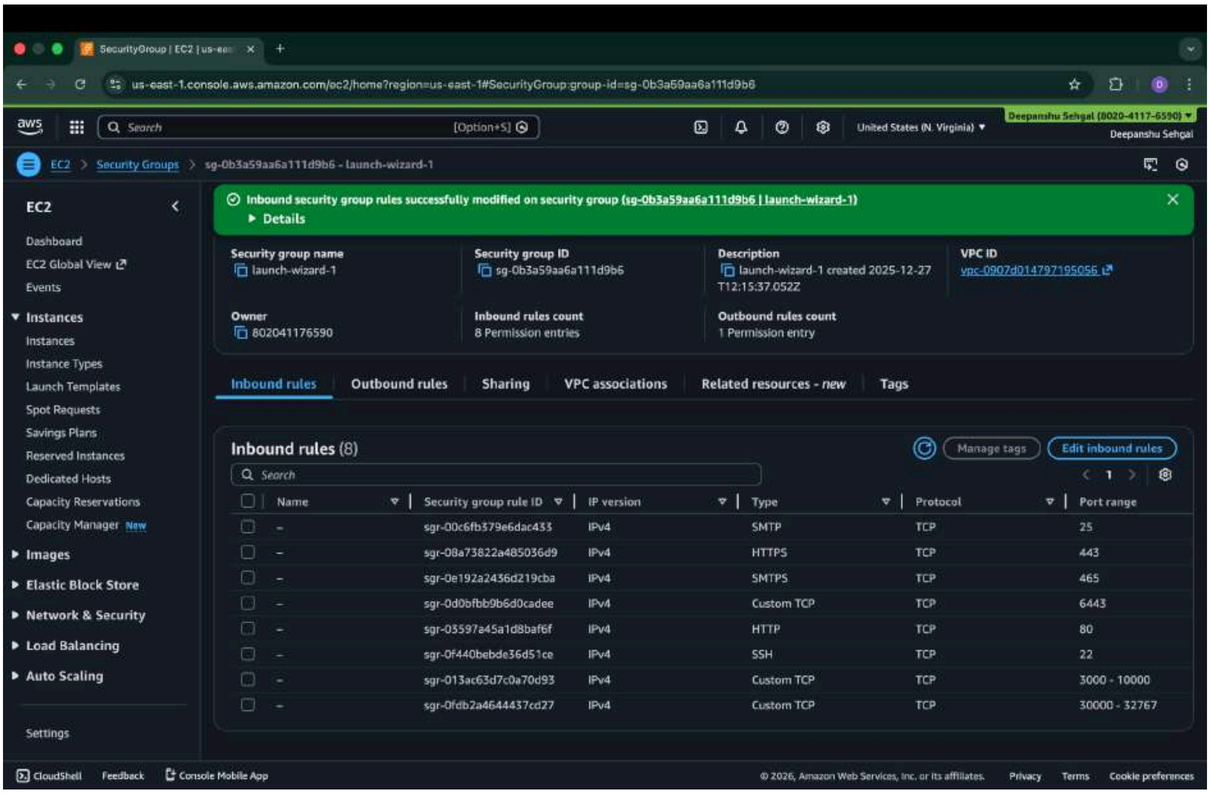
SecurityGroup | EC2 | us-east
us-east-1.console.aws.amazon.com/ec2/home?region=us-east-1#SecurityGroup:group-id=sg-0b3a59aa6a111d9b6
EC2 > Security Groups > sg-0b3a59aa6a111d9b6 - launch-wizard-1
Inbound security group rules successfully modified on security group (sg-0b3a59aa6a111d9b6 | launch-wizard-1)
Details
Security group name
launch-wizard-1
Security group ID
sg-0b3a59aa6a111d9b6
Description
launch-wizard-1 created 2025-12-27T12:15:37.052Z
VPC ID
vpc-0907d014797195056
Owner
802041176590
Inbound rules count
8 Permission entries
Outbound rules count
1 Permission entry
EC2
Dashboard
EC2 Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
Elastic Block Store
Network & Security
Load Balancing
Auto Scaling
Settings
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (8)
Manage tags
Edit inbound rules
Search
Name
Security group rule ID
IP version
Type
Protocol
Port range
sgr-00c6fb379e6dac433 IPv4 SMTP TCP 25
sgr-08a73822a485036d9 IPv4 HTTPS TCP 443
sgr-0e192a2436d219cba IPv4 SMTPS TCP 465
sgr-0d0bfbb9b6d0cadee IPv4 Custom TCP TCP 6443
sgr-03597a45a1d8baf6f IPv4 HTTP TCP 80
sgr-0f440bebde36d51ce IPv4 SSH TCP 22
sgr-013ac63d7c0a70d93 IPv4 Custom TCP TCP 3000 - 10000
sgr-0fdb2a4644437cd27 IPv4 Custom TCP TCP 30000 - 32767
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# Step 2: Tools Installation

```
deepanshusehgal~$ssh -i "jenkins machine.pem" ubuntu@ec2-3-82-60-56.compute-1.amazonaws.com
The authenticity of host 'ec2-3-82-60-56.compute-1.amazonaws.com (3.82.60.56)' can't be established.
ED25519 key fingerprint is SHA256:hZcAokRXOm471Yxjo4huueYK7c52Yry2MxXsRL9NdFM.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-3-82-60-56.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Tue Feb  3 08:03:44 UTC 2026

  System load:  0.08               Processes:             119
  Usage of /:   6.7% of 26.08GB    Users logged in:       0
  Memory usage: 2%                 IPv4 address for enX0: 172.31.23.238
  Swap usage:   0%

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.

   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-23-238:~$ clear
```

## Jenkins Installation

Install:

- OpenJDK 17
- Jenkins repo
- Jenkins package

Start Jenkins & open **port 8080**.

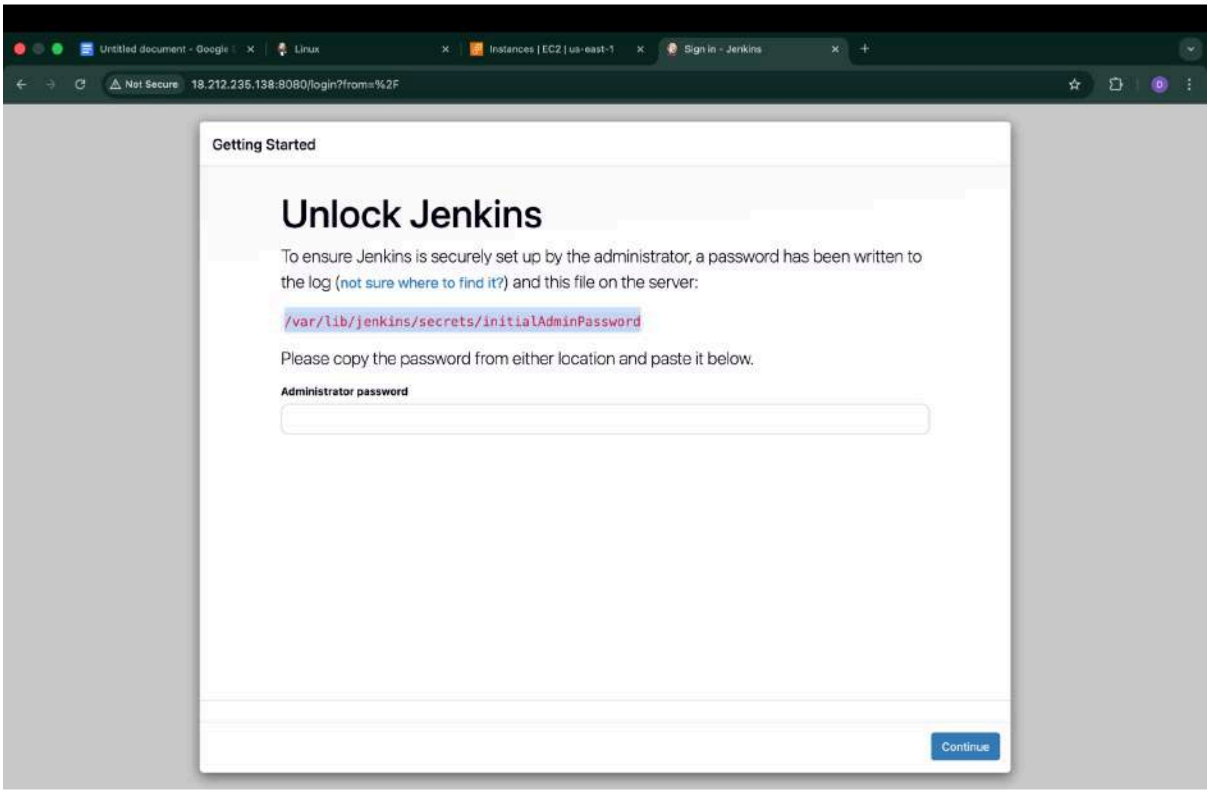
```
ubuntu@ip-172-31-23-238:~$ vi jenkins.sh
```

```
#!/bin/bash

# Install OpenJDK 17 JRE Headless
sudo apt install fontconfig openjdk-21-jre -y

java -version

# Download Jenkins GPG key
sudo wget -O /usr/share/keyrings/jenkins-keyring.asc \
  https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key

# Add Jenkins repository to package manager sources
echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] \
  https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
  /etc/apt/sources.list.d/jenkins.list > /dev/null

# Update package manager repositories
sudo apt-get update

# Install Jenkins
sudo apt-get install jenkins -y
~
~
~
```

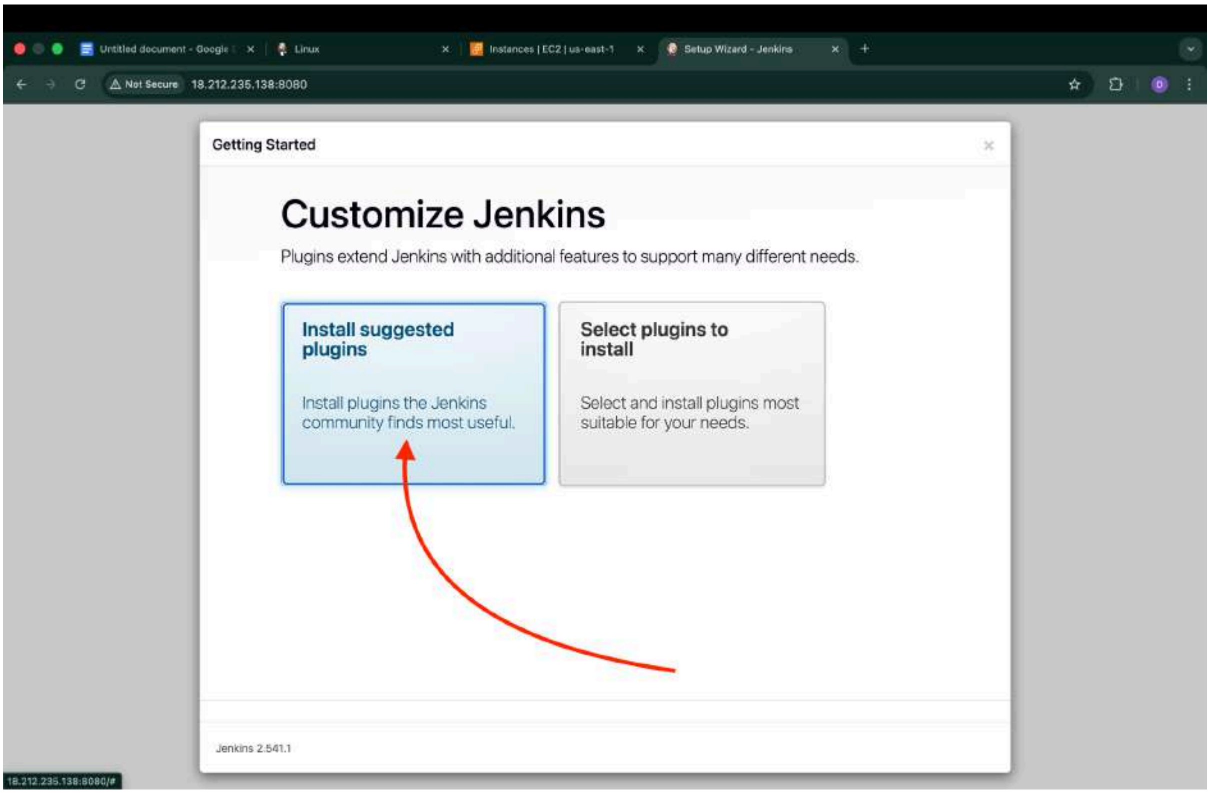
```
ubuntu@ip-172-31-23-238:~$ vi jenkins.sh
ubuntu@ip-172-31-23-238:~$ ls
jenkins.sh
ubuntu@ip-172-31-23-238:~$ ll
total 36
drwxr-x--- 4 ubuntu ubuntu 4096 Feb  3 08:11 ./
drwxr-xr-x 3 root   root   4096 Feb  3 07:54 ../
-rw-r--r-- 1 ubuntu ubuntu  220 Mar 31  2024 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Mar 31  2024 .bashrc
drwx------ 2 ubuntu ubuntu 4096 Feb  3 08:03 .cache/
-rw-r--r-- 1 ubuntu ubuntu  807 Mar 31  2024 .profile
drwx------ 2 ubuntu ubuntu 4096 Feb  3 07:54 .ssh/
-rw-r--r-- 1 ubuntu ubuntu    0 Feb  3 08:06 .sudo_as_admin_successful
-rw------- 1 ubuntu ubuntu  811 Feb  3 08:11 .viminfo
-rw-rw-r-- 1 ubuntu ubuntu  570 Feb  3 08:11 jenkins.sh
ubuntu@ip-172-31-23-238:~$ sudo chmod +x jenkins.sh
ubuntu@ip-172-31-23-238:~$ ll
total 36
drwxr-x--- 4 ubuntu ubuntu 4096 Feb  3 08:11 ./
drwxr-xr-x 3 root   root   4096 Feb  3 07:54 ../
-rw-r--r-- 1 ubuntu ubuntu  220 Mar 31  2024 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Mar 31  2024 .bashrc
drwx------ 2 ubuntu ubuntu 4096 Feb  3 08:03 .cache/
-rw-r--r-- 1 ubuntu ubuntu  807 Mar 31  2024 .profile
drwx------ 2 ubuntu ubuntu 4096 Feb  3 07:54 .ssh/
-rw-r--r-- 1 ubuntu ubuntu    0 Feb  3 08:06 .sudo_as_admin_successful
-rw------- 1 ubuntu ubuntu  811 Feb  3 08:11 .viminfo
-rwxrwxr-x 1 ubuntu ubuntu  570 Feb  3 08:11 jenkins.sh*
ubuntu@ip-172-31-23-238:~$
```

```
ubuntu@ip-172-31-23-238:~$ ls
jenkins.sh
ubuntu@ip-172-31-23-238:~$ ./jenkins.sh
```

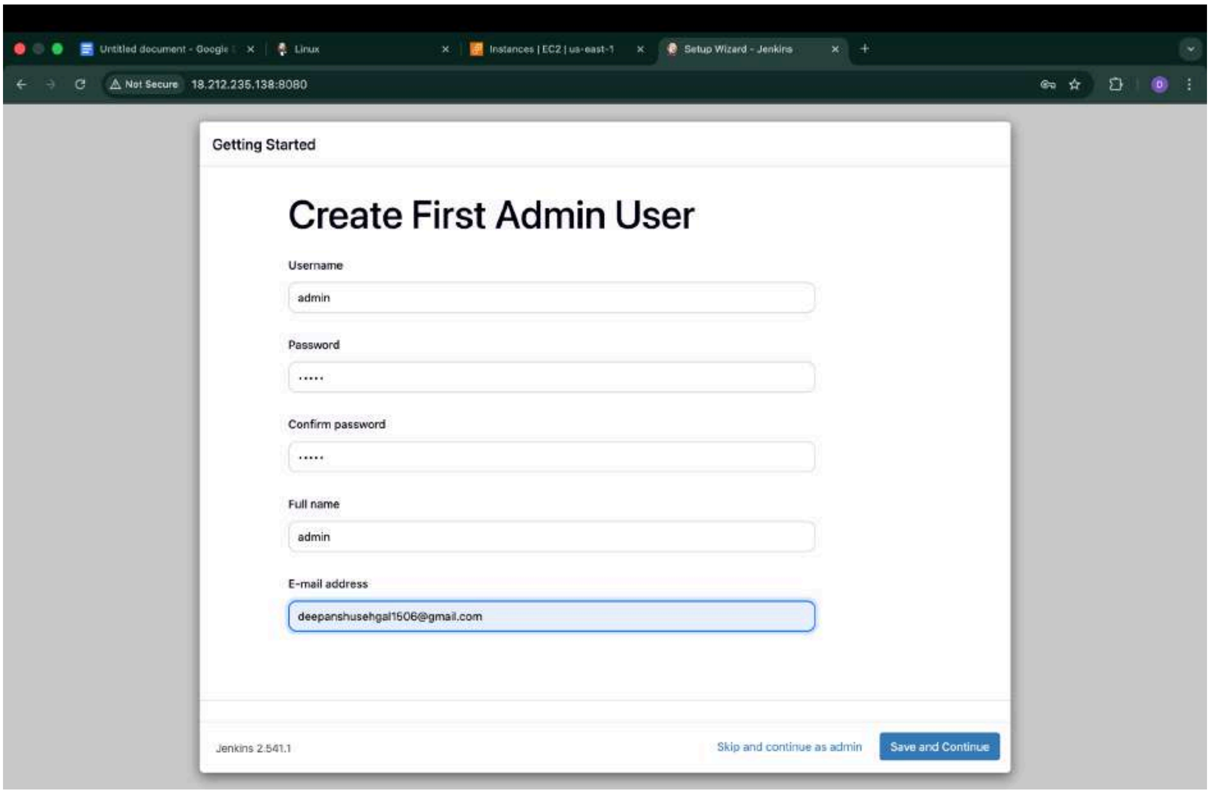
```
ubuntu@ip-172-31-18-170:~$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
     Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-02-04 07:12:38 UTC; 26s ago
   Main PID: 4030 (java)
      Tasks: 50 (limit: 9498)
     Memory: 657.0M (peak: 662.3M)
        CPU: 15.264s
     CGroup: /system.slice/jenkins.service
             └─4030 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Feb 04 07:12:25 ip-172-31-18-170 jenkins[4030]: [LF]> This may also be found at: /var/lib/jenkins/secrets/initialAdminPassword
Feb 04 07:12:25 ip-172-31-18-170 jenkins[4030]: [LF]>
Feb 04 07:12:25 ip-172-31-18-170 jenkins[4030]: [LF]> *************************************************************
Feb 04 07:12:25 ip-172-31-18-170 jenkins[4030]: [LF]> *************************************************************
Feb 04 07:12:25 ip-172-31-18-170 jenkins[4030]: [LF]> *************************************************************
Feb 04 07:12:38 ip-172-31-18-170 jenkins[4030]: 2026-02-04 07:12:38.245+0000 [id=37]       INFO       jenkins.InitReactorRunner$1#onAttained: Completed initia>
Feb 04 07:12:38 ip-172-31-18-170 jenkins[4030]: 2026-02-04 07:12:38.271+0000 [id=30]       INFO       hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up >
Feb 04 07:12:38 ip-172-31-18-170 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Feb 04 07:12:38 ip-172-31-18-170 jenkins[4030]: 2026-02-04 07:12:38.591+0000 [id=56]       INFO       h.m.DownloadService$Downloadable#load: Obtained the upda>
Feb 04 07:12:38 ip-172-31-18-170 jenkins[4030]: 2026-02-04 07:12:38.593+0000 [id=56]       INFO       hudson.util.Retrier#start: Performed the action check up>
lines 1-20/20 (END)
```


Untitled document - Google
Linux
Instances | EC2 | us-east-1
Sign in - Jenkins
Not Secure
18.212.235.138:8080/login?from=%2F
Getting Started
Unlock Jenkins
To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:
/var/lib/jenkins/secrets/initialAdminPassword
Please copy the password from either location and paste it below.
Administrator password
Continue

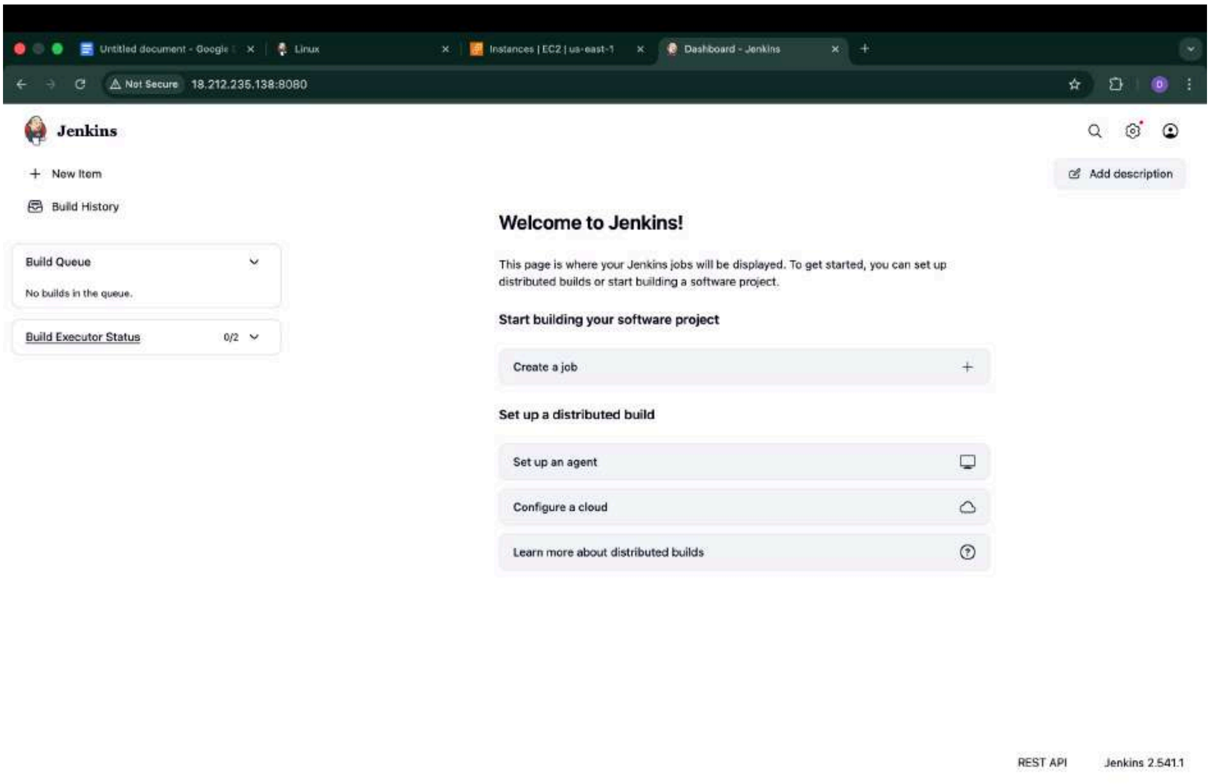

ubuntu@ip-172-31-18-170:~$ sudo cat /var/lib/jenkins/secrets/initialAdminPassword
a3f3a4be198c4bb99e9bc563ee17c2e1
ubuntu@ip-172-31-18-170:~$



Untitled document - Google
Linux
Instances | EC2 | us-east-1
Setup Wizard - Jenkins
Not Secure
18.212.235.138:8080
Getting Started
Customize Jenkins
Plugins extend Jenkins with additional features to support many different needs.
Install suggested plugins
Install plugins the Jenkins community finds most useful.
Select plugins to install
Select and install plugins most suitable for your needs.
Jenkins 2.541.1
18.212.235.138:8080/#

Untitled document - Google
Linux
Instances | EC2 | us-east-1
Setup Wizard - Jenkins
Not Secure
18.212.235.138:8080
Getting Started
Create First Admin User
Username
admin
Password
•••••
Confirm password
•••••
Full name
admin
E-mail address
deepanshusehgal1506@gmail.com
Jenkins 2.541.1
Skip and continue as admin
Save and Continue




## Docker Installation

Install Docker CE, CLI, Buildx & Compose.

```
ubuntu@ip-172-31-18-170:~$ vi docker.sh
```

```
#!/bin/bash

# Update package manager repositories
sudo apt-get update

# Install necessary dependencies
sudo apt-get install -y ca-certificates curl

# Create directory for Docker GPG key
sudo install -m 0755 -d /etc/apt/keyrings

# Download Docker's GPG key
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc

# Ensure proper permissions for the key
sudo chmod a+r /etc/apt/keyrings/docker.asc

# Add Docker repository to Apt sources
echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu \
$(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null

# Update package manager repositories
sudo apt-get update

sudo apt-get install -y docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
~
~
~
~
~
~
```

```
ubuntu@ip-172-31-18-170:~$ ls
docker.sh  jenkins.sh
ubuntu@ip-172-31-18-170:~$ sudo chmod +x docker.sh
ubuntu@ip-172-31-18-170:~$ ls
docker.sh  jenkins.sh
ubuntu@ip-172-31-18-170:~$ ./docker.sh
```

```
ubuntu@ip-172-31-18-170:~$ docker --version
Docker version 29.2.1, build a5c7197
ubuntu@ip-172-31-18-170:~$
```

```
ubuntu@ip-172-31-18-170:~$ docker pull hello-world
Using default tag: latest
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
ubuntu@ip-172-31-18-170:~$ sudo chmod 666 /var/run/docker.sock
ubuntu@ip-172-31-18-170:~$ docker pull hello-world
Using default tag: latest
latest: Pulling from library/hello-world
17eec7bbc9d7: Pull complete
ea52d2000f90: Download complete
Digest: sha256:05813aedc15fb7b4d732e1be879d3252c1c9c25d885824f6295cab4538cb85cd
Status: Downloaded newer image for hello-world:latest
docker.io/library/hello-world:latest
ubuntu@ip-172-31-18-170:~$
```

```
ubuntu@ip-172-31-18-170:~$ docker login -u deepanshu112

i Info → A Personal Access Token (PAT) can be used instead.
         To create a PAT, visit https://app.docker.com/settings


Password:

WARNING! Your credentials are stored unencrypted in '/home/ubuntu/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
ubuntu@ip-172-31-18-170:~$
```

## Trivy Installation

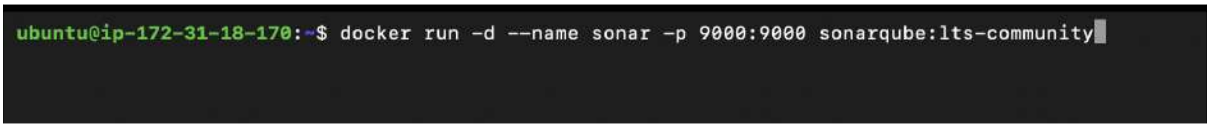
```
ubuntu@ip-172-31-18-170:~$ vi trivy.sh
```

```
#!/bin/bash
sudo apt-get install wget apt-transport-https gnupg
wget -qO - https://aquasecurity.github.io/trivy-repo/deb/public.key | gpg --dearmor | sudo tee /usr/share/keyrings/trivy.gpg > /dev/null
echo "deb [signed-by=/usr/share/keyrings/trivy.gpg] https://aquasecurity.github.io/trivy-repo/deb generic main" | sudo tee -a /etc/apt/sources.list.d/trivy.list
sudo apt-get update
sudo apt-get install trivy
~
~
~
~
~
~
~
~
```

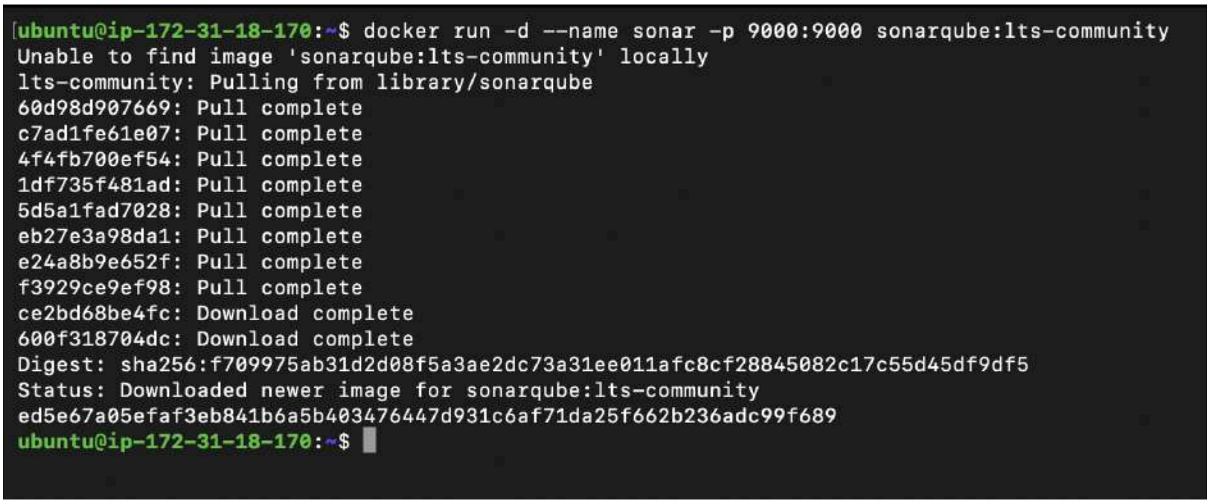
```
[ubuntu@ip-172-31-18-170:~$ vi trivy.sh
[ubuntu@ip-172-31-18-170:~$ ls
 docker.sh  jenkins.sh  trivy.sh
[ubuntu@ip-172-31-18-170:~$ sudo chmod +x trivy.sh
[ubuntu@ip-172-31-18-170:~$ ls
 docker.sh  jenkins.sh  trivy.sh
[ubuntu@ip-172-31-18-170:~$ ./trivy.sh
```

```
[ubuntu@ip-172-31-18-170:~$ trivy --version
 Version: 0.69.0
 ubuntu@ip-172-31-18-170:~$
```

## SonarQube Setup

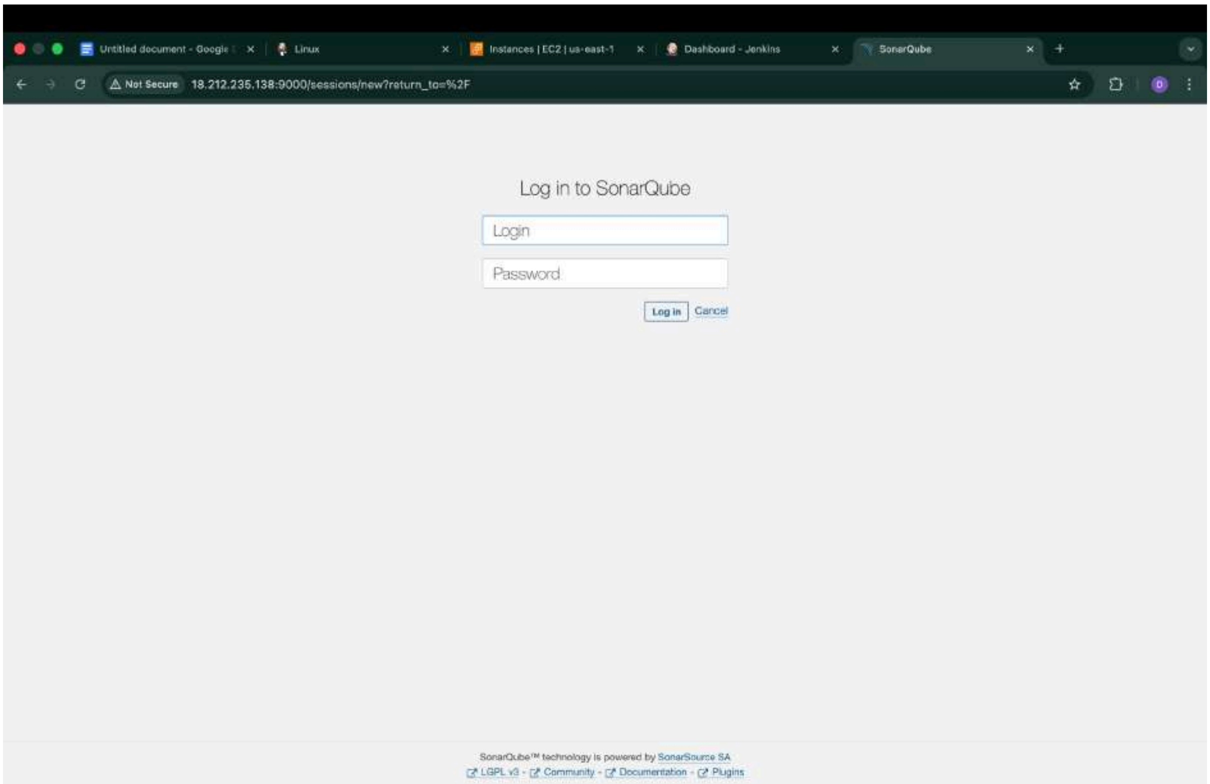


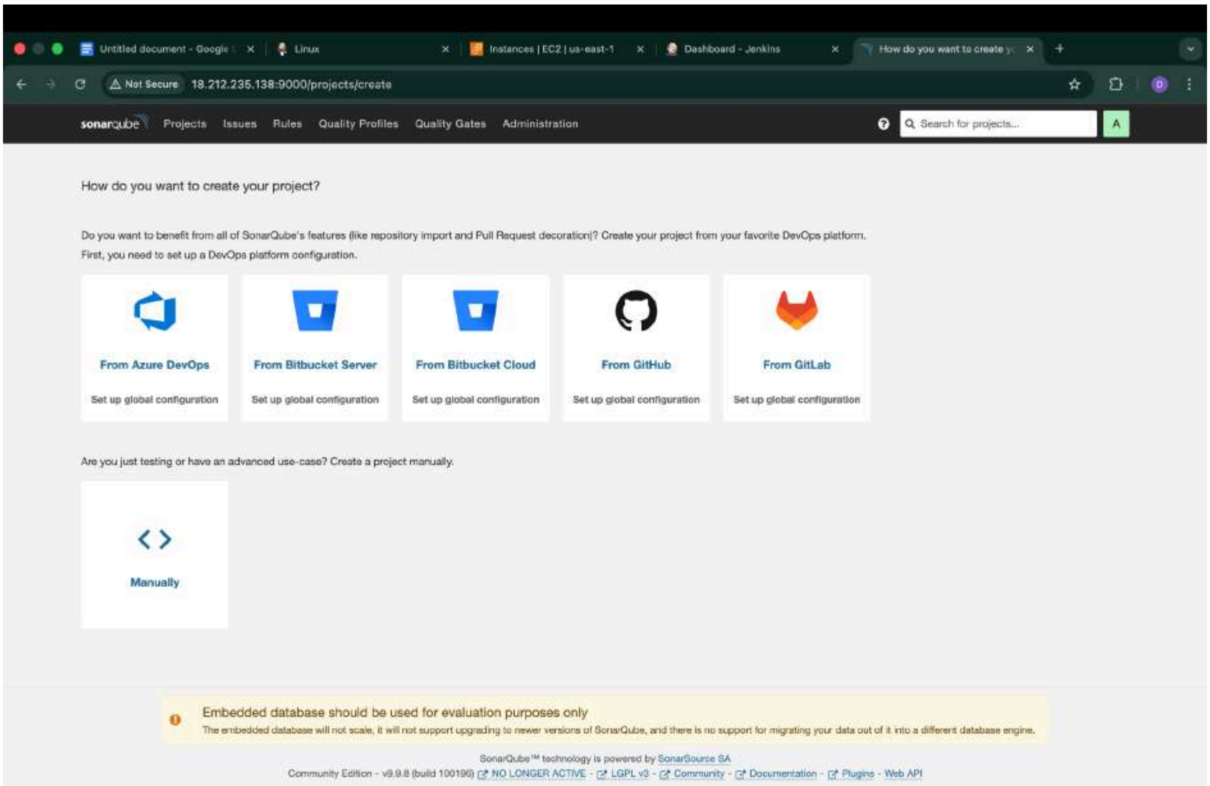
```
ubuntu@ip-172-31-18-170:~$ docker run -d --name sonar -p 9000:9000 sonarqube:lts-community
```



```
ubuntu@ip-172-31-18-170:~$ docker run -d --name sonar -p 9000:9000 sonarqube:lts-community
Unable to find image 'sonarqube:lts-community' locally
lts-community: Pulling from library/sonarqube
60d98d907669: Pull complete
c7ad1fe61e07: Pull complete
4f4fb700ef54: Pull complete
1df735f481ad: Pull complete
5d5a1fad7028: Pull complete
eb27e3a98da1: Pull complete
e24a8b9e652f: Pull complete
f3929ce9ef98: Pull complete
ce2bd68be4fc: Download complete
600f318704dc: Download complete
Digest: sha256:f709975ab31d2d08f5a3ae2dc73a31ee011afc8cf28845082c17c55d45df9df5
Status: Downloaded newer image for sonarqube:lts-community
ed5e67a05efaf3eb841b6a5b403476447d931c6af71da25f662b236adc99f689
ubuntu@ip-172-31-18-170:~$
```

```
ubuntu@ip-172-31-18-170:~$ docker images                                               i Info →  U In Use

IMAGE                    ID             DISK USAGE   CONTENT SIZE   EXTRA
sonarqube:lts-community  f709975ab31d       1.02GB          398MB     U
ubuntu@ip-172-31-18-170:~$ docker ps
CONTAINER ID   IMAGE                     COMMAND                  CREATED          STATUS          PORTS                                         NAMES
ed5e67a05efa   sonarqube:lts-community   "/opt/sonarqube/dock…"   35 seconds ago   Up 31 seconds   0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp   sonar
ubuntu@ip-172-31-18-170:~$
```





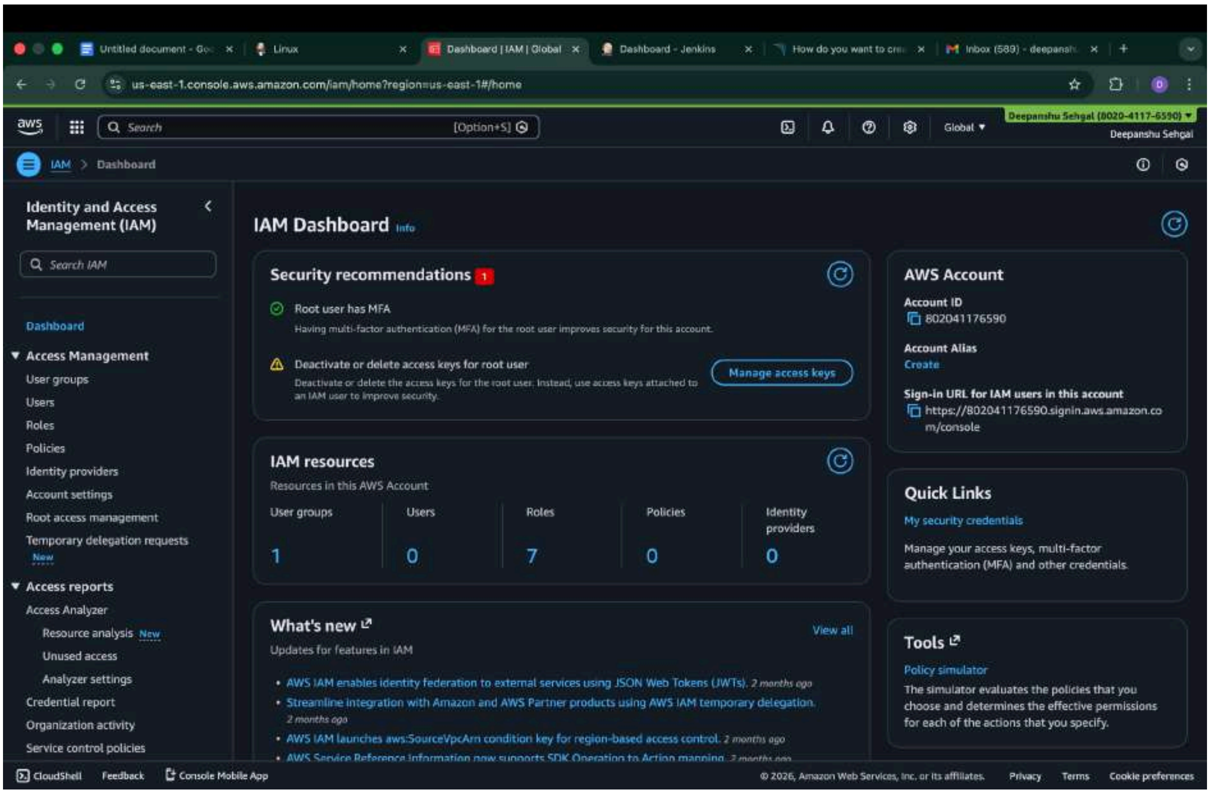
Untitled document - Google
Linux
Instances | EC2 | us-east-1
Dashboard - Jenkins
How do you want to create y
Not Secure
18.212.235.138:9000/projects/create
sonarqube
Projects
Issues
Rules
Quality Profiles
Quality Gates
Administration
Search for projects...
How do you want to create your project?
Do you want to benefit from all of SonarQube's features (like repository import and Pull Request decoration)? Create your project from your favorite DevOps platform.
First, you need to set up a DevOps platform configuration.
From Azure DevOps
Set up global configuration
From Bitbucket Server
Set up global configuration
From Bitbucket Cloud
Set up global configuration
From GitHub
Set up global configuration
From GitLab
Set up global configuration
Are you just testing or have an advanced use-case? Create a project manually.
Manually
Embedded database should be used for evaluation purposes only
The embedded database will not scale, it will not support upgrading to newer versions of SonarQube, and there is no support for migrating your data out of it into a different database engine.
SonarQube™ technology is powered by SonarSource SA
Community Edition - v9.9.8 (build 100196) - NO LONGER ACTIVE - LGPL v3 - Community - Documentation - Plugins - Web API

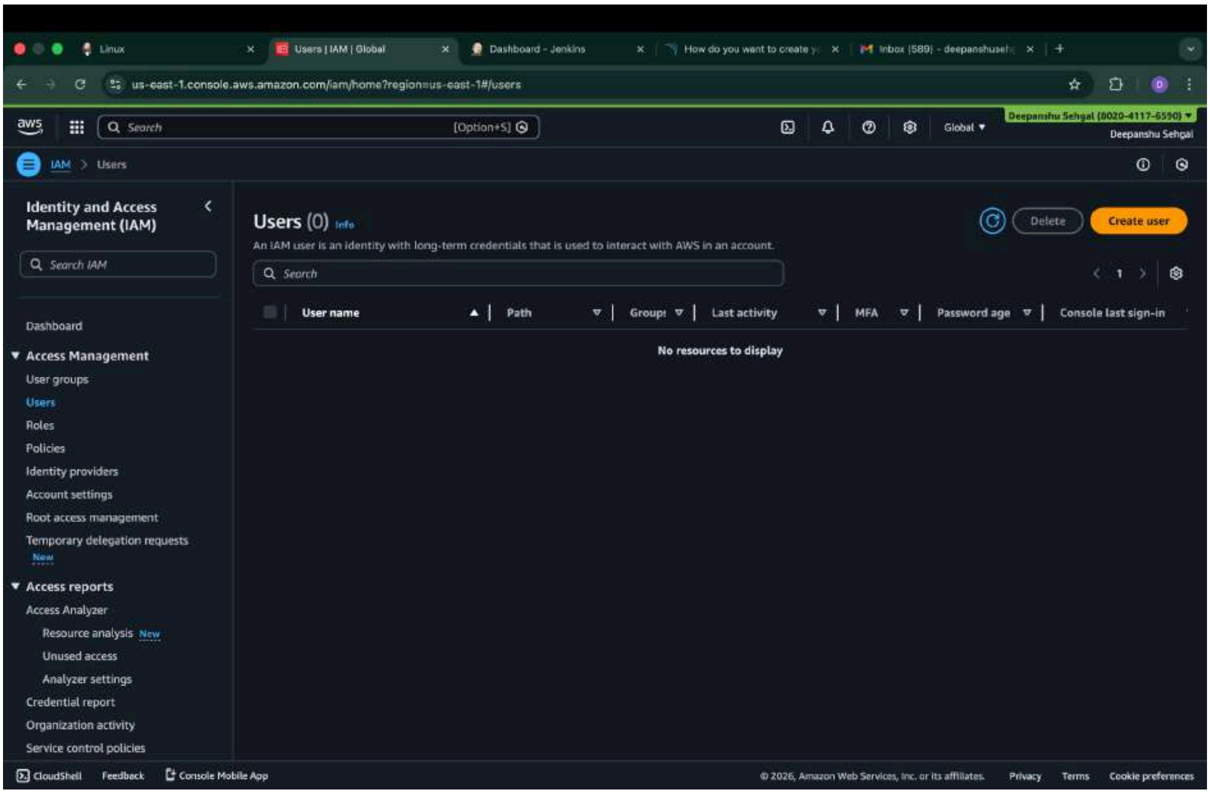
## 1.2 IAM User for EKS

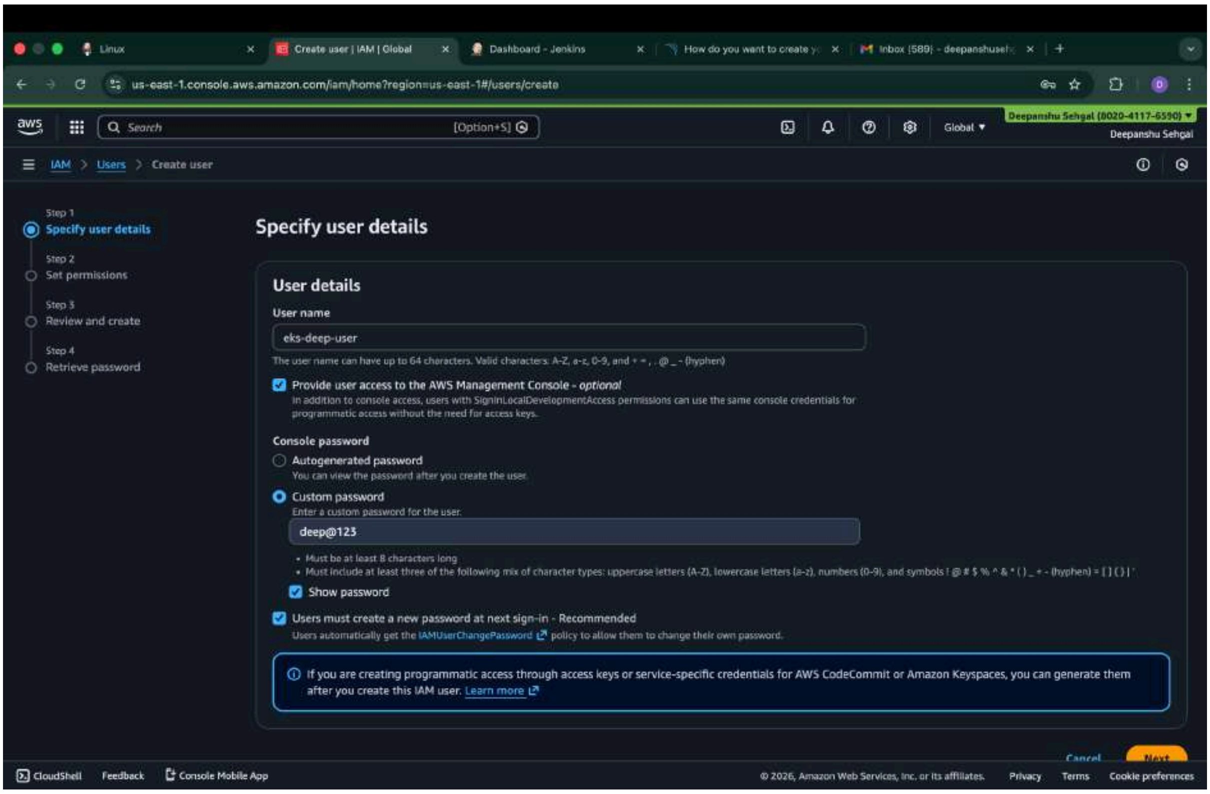
Create IAM user (avoid root account).

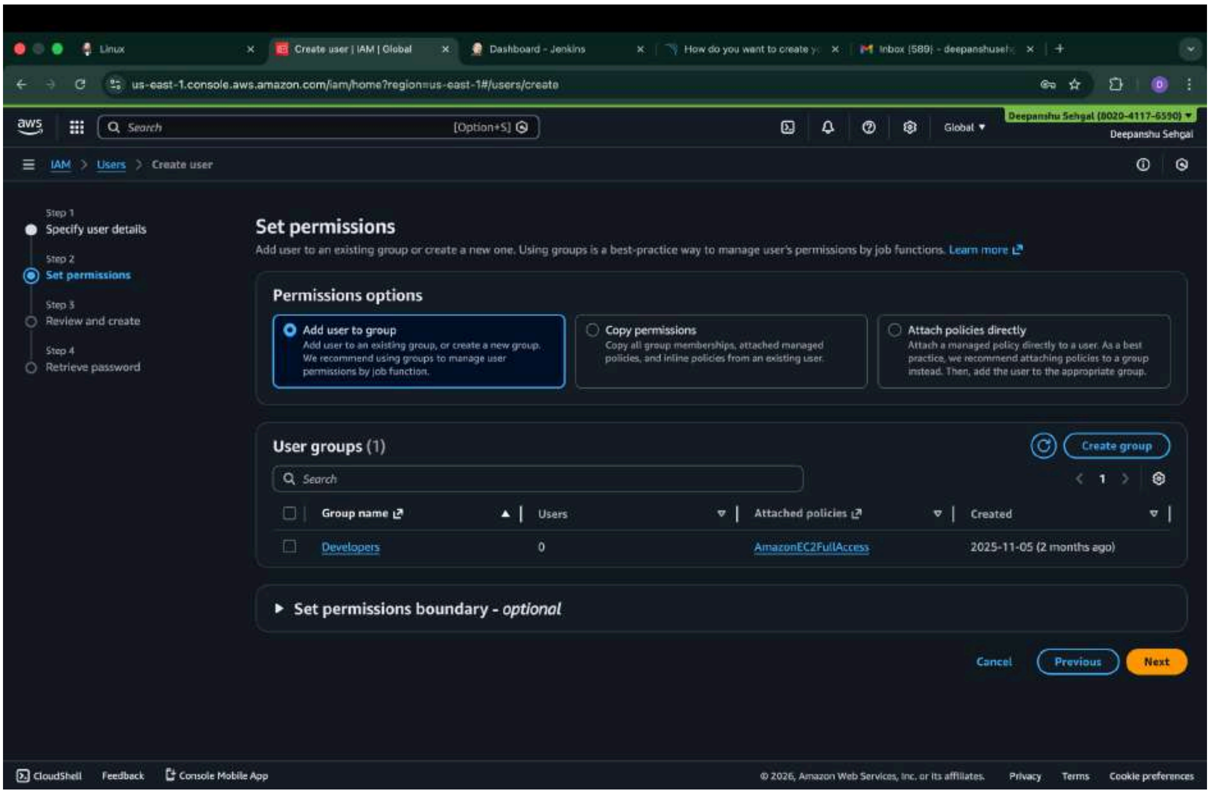
### Attach Policies

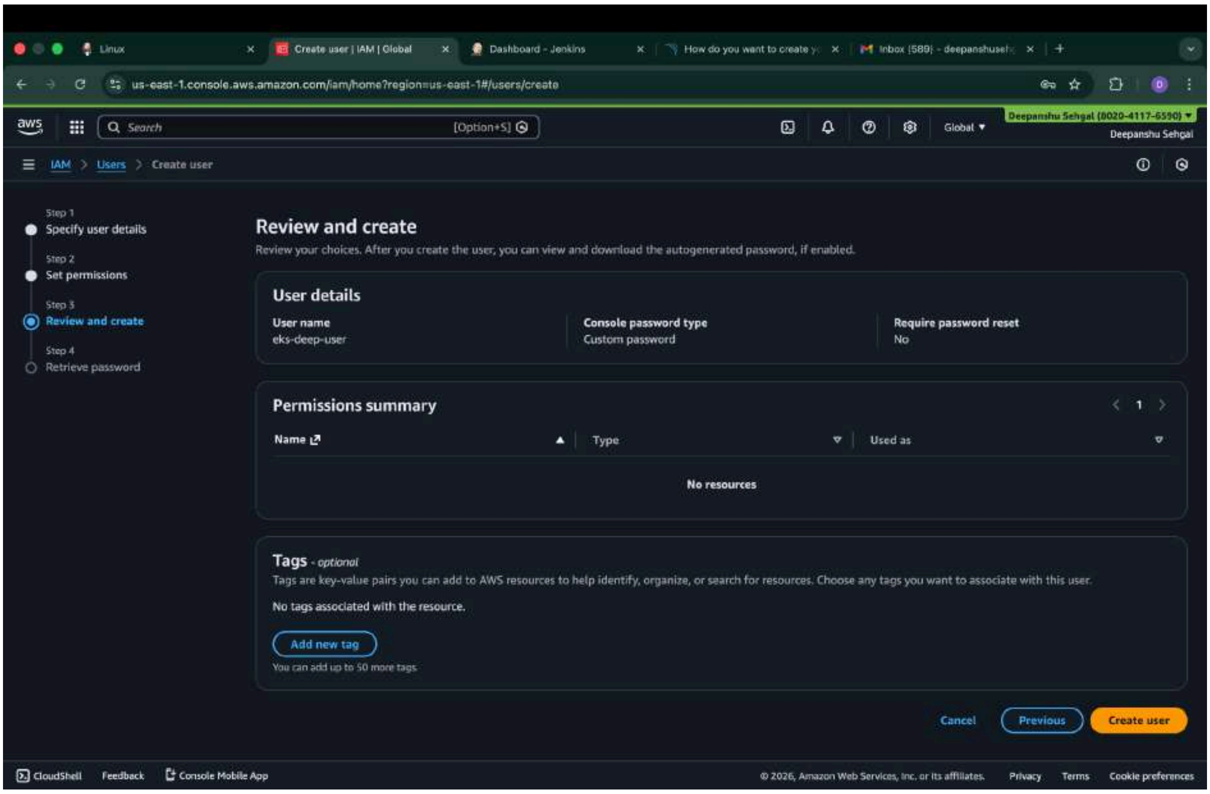
- AmazonEC2FullAccess
- AmazonEKSClusterPolicy
- AmazonEKSWorkerNodePolicy
- AmazonEKS_CNI_Policy
- AWSCloudFormationFullAccess
- IAMFullAccess

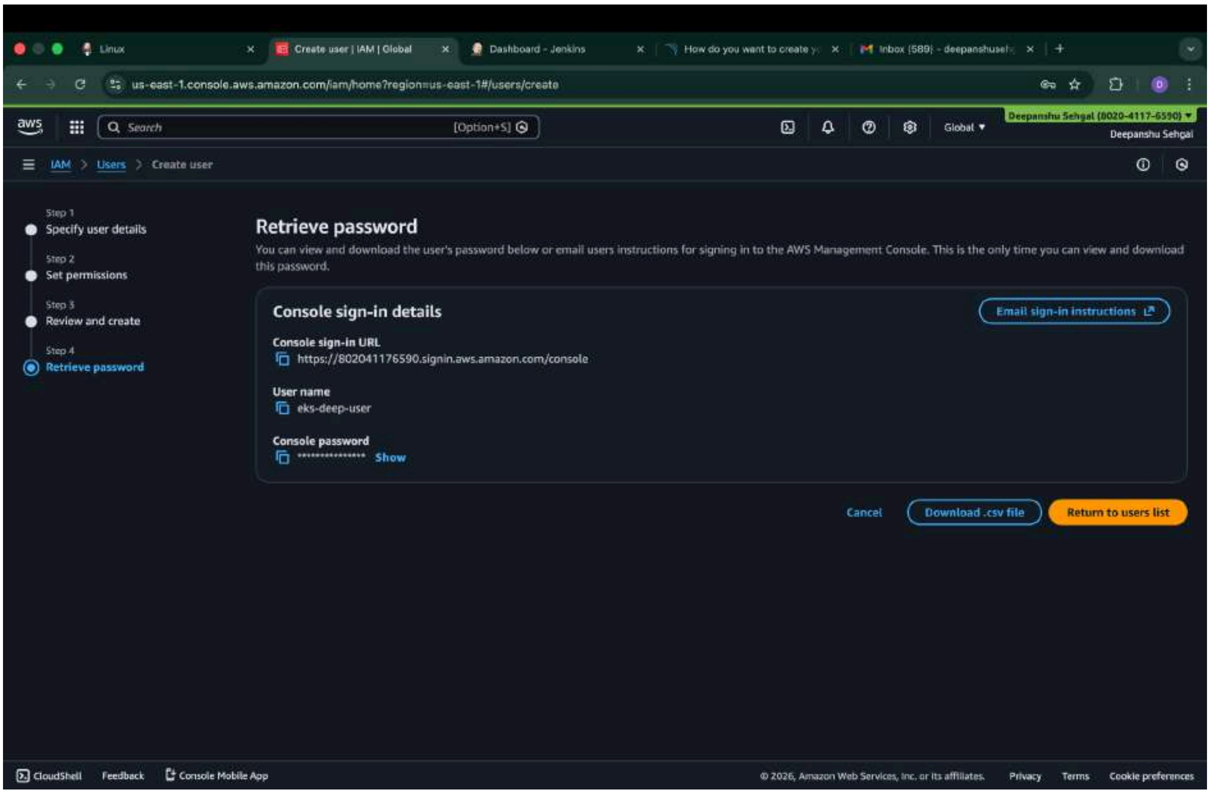


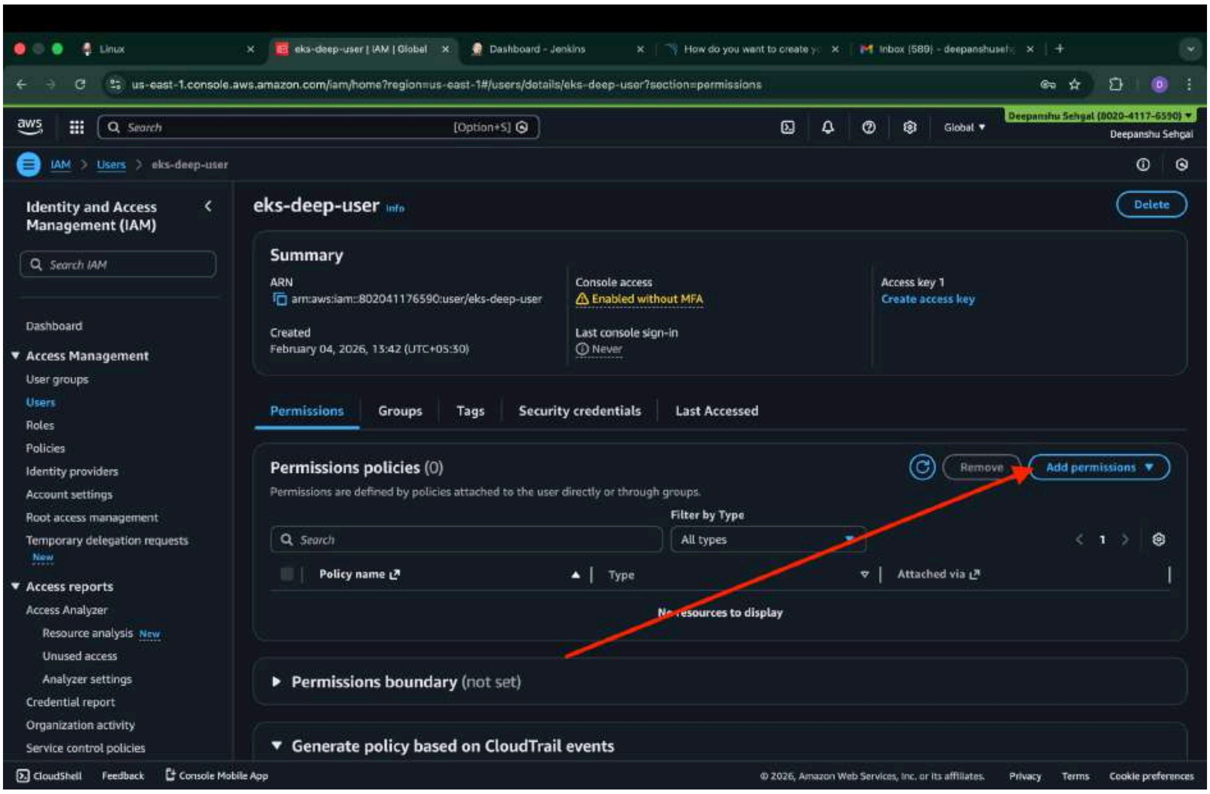

Linux
Users | IAM | Global
Dashboard - Jenkins
How do you want to create y
Inbox (589) - deepanshusehg
us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users
Search
[Option+S]
Global
Deepanshu Sehgal (8020-4117-6590)
Deepanshu Sehgal
IAM
Users
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
User groups
Users
Roles
Policies
Identity providers
Account settings
Root access management
Temporary delegation requests
New
Access reports
Access Analyzer
Resource analysis New
Unused access
Analyzer settings
Credential report
Organization activity
Service control policies
Users (0) Info
An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.
Delete
Create user
Search
User name
Path
Groups
Last activity
MFA
Password age
Console last sign-in
No resources to display
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

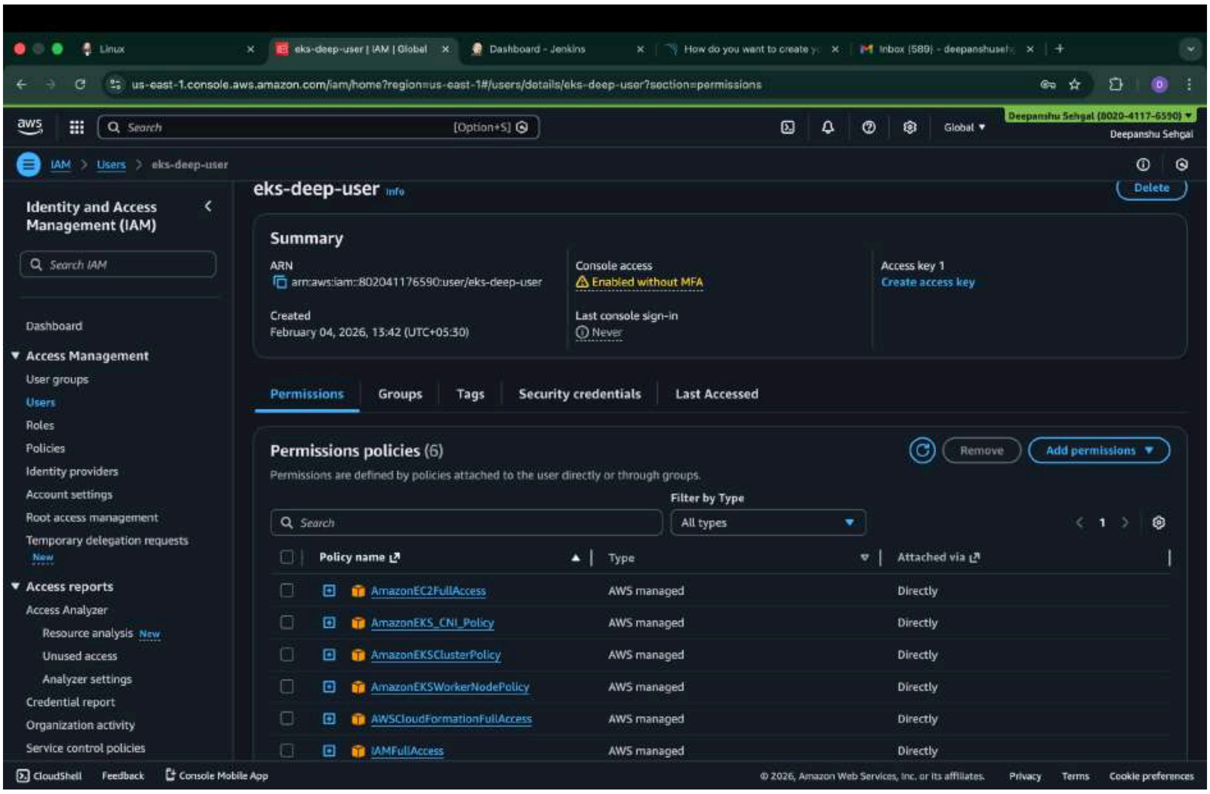

Linux
Create user | IAM | Global
Dashboard - Jenkins
How do you want to create y
Inbox (589) - deepanshusehg
us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/create
Search
[Option+S]
Global
Deepanshu Sehgal (8020-4117-6590)
Deepanshu Sehgal
IAM
Users
Create user
Step 1
Specify user details
Step 2
Set permissions
Step 3
Review and create
Step 4
Retrieve password
Specify user details
User details
User name
eks-deep-user
The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)
Provide user access to the AWS Management Console - optional
In addition to console access, users with SigninLocalDevelopmentAccess permissions can use the same console credentials for programmatic access without the need for access keys.
Console password
Autogenerated password
You can view the password after you create the user.
Custom password
Enter a custom password for the user.
deep@123
Must be at least 8 characters long
Show password
Users must create a new password at next sign-in - Recommended
Users automatically get the IAMUserChangePassword policy to allow them to change their own password.
If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. Learn more
Cancel
Next
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Set permissions
Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. Learn more
Permissions options
Add user to group
Copy permissions
Attach policies directly
User groups (1)
Create group
Group name
Users
Attached policies
Created
Developers
0
AmazonEC2FullAccess
2025-11-05 (2 months ago)
Set permissions boundary - optional
Cancel
Previous
Next


Review and create
Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.
User details
User name
eks-deep-user
Console password type
Custom password
Require password reset
No
Permissions summary
No resources
Tags - optional
No tags associated with the resource.
Add new tag
Cancel
Previous
Create user






us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/details/eks-deep-user?section=permissions
IAM > Users > eks-deep-user
eks-deep-user
Delete
Identity and Access Management (IAM)
Summary
ARN
arn:aws:iam::802041176590:user/eks-deep-user
Console access
Enabled without MFA
Access key 1
Create access key
Created
February 04, 2026, 13:42 (UTC+05:30)
Last console sign-in
Never
Dashboard
Access Management
User groups
Users
Roles
Policies
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Unused access
Analyzer settings
Credential report
Organization activity
Service control policies
Permissions
Groups
Tags
Security credentials
Last Accessed
Permissions policies (6)
Remove
Add permissions
Permissions are defined by policies attached to the user directly or through groups.
Filter by Type
All types
Policy name
Type
Attached via
AmazonEC2FullAccess
AWS managed
Directly
AmazonEKS_CNI_Policy
AWS managed
Directly
AmazonEKSClusterPolicy
AWS managed
Directly
AmazonEKSWorkerNodePolicy
AWS managed
Directly
AWSCloudFormationFullAccess
AWS managed
Directly
IAMFullAccess
AWS managed
Directly
CloudShell
Feedback
Console Mobile App
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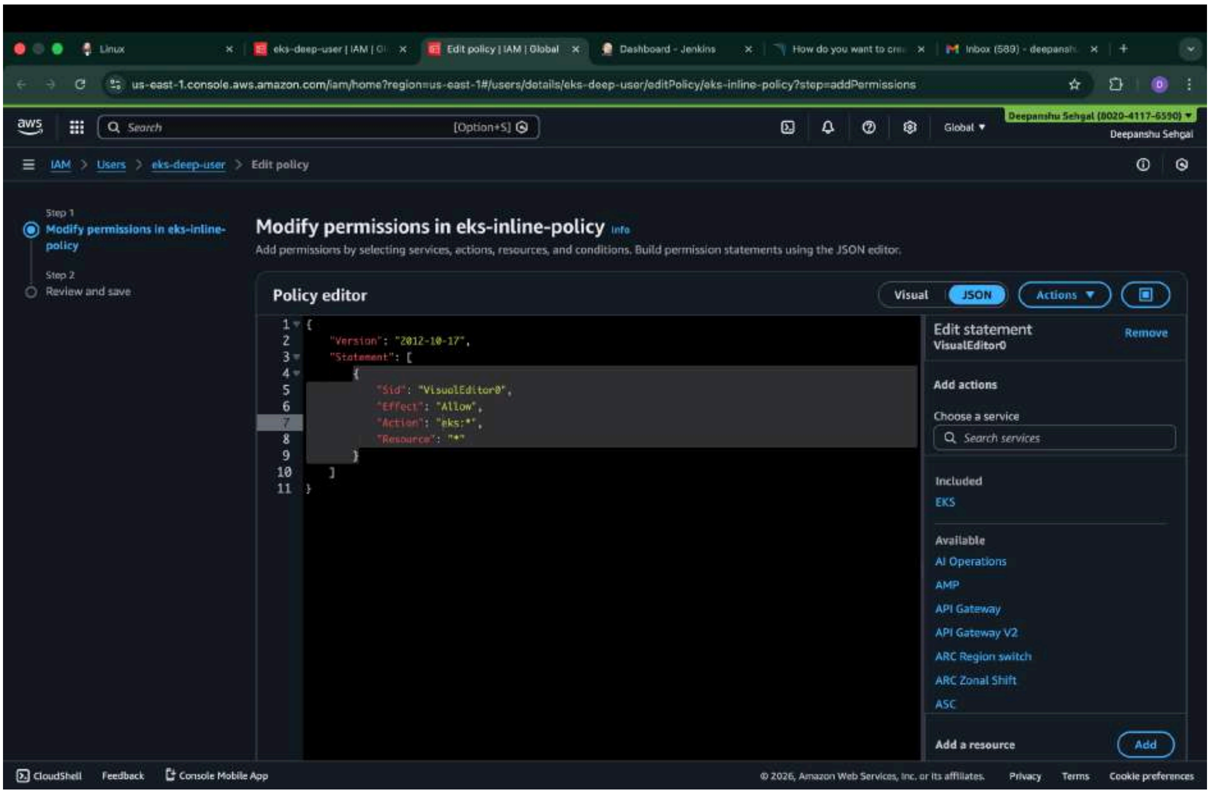
us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/details/eks-deep-user/editPolicy/eks-inline-policy?step=addPermissions
IAM > Users > eks-deep-user > Edit policy
Step 1
Modify permissions in eks-inline-policy
Step 2
Review and save
Modify permissions in eks-inline-policy
Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.
Policy editor
Visual
JSON
Actions
{
"Version": "2012-10-17",
"Statement": [
{
"Sid": "VisualEditor0",
"Effect": "Allow",
"Action": "eks:*",
"Resource": "*"
}
]
}
Edit statement
VisualEditor0
Remove
Add actions
Choose a service
Included
EKS
Available
AI Operations
AMP
API Gateway
API Gateway V2
ARC Region switch
ARC Zonal Shift
ASC
Add a resource
Add
CloudShell
Feedback
Console Mobile App
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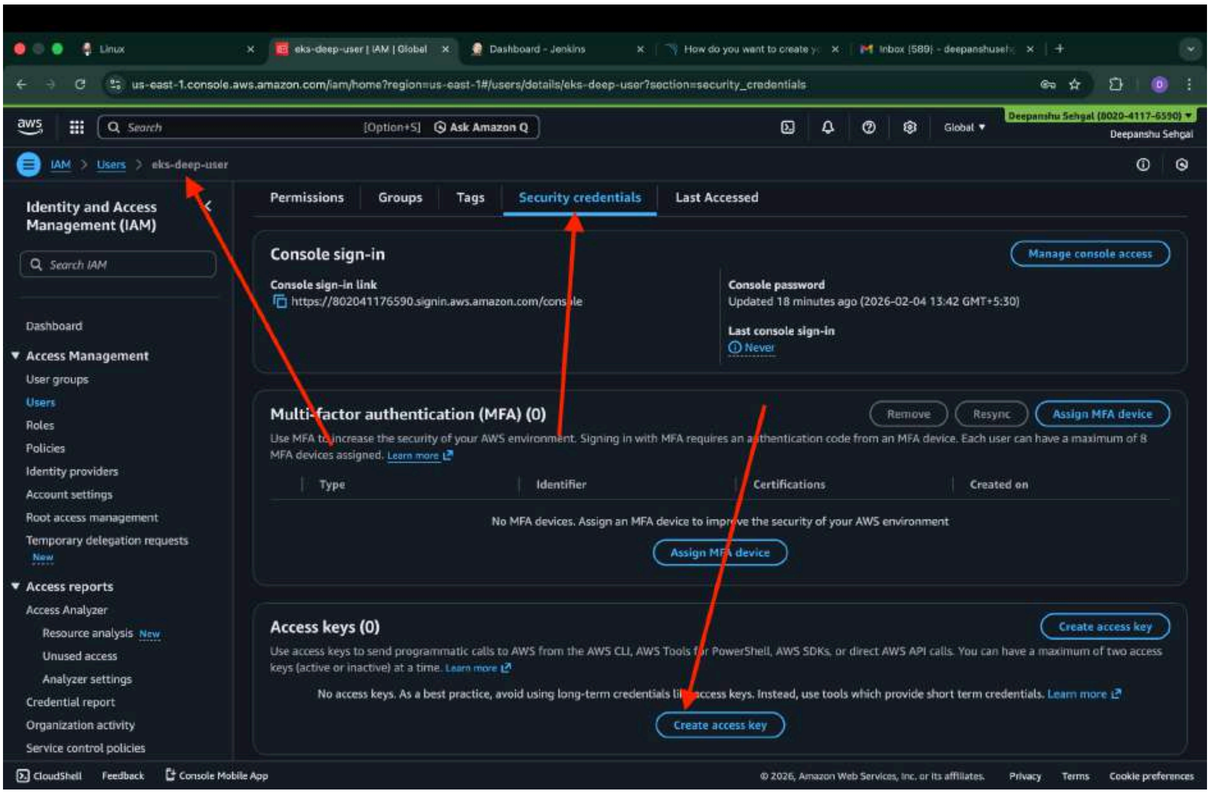
us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/details/eks-deep-user?section=security_credentials
IAM > Users > eks-deep-user
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
User groups
Users
Roles
Policies
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Unused access
Analyzer settings
Credential report
Organization activity
Service control policies
Permissions
Groups
Tags
Security credentials
Last Accessed
Console sign-in
Manage console access
Console sign-in link
https://802041176590.signin.aws.amazon.com/console
Console password
Updated 18 minutes ago (2026-02-04 13:42 GMT+5:30)
Last console sign-in
Never
Multi-factor authentication (MFA) (0)
Remove
Resync
Assign MFA device
Use MFA to increase the security of your AWS environment. Signing in with MFA requires an authentication code from an MFA device. Each user can have a maximum of 8 MFA devices assigned. Learn more
Type
Identifier
Certifications
Created on
No MFA devices. Assign an MFA device to improve the security of your AWS environment
Assign MFA device
Access keys (0)
Create access key
Use access keys to send programmatic calls to AWS from the AWS CLI, AWS Tools for PowerShell, AWS SDKs, or direct AWS API calls. You can have a maximum of two access keys (active or inactive) at a time. Learn more
No access keys. As a best practice, avoid using long-term credentials like access keys. Instead, use tools which provide short term credentials. Learn more
Create access key
CloudShell
Feedback
Console Mobile App
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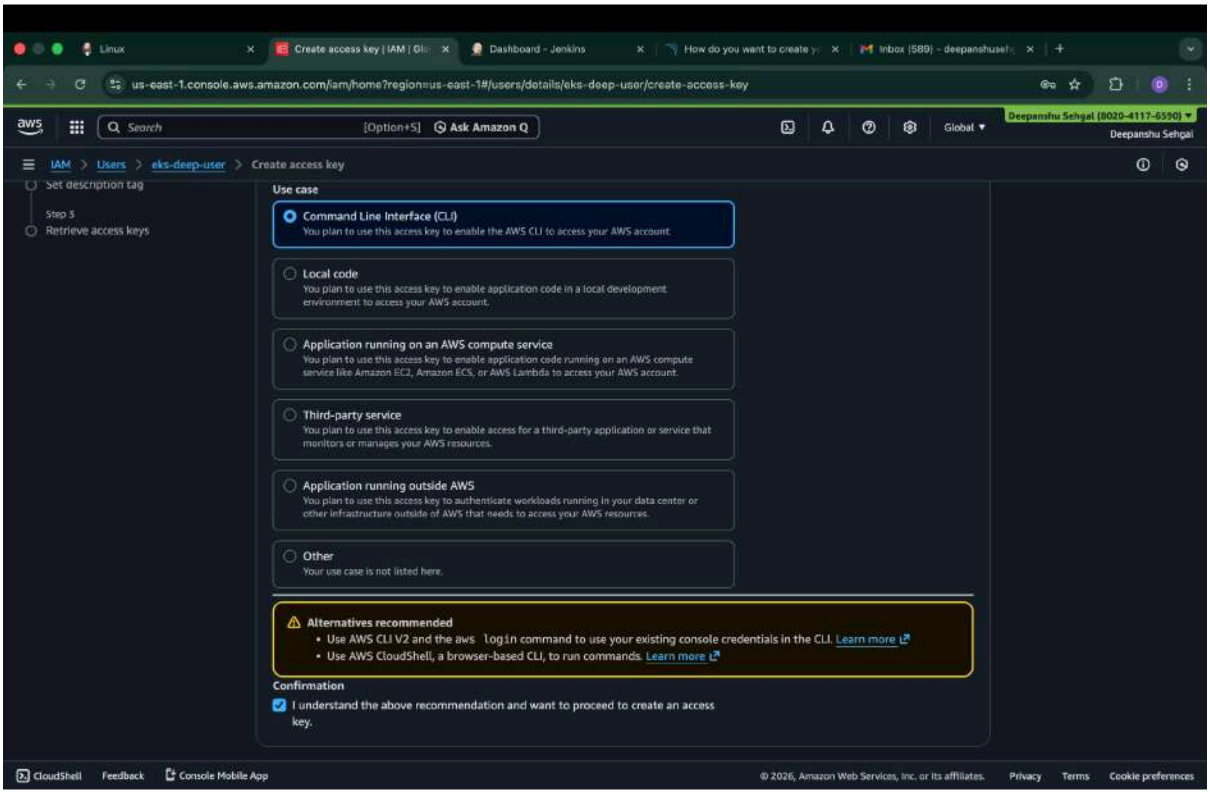
us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/users/details/eks-deep-user/create-access-key
IAM > Users > eks-deep-user > Create access key
Set description tag
Step 3
Retrieve access keys
Use case
Command Line Interface (CLI)
You plan to use this access key to enable the AWS CLI to access your AWS account.
Local code
You plan to use this access key to enable application code in a local development environment to access your AWS account.
Application running on an AWS compute service
You plan to use this access key to enable application code running on an AWS compute service like Amazon EC2, Amazon ECS, or AWS Lambda to access your AWS account.
Third-party service
You plan to use this access key to enable access for a third-party application or service that monitors or manages your AWS resources.
Application running outside AWS
You plan to use this access key to authenticate workloads running in your data center or other infrastructure outside of AWS that needs to access your AWS resources.
Other
Your use case is not listed here.
Alternatives recommended
Use AWS CLI V2 and the aws login command to use your existing console credentials in the CLI. Learn more
Use AWS CloudShell, a browser-based CLI, to run commands. Learn more
Confirmation
I understand the above recommendation and want to proceed to create an access key.
CloudShell
Feedback
Console Mobile App
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Privacy
Terms
Cookie preferences

# Step 3 EKS Cluster Setup

## Install Tools

### AWS CLI

```
ubuntu@ip-172-31-18-170:~$ curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
sudo apt install unzip
unzip awscliv2.zip
sudo ./aws/install
aws configure
```

### kubectl

```
ubuntu@ip-172-31-18-170:~$ vi kubectl.sh
```

```
curl -o kubectl https://amazon-eks.s3.us-west-2.amazonaws.com/1.19.6/2021-01-05/bin/linux/amd64/kubectl
chmod +x ./kubectl
sudo mv ./kubectl /usr/local/bin
kubectl version --short --client
~
~
~
```

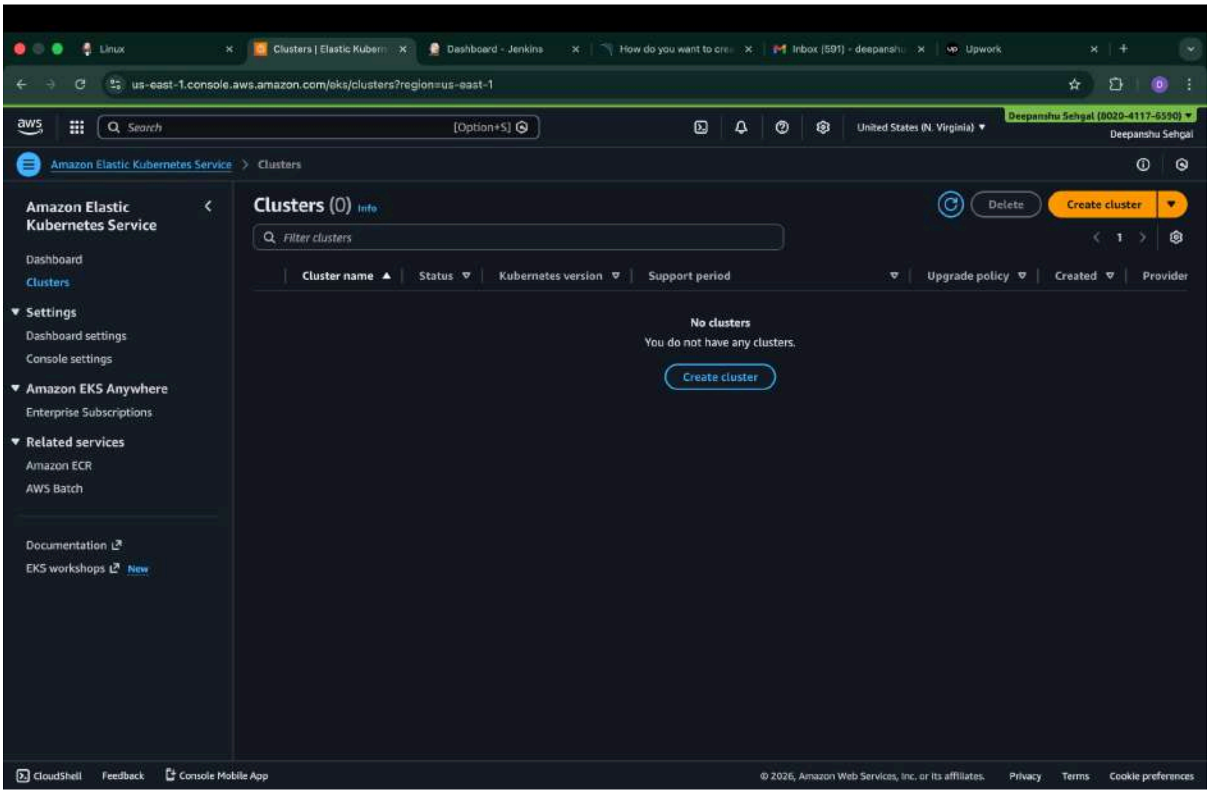
```
ubuntu@ip-172-31-18-170:~$ ls
aws  aws.sh  awscliv2.zip  docker.sh  jenkins.sh  kubectl.sh  trivy.sh
ubuntu@ip-172-31-18-170:~$ sudo chmod +x kubectl.sh
ubuntu@ip-172-31-18-170:~$ ls
aws  aws.sh  awscliv2.zip  docker.sh  jenkins.sh  kubectl.sh  trivy.sh
ubuntu@ip-172-31-18-170:~$ ./kubectl.sh
```

### Eksctl

```
curl --silent --location "https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_$(uname -s)_amd64.tar.gz" | tar xz -C /tmp
sudo mv /tmp/eksctl /usr/local/bin
eksctl version
~
~
```

```
ubuntu@ip-172-31-18-170:~$ ls
aws  aws.sh  awscliv2.zip  docker.sh  eksctl.sh  jenkins.sh  kubectl.sh  trivy.sh
ubuntu@ip-172-31-18-170:~$ sudo chmod +x eksctl.sh
ubuntu@ip-172-31-18-170:~$ ls
aws  aws.sh  awscliv2.zip  docker.sh  eksctl.sh  jenkins.sh  kubectl.sh  trivy.sh
ubuntu@ip-172-31-18-170:~$ ./eksctl.sh
```

## Create Cluster




```
ubuntu@ip-172-31-18-170:~$ eksctl create cluster --name=deepanshu-eks \
                          --region=us-east-1 \
                          --zones=us-east-1a,us-east-1b \
                          --version=1.30 \
                          --without-nodegroup
```

This process may take around 10–15 minutes to complete. In the meantime, we will proceed with the Jenkins setup and plugin installation.

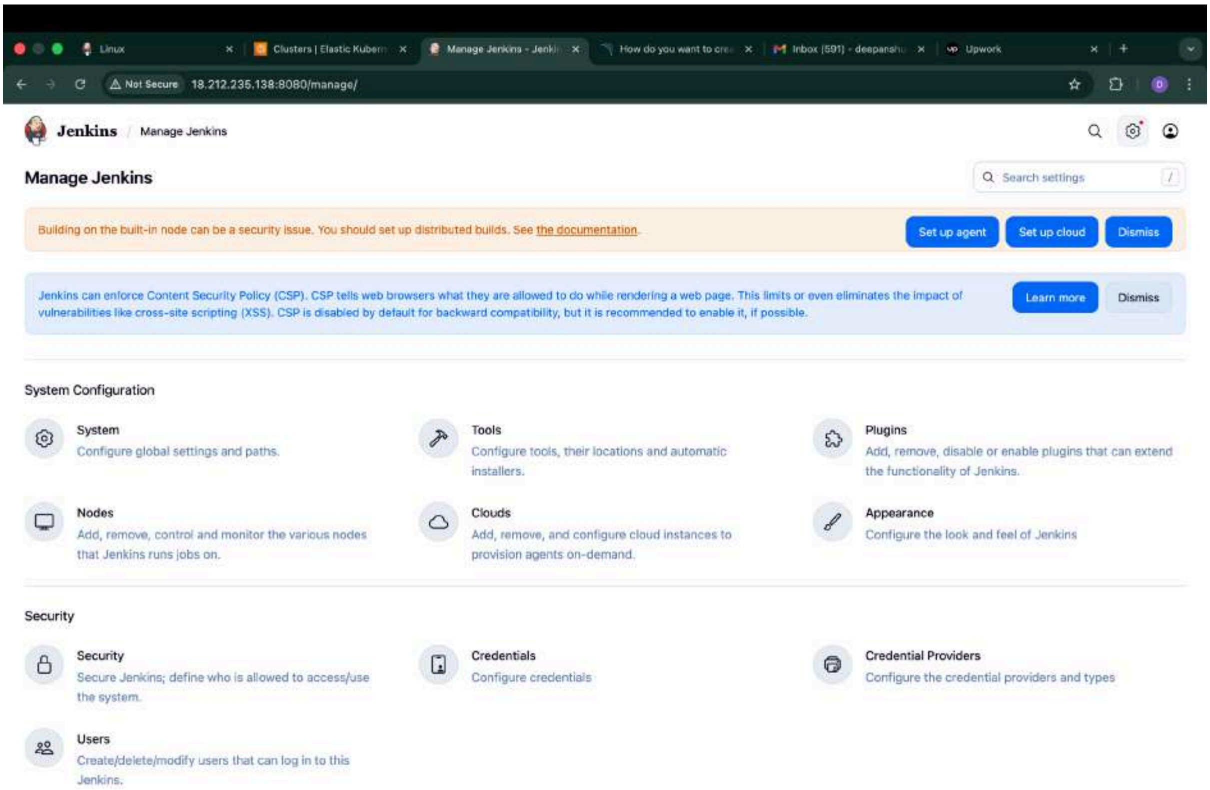
# Step 3: Jenkins Configuration

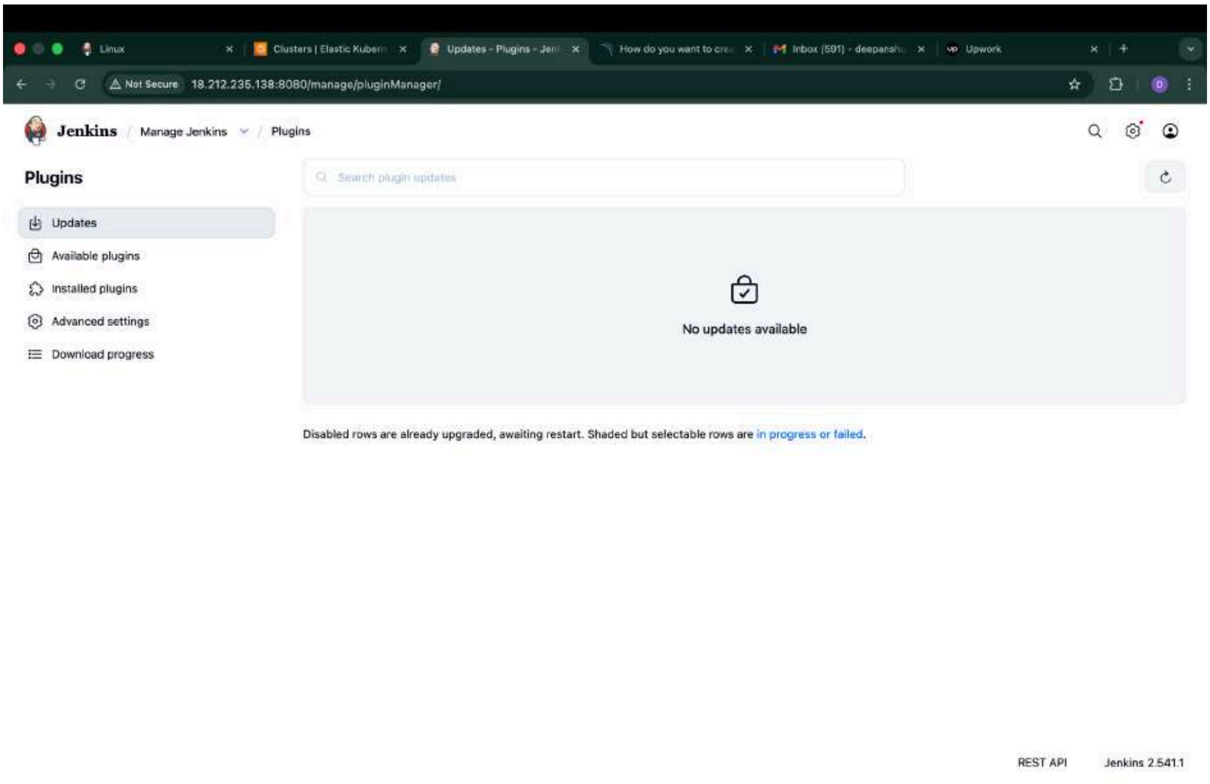
## Install Plugins:

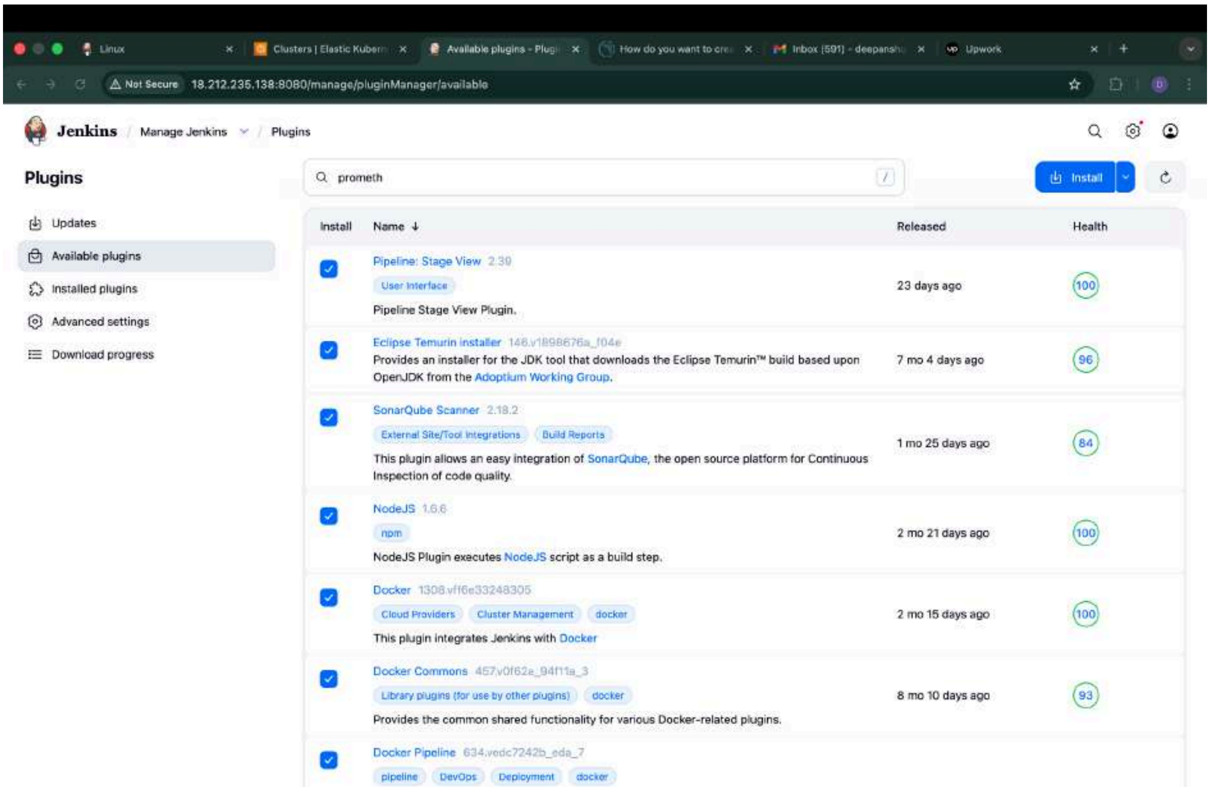
- Docker
- NodeJS
- SonarQube Scanner
- OWASP Dependency Check
- Kubernetes plugins
- Email Extension
- Prometheus metrics
- Pipeline Stage View

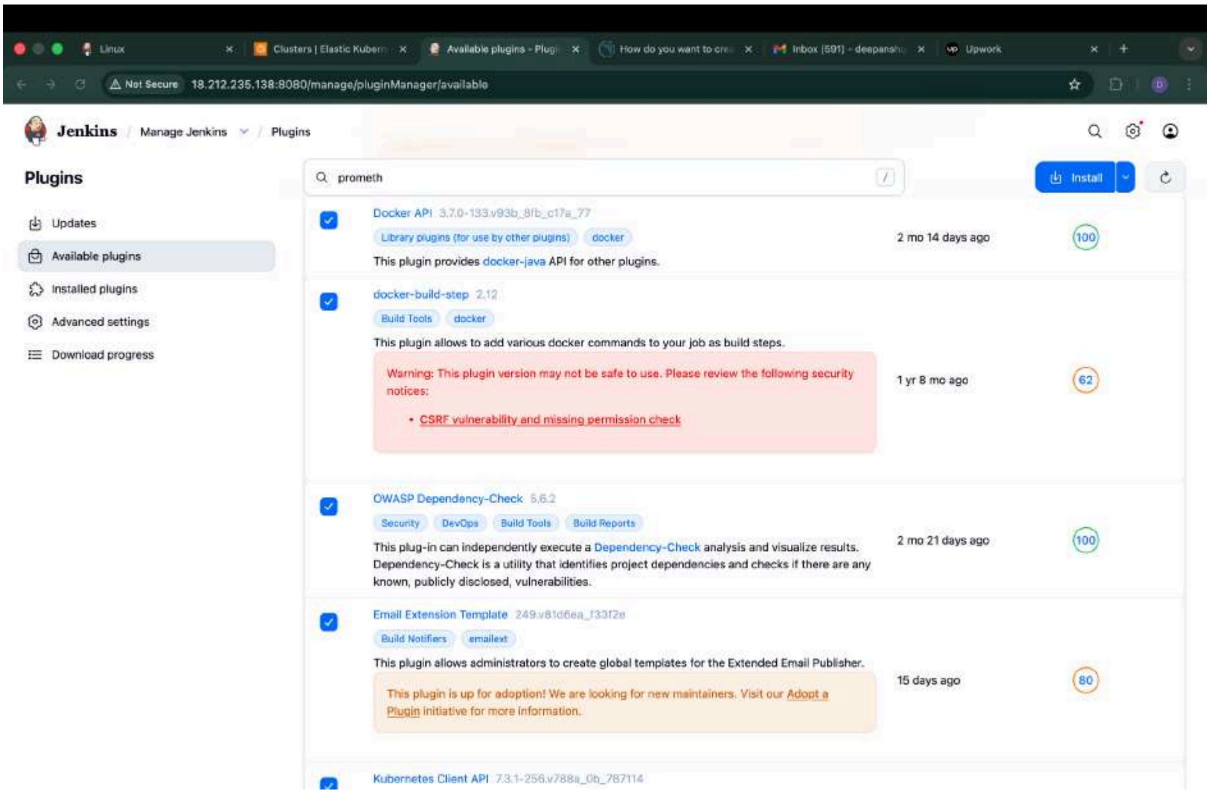
## Configure:

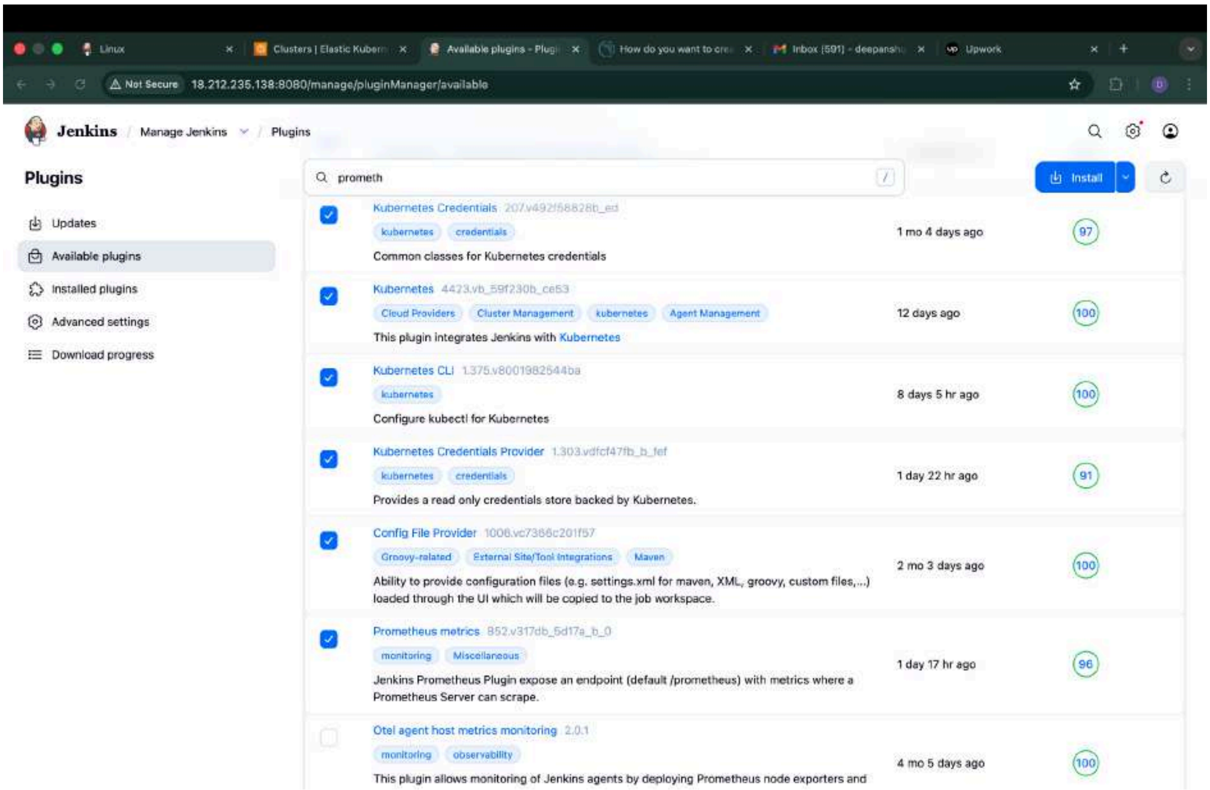
- SonarQube server
- Credentials
- Tools (JDK, NodeJS, Scanner)

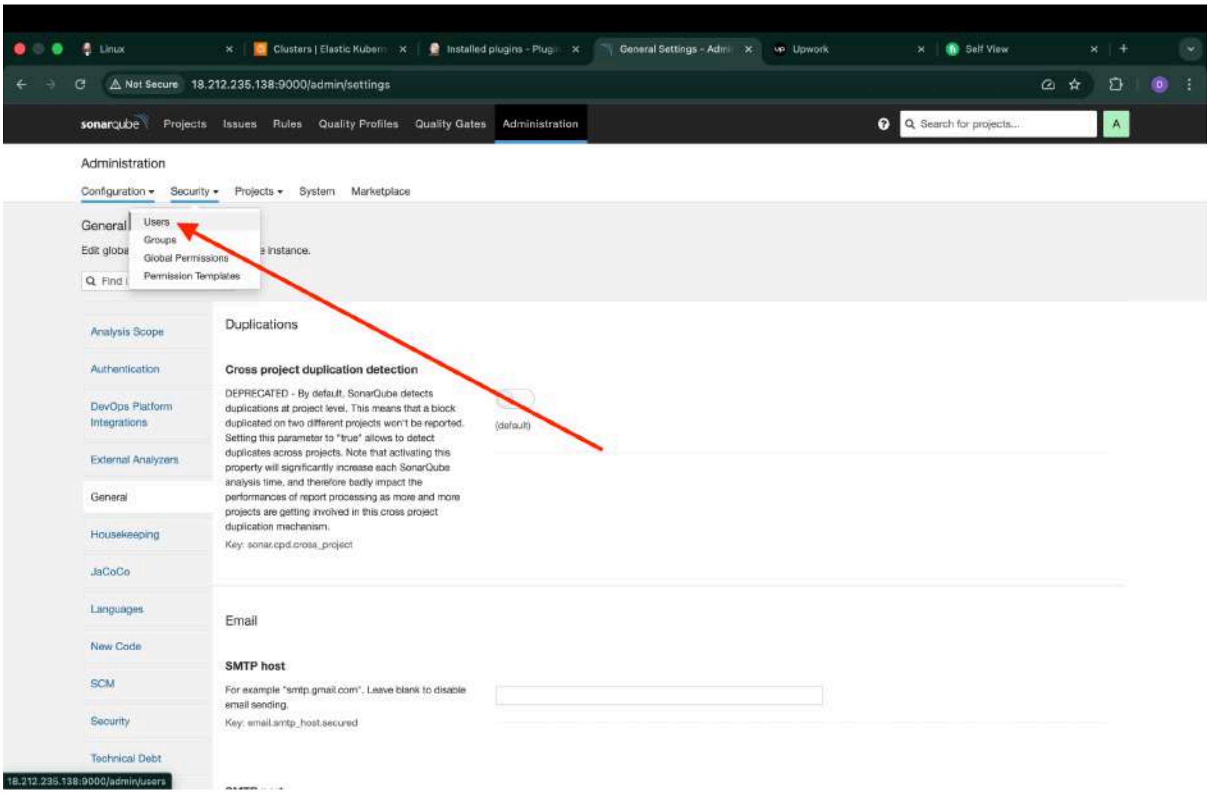



Linux
Clusters | Elastic Kubern
Updates - Plugins - Jen
How do you want to cre
Inbox (591) - deepansh
Upwork
Not Secure
18.212.235.138:8080/manage/pluginManager/
Jenkins
Manage Jenkins
Plugins
Plugins
Search plugin updates
Updates
Available plugins
Installed plugins
Advanced settings
Download progress
No updates available
Disabled rows are already upgraded, awaiting restart. Shaded but selectable rows are in progress or failed.
REST API
Jenkins 2.541.1

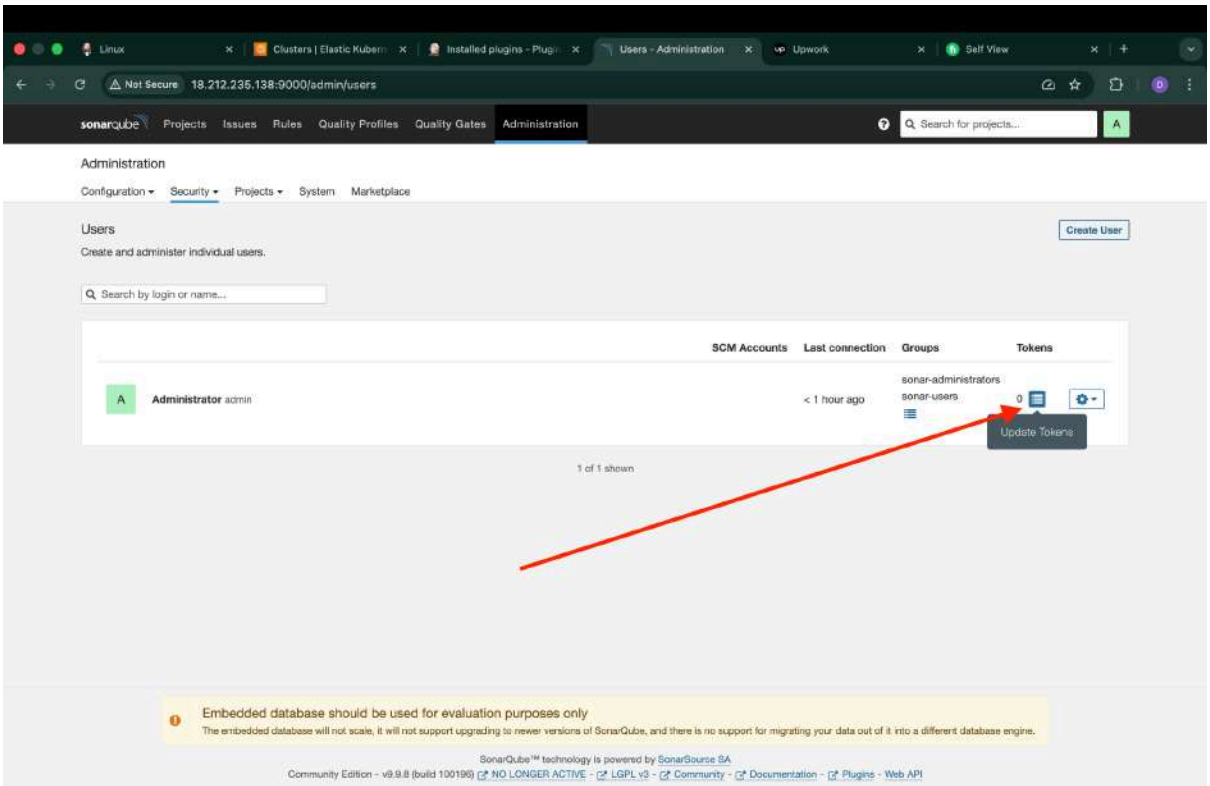

Linux
Clusters | Elastic Kubern
Available plugins - Plug
How do you want to cre
Inbox (591) - deepansh
Upwork
Not Secure
18.212.235.138:8080/manage/pluginManager/available
Jenkins
Manage Jenkins
Plugins
Plugins
prometh
Install
Updates
Available plugins
Installed plugins
Advanced settings
Download progress
Install
Name
Released
Health
Pipeline: Stage View 2.39
User Interface
Pipeline Stage View Plugin.
23 days ago
100
Eclipse Temurin installer 146.v1898676a_f04e
Provides an installer for the JDK tool that downloads the Eclipse Temurin™ build based upon OpenJDK from the Adoptium Working Group.
7 mo 4 days ago
96
SonarQube Scanner 2.18.2
External Site/Tool Integrations
Build Reports
This plugin allows an easy integration of SonarQube, the open source platform for Continuous Inspection of code quality.
1 mo 25 days ago
84
NodeJS 1.6.6
npm
NodeJS Plugin executes NodeJS script as a build step.
2 mo 21 days ago
100
Docker 1308.vf16e33248305
Cloud Providers
Cluster Management
docker
This plugin integrates Jenkins with Docker
2 mo 15 days ago
100
Docker Commons 457.v0f62a_94f11a_3
Library plugins (for use by other plugins)
docker
Provides the common shared functionality for various Docker-related plugins.
8 mo 10 days ago
93
Docker Pipeline 634.vedc7242b_eda_7
pipeline
DevOps
Deployment
docker


Not Secure 18.212.235.138:8080/manage/pluginManager/available
Jenkins
Manage Jenkins
Plugins
Plugins
prometh
Install
Updates
Available plugins
Installed plugins
Advanced settings
Download progress
Docker API
Library plugins (for use by other plugins)
docker
2 mo 14 days ago
100
This plugin provides docker-java API for other plugins.
docker-build-step 2.12
Build Tools
docker
This plugin allows to add various docker commands to your job as build steps.
Warning: This plugin version may not be safe to use. Please review the following security notices:
CSRF vulnerability and missing permission check
1 yr 8 mo ago
62
OWASP Dependency-Check 5.6.2
Security
DevOps
Build Tools
Build Reports
This plug-in can independently execute a Dependency-Check analysis and visualize results. Dependency-Check is a utility that identifies project dependencies and checks if there are any known, publicly disclosed, vulnerabilities.
2 mo 21 days ago
100
Email Extension Template
Build Notifiers
emailext
This plugin allows administrators to create global templates for the Extended Email Publisher.
This plugin is up for adoption! We are looking for new maintainers. Visit our Adopt a Plugin initiative for more information.
15 days ago
80
Kubernetes Client API

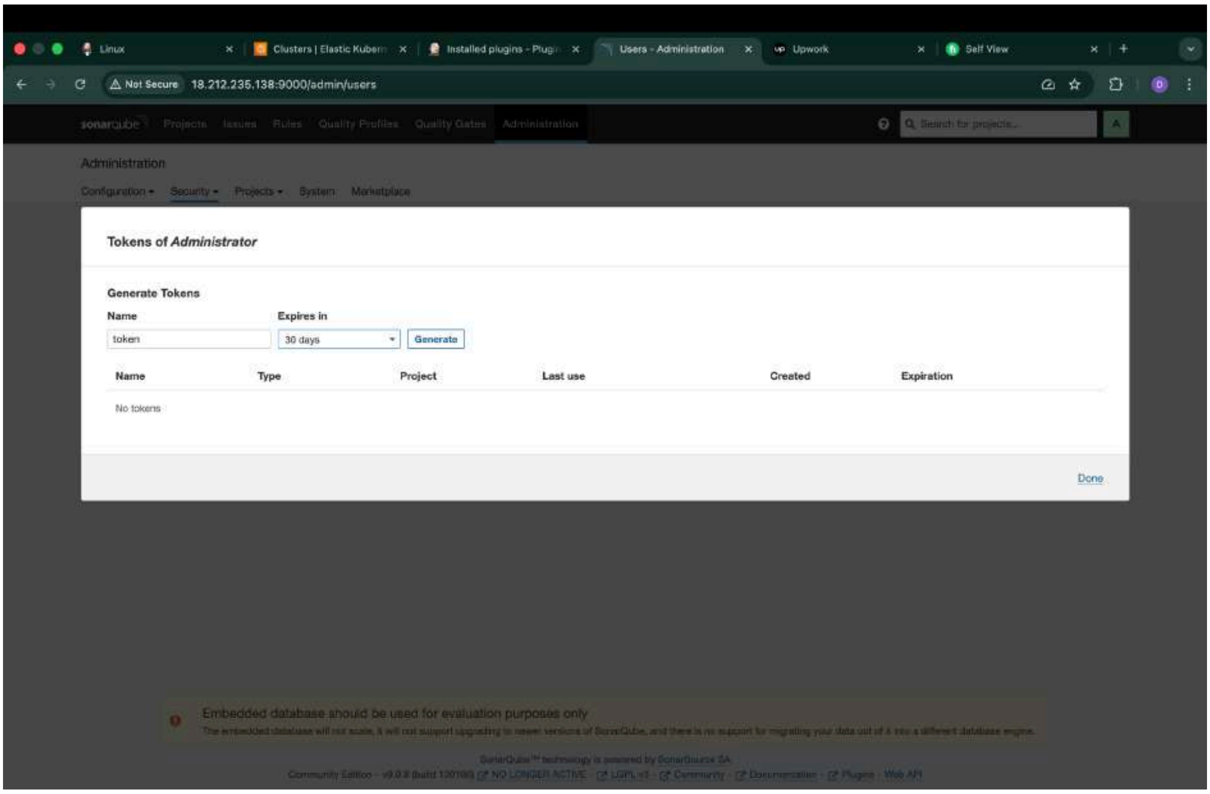

Not Secure 18.212.235.138:8080/manage/pluginManager/available
Jenkins
Manage Jenkins
Plugins
Plugins
prometh
Install
Updates
Available plugins
Installed plugins
Advanced settings
Download progress
Kubernetes Credentials
kubernetes
credentials
Common classes for Kubernetes credentials
1 mo 4 days ago
97
Kubernetes
Cloud Providers
Cluster Management
kubernetes
Agent Management
This plugin integrates Jenkins with Kubernetes
12 days ago
100
Kubernetes CLI
kubernetes
Configure kubectl for Kubernetes
8 days 5 hr ago
100
Kubernetes Credentials Provider
kubernetes
credentials
Provides a read only credentials store backed by Kubernetes.
1 day 22 hr ago
91
Config File Provider
Groovy-related
External Site/Tool Integrations
Maven
Ability to provide configuration files (e.g. settings.xml for maven, XML, groovy, custom files,...) loaded through the UI which will be copied to the job workspace.
2 mo 3 days ago
100
Prometheus metrics
monitoring
Miscellaneous
Jenkins Prometheus Plugin expose an endpoint (default /prometheus) with metrics where a Prometheus Server can scrape.
1 day 17 hr ago
96
Otel agent host metrics monitoring 2.0.1
monitoring
observability
This plugin allows monitoring of Jenkins agents by deploying Prometheus node exporters and
4 mo 5 days ago
100

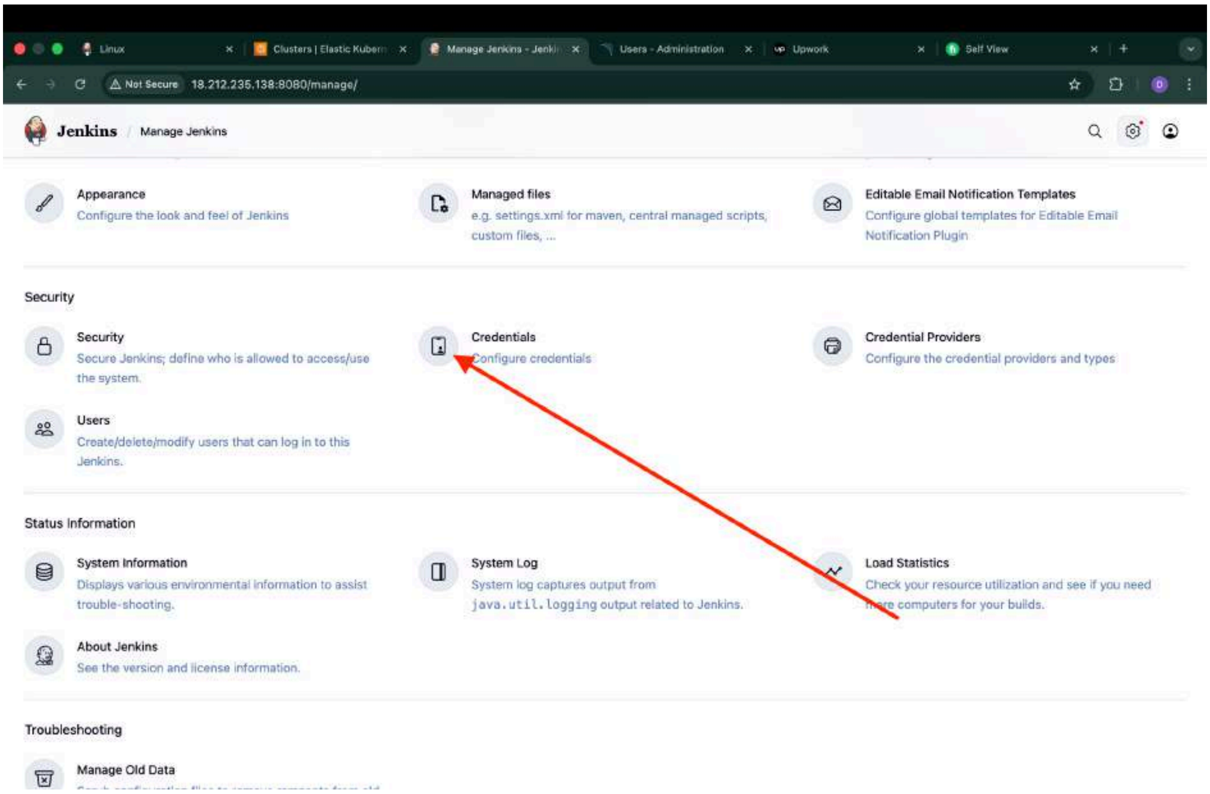


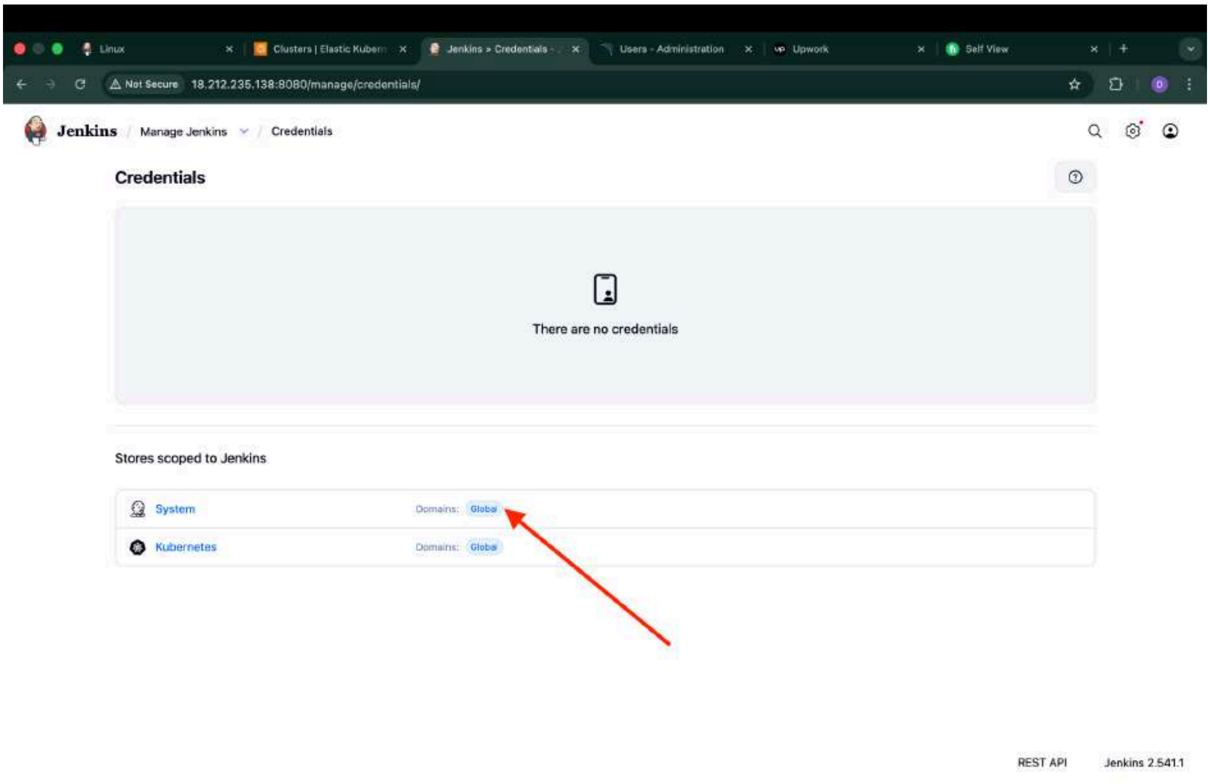
Not Secure 18.212.235.138:9000/admin/settings
sonarqube Projects Issues Rules Quality Profiles Quality Gates Administration
Search for projects...
Administration
Configuration Security Projects System Marketplace
Users
Groups
Global Permissions
Permission Templates
Analysis Scope
Authentication
DevOps Platform Integrations
External Analyzers
General
Housekeeping
JaCoCo
Languages
New Code
SCM
Security
Technical Debt
Duplications
Cross project duplication detection
DEPRECATED - By default, SonarQube detects duplications at project level. This means that a block duplicated on two different projects won't be reported. Setting this parameter to "true" allows to detect duplicates across projects. Note that activating this property will significantly increase each SonarQube analysis time, and therefore badly impact the performances of report processing as more and more projects are getting involved in this cross project duplication mechanism.
Key: sonar.cpd.cross_project
(default)
Email
SMTP host
For example "smtp.gmail.com". Leave blank to disable email sending.
Key: email.smtp_host.secured
18.212.235.138:9000/admin/users

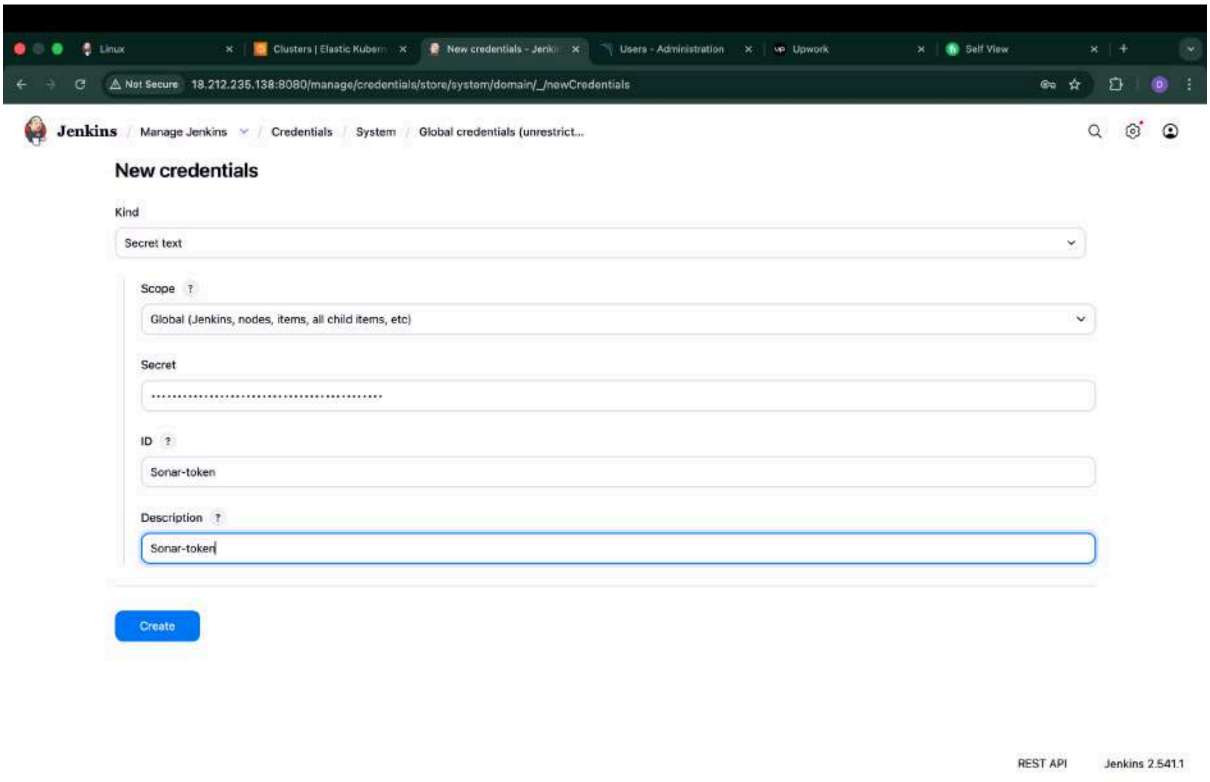
Not Secure 18.212.235.138:9000/admin/users
sonarqube Projects Issues Rules Quality Profiles Quality Gates Administration
Search for projects...
Administration
Configuration Security Projects System Marketplace
Users
Create and administer individual users.
Create User
Search by login or name...
SCM Accounts Last connection Groups Tokens
Administrator admin
< 1 hour ago
sonar-administrators
sonar-users
0
Update Tokens
1 of 1 shown
Embedded database should be used for evaluation purposes only
The embedded database will not scale, it will not support upgrading to newer versions of SonarQube, and there is no support for migrating your data out of it into a different database engine.
SonarQube™ technology is powered by SonarSource SA
Community Edition - v9.9.8 (build 100196) NO LONGER ACTIVE - LGPL v3 - Community - Documentation - Plugins - Web API


Not Secure 18.212.235.138:9000/admin/users
Administration
Configuration Security Projects System Marketplace
Tokens of Administrator
Generate Tokens
Name
Expires in
token
30 days
Generate
Name
Type
Project
Last use
Created
Expiration
No tokens
Done
Embedded database should be used for evaluation purposes only

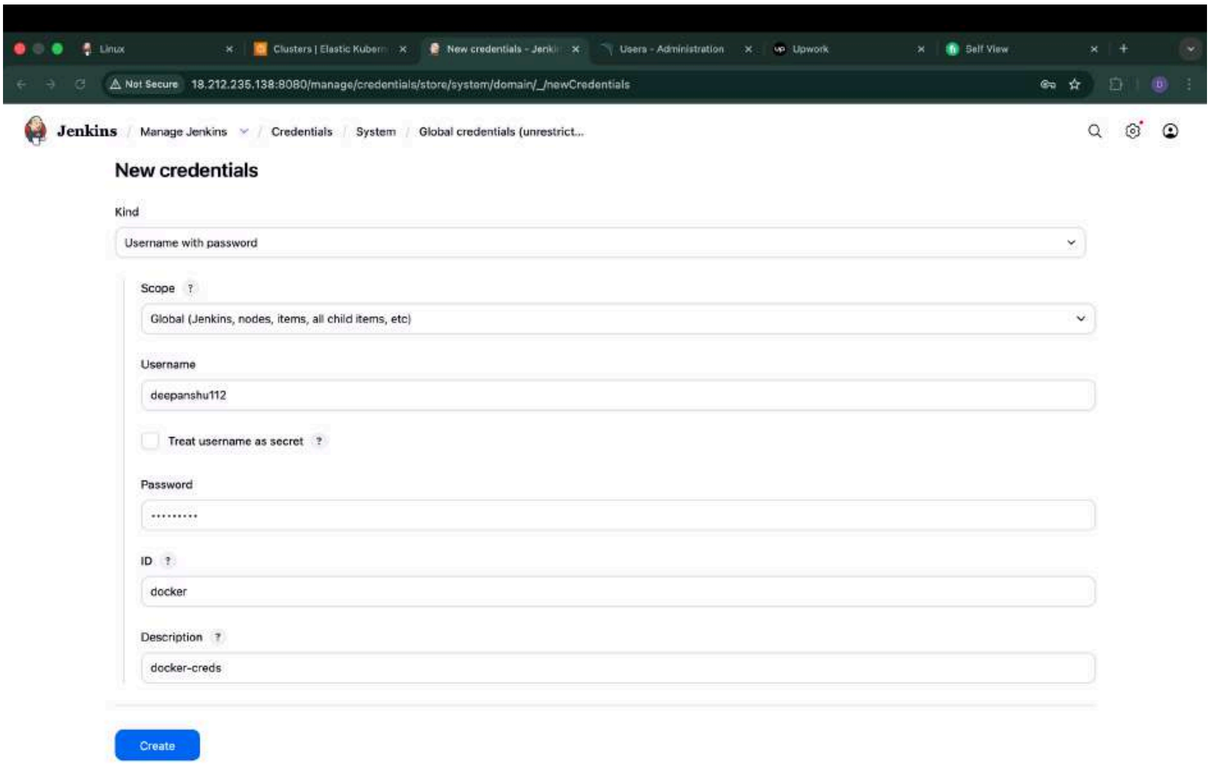

Not Secure 18.212.235.138:8080/manage/
Jenkins
Manage Jenkins
Appearance
Configure the look and feel of Jenkins
Managed files
e.g. settings.xml for maven, central managed scripts, custom files, ...
Editable Email Notification Templates
Configure global templates for Editable Email Notification Plugin
Security
Security
Secure Jenkins; define who is allowed to access/use the system.
Credentials
Configure credentials
Credential Providers
Configure the credential providers and types
Users
Create/delete/modify users that can log in to this Jenkins.
Status Information
System Information
Displays various environmental information to assist trouble-shooting.
System Log
System log captures output from java.util.logging output related to Jenkins.
Load Statistics
Check your resource utilization and see if you need more computers for your builds.
About Jenkins
See the version and license information.
Troubleshooting
Manage Old Data



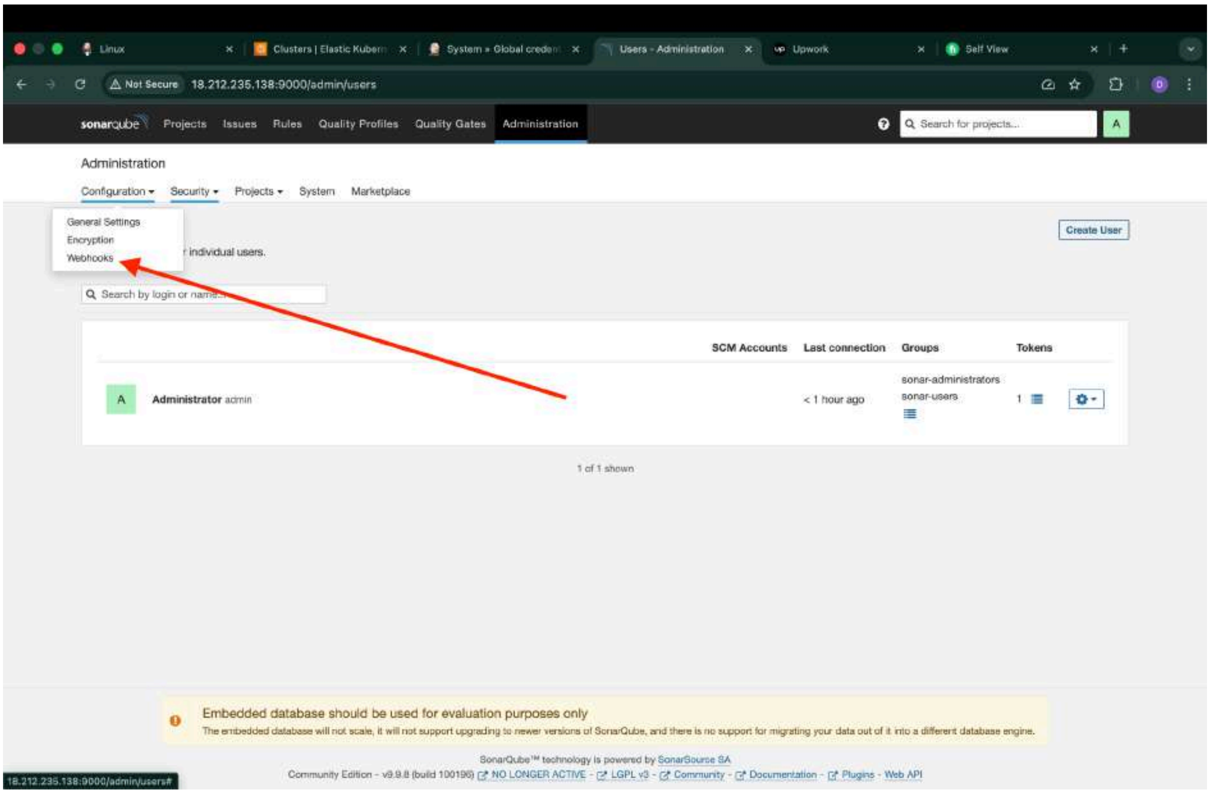
18.212.235.138:8080/manage/credentials/
Jenkins
Manage Jenkins
Credentials
Credentials
There are no credentials
Stores scoped to Jenkins
System
Domains: Global
Kubernetes
Domains: Global
REST API
Jenkins 2.541.1

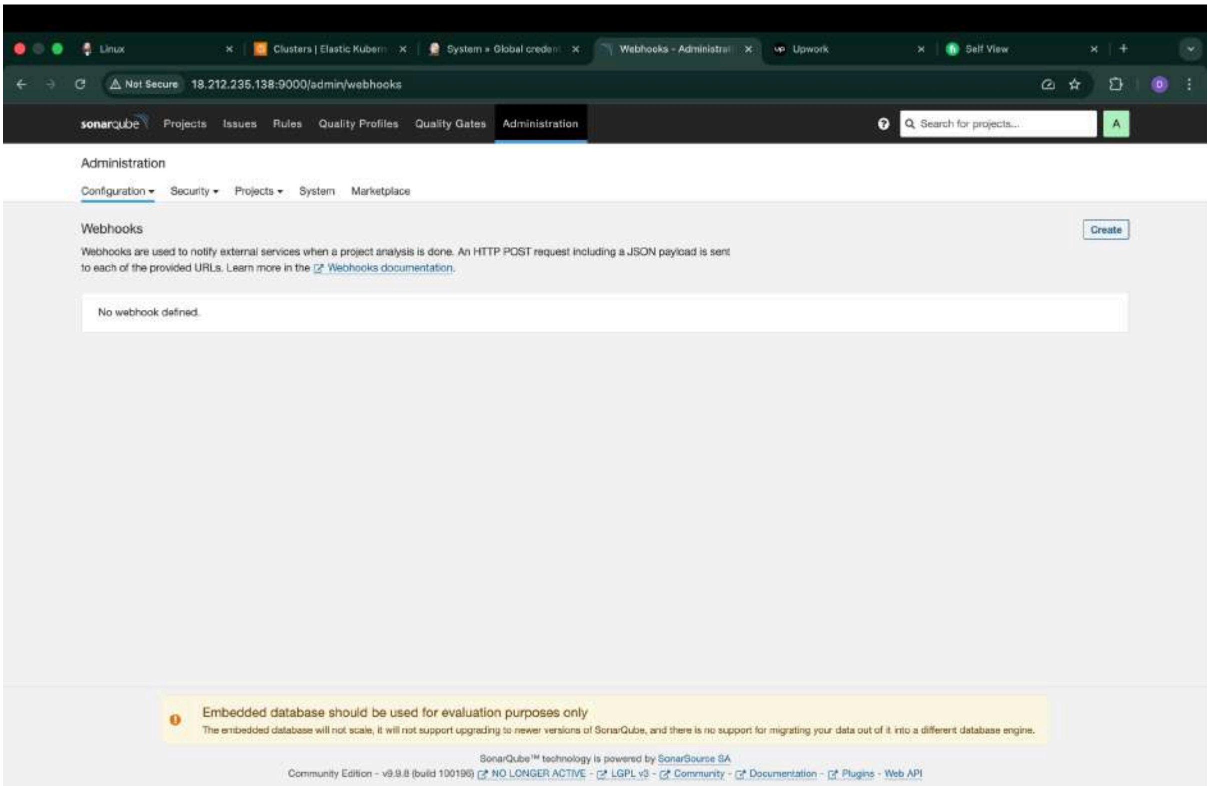
18.212.235.138:8080/manage/credentials/store/system/domain/_/newCredentials
Jenkins
Manage Jenkins
Credentials
System
Global credentials (unrestrict...
New credentials
Kind
Secret text
Scope
Global (Jenkins, nodes, items, all child items, etc)
Secret
ID
Sonar-token
Description
Sonar-token
Create
REST API
Jenkins 2.541.1

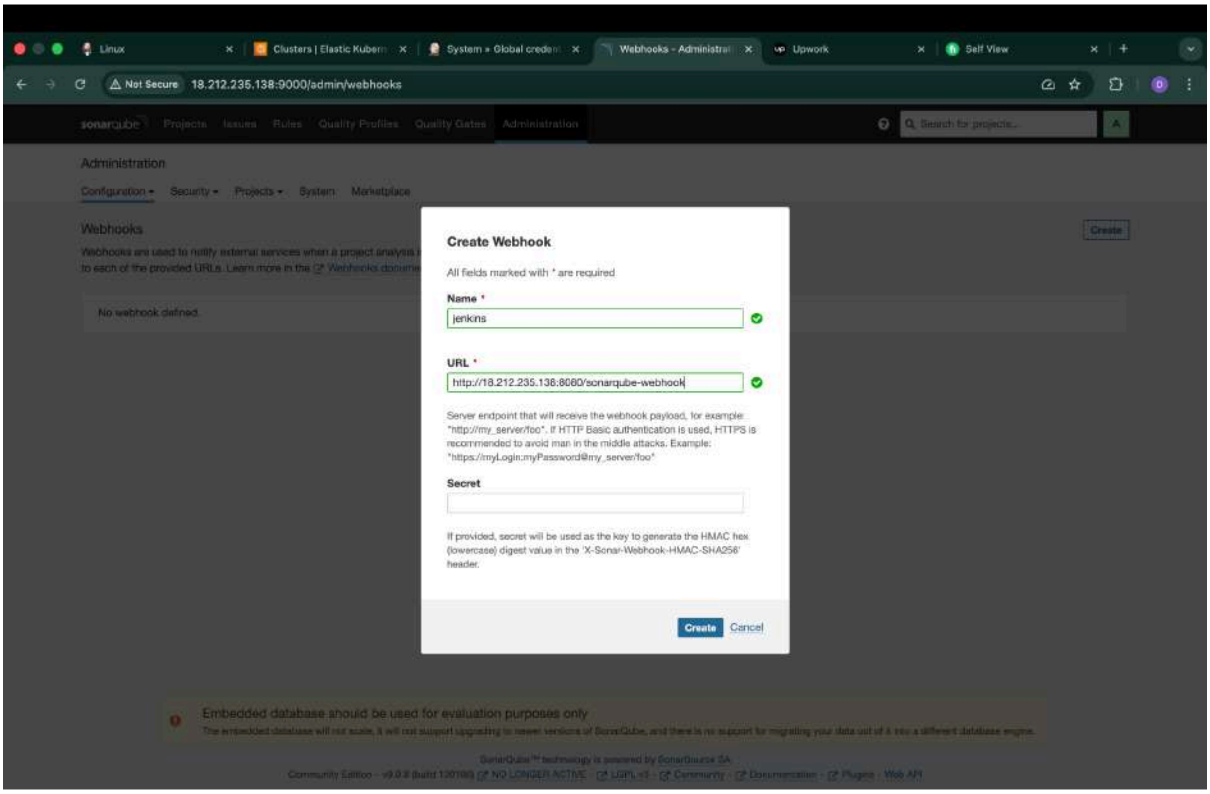


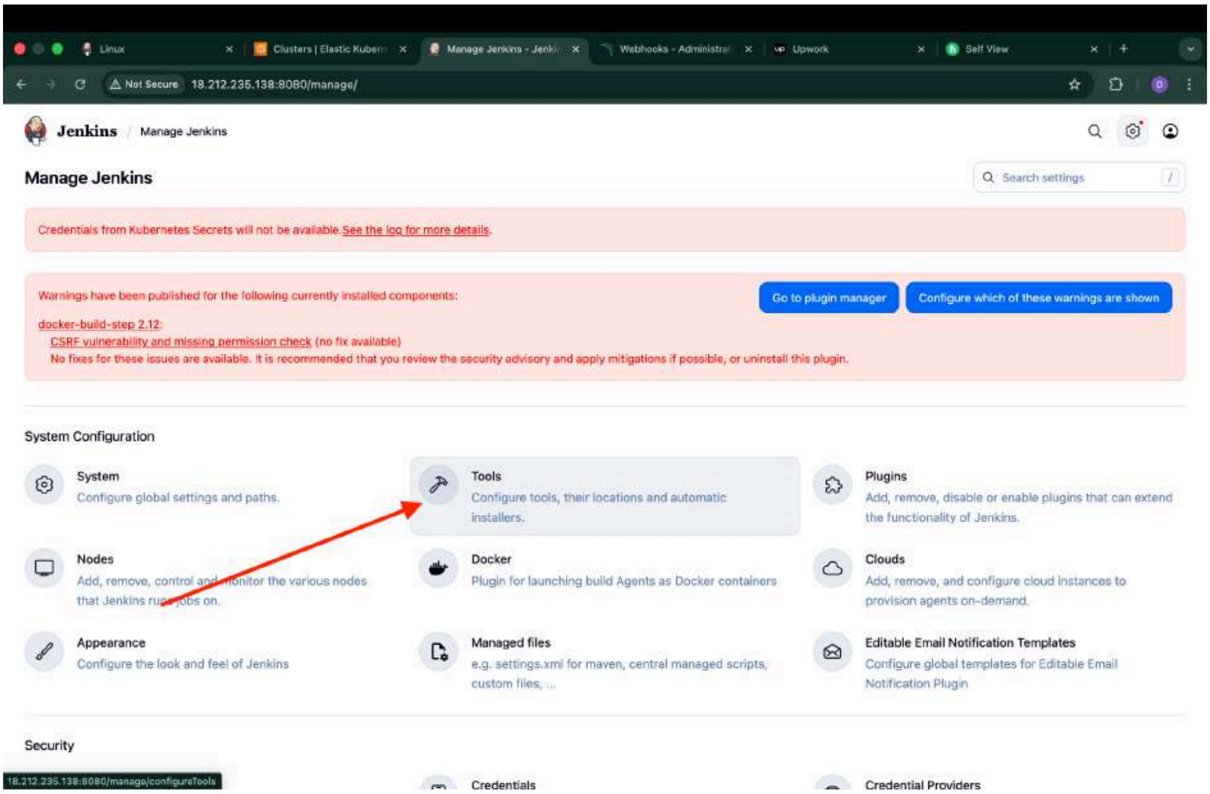
18.212.235.138:8080/manage/credentials/store/system/domain/_/newCredentials
Jenkins
Manage Jenkins
Credentials
System
Global credentials (unrestrict...
New credentials
Kind
Username with password
Scope
Global (Jenkins, nodes, items, all child items, etc)
Username
deepanshu112
Treat username as secret
Password
ID
docker
Description
docker-creds
Create

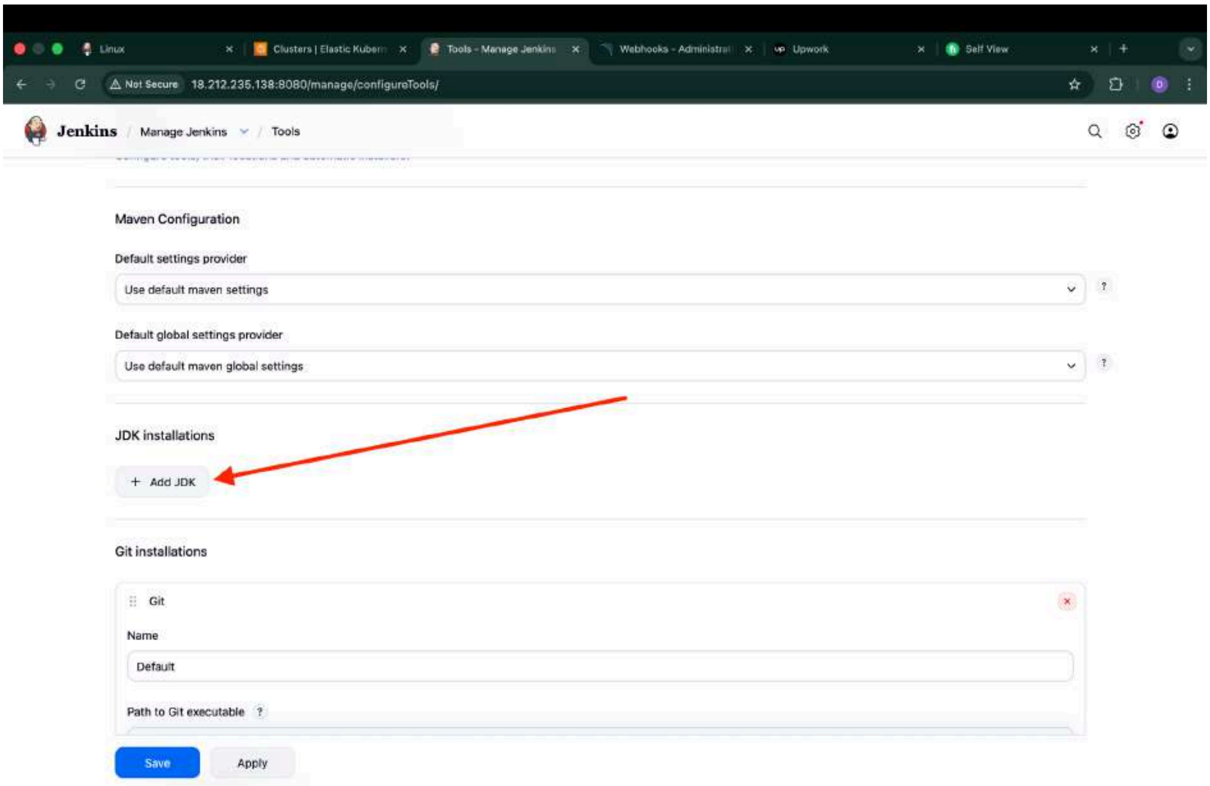
18.212.235.138:9000/admin/users
Projects
Issues
Rules
Quality Profiles
Quality Gates
Administration
Search for projects...
Administration
Configuration
Security
Projects
System
Marketplace
General Settings
Encryption
Webhooks
Create User
SCM Accounts
Last connection
Groups
Tokens
Administrator admin
< 1 hour ago
sonar-administrators
sonar-users
1 of 1 shown
Embedded database should be used for evaluation purposes only


Not Secure 18.212.235.138:9000/admin/webhooks
sonarqube Projects Issues Rules Quality Profiles Quality Gates Administration
Search for projects...
Administration
Configuration Security Projects System Marketplace
Webhooks
Create
Webhooks are used to notify external services when a project analysis is done. An HTTP POST request including a JSON payload is sent to each of the provided URLs. Learn more in the Webhooks documentation.
No webhook defined.
Embedded database should be used for evaluation purposes only
The embedded database will not scale, it will not support upgrading to newer versions of SonarQube, and there is no support for migrating your data out of it into a different database engine.
SonarQube™ technology is powered by SonarSource SA
Community Edition - v9.9.8 (build 100196) NO LONGER ACTIVE - LGPL v3 - Community - Documentation - Plugins - Web API

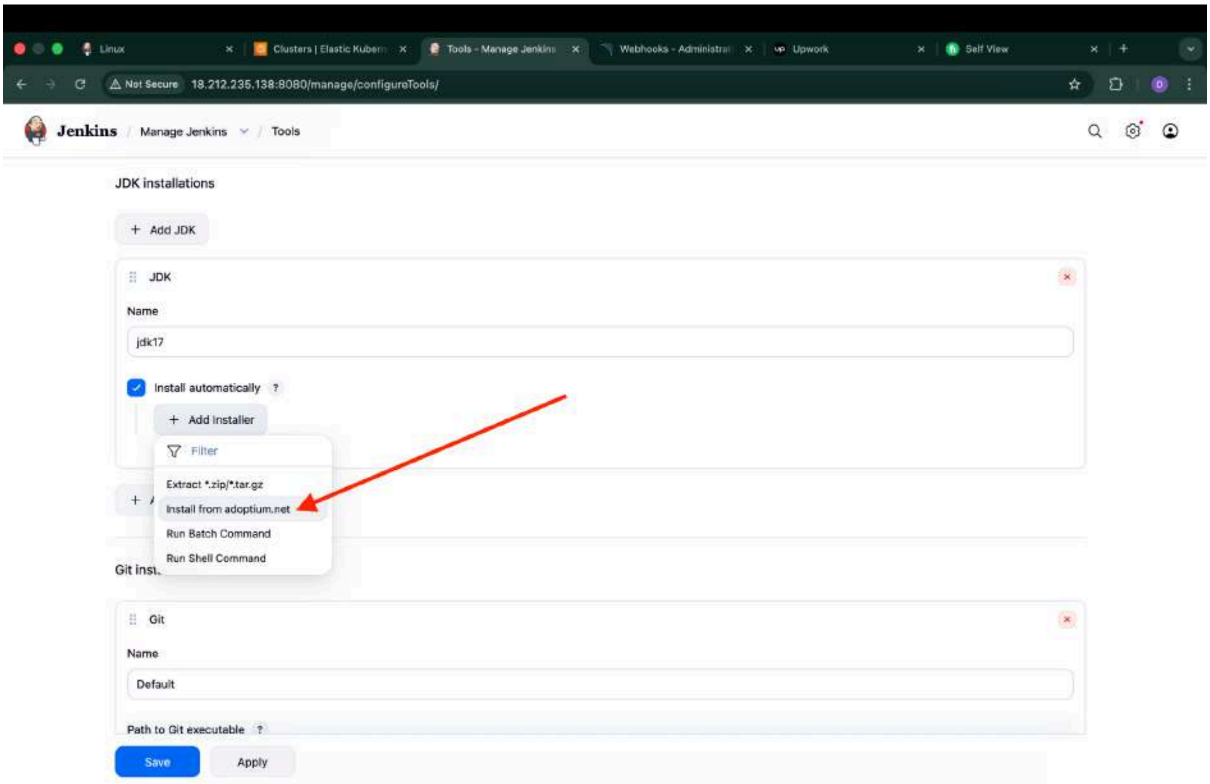

Not Secure 18.212.235.138:9000/admin/webhooks
Create Webhook
All fields marked with * are required
Name *
jenkins
URL *
http://18.212.235.138:8080/sonarqube-webhook
Server endpoint that will receive the webhook payload, for example: "http://my_server/foo". If HTTP Basic authentication is used, HTTPS is recommended to avoid man in the middle attacks. Example: "https://myLogin:myPassword@my_server/foo"
Secret
If provided, secret will be used as the key to generate the HMAC hex (lowercase) digest value in the 'X-Sonar-Webhook-HMAC-SHA256' header.
Create
Cancel



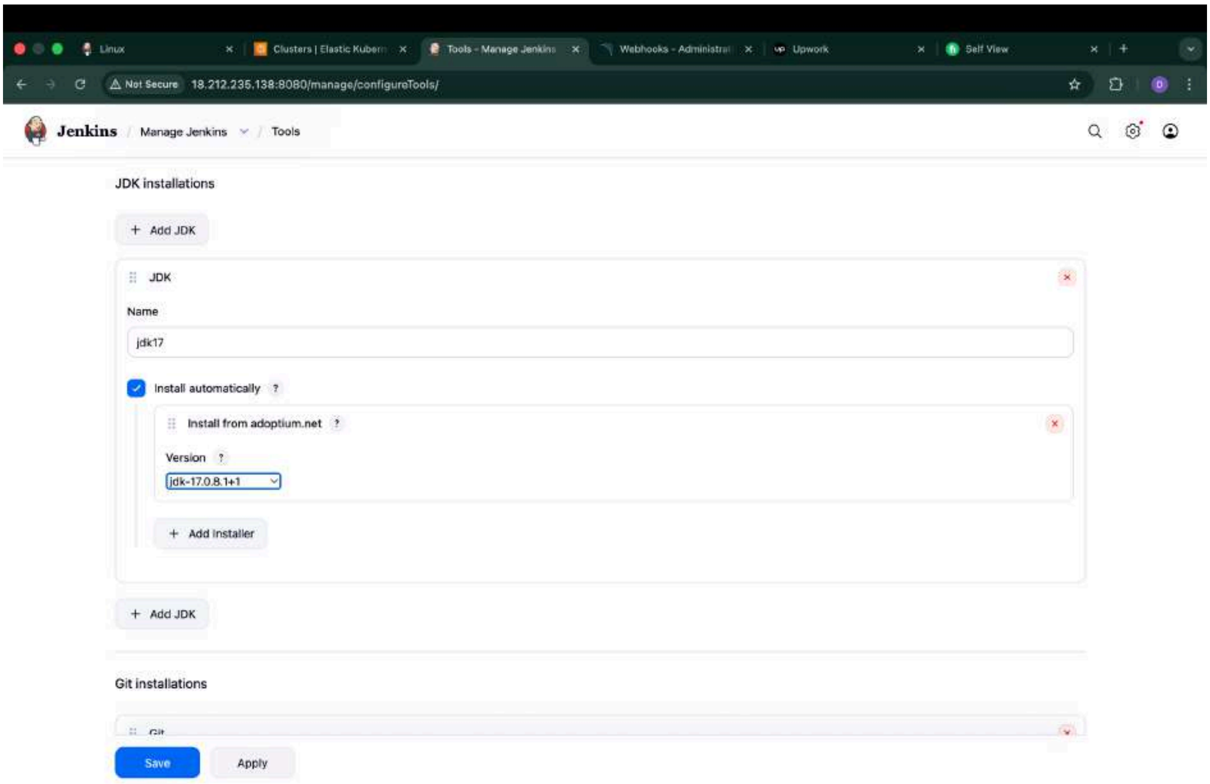
Linux
Clusters | Elastic Kuberr
Manage Jenkins - Jenki
Webhooks - Administrat
Upwork
Self View
Not Secure
18.212.235.138:8080/manage/
Jenkins
Manage Jenkins
Manage Jenkins
Search settings
Credentials from Kubernetes Secrets will not be available. See the log for more details.
Warnings have been published for the following currently installed components:
Go to plugin manager
Configure which of these warnings are shown
docker-build-step 2.12:
CSRF vulnerability and missing permission check (no fix available)
No fixes for these issues are available. It is recommended that you review the security advisory and apply mitigations if possible, or uninstall this plugin.
System Configuration
System
Configure global settings and paths.
Tools
Configure tools, their locations and automatic installers.
Plugins
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.
Nodes
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.
Docker
Plugin for launching build Agents as Docker containers
Clouds
Add, remove, and configure cloud instances to provision agents on-demand.
Appearance
Configure the look and feel of Jenkins
Managed files
e.g. settings.xml for maven, central managed scripts, custom files, ...
Editable Email Notification Templates
Configure global templates for Editable Email Notification Plugin
Security
18.212.235.138:8080/manage/configureTools
Credentials
Credential Providers

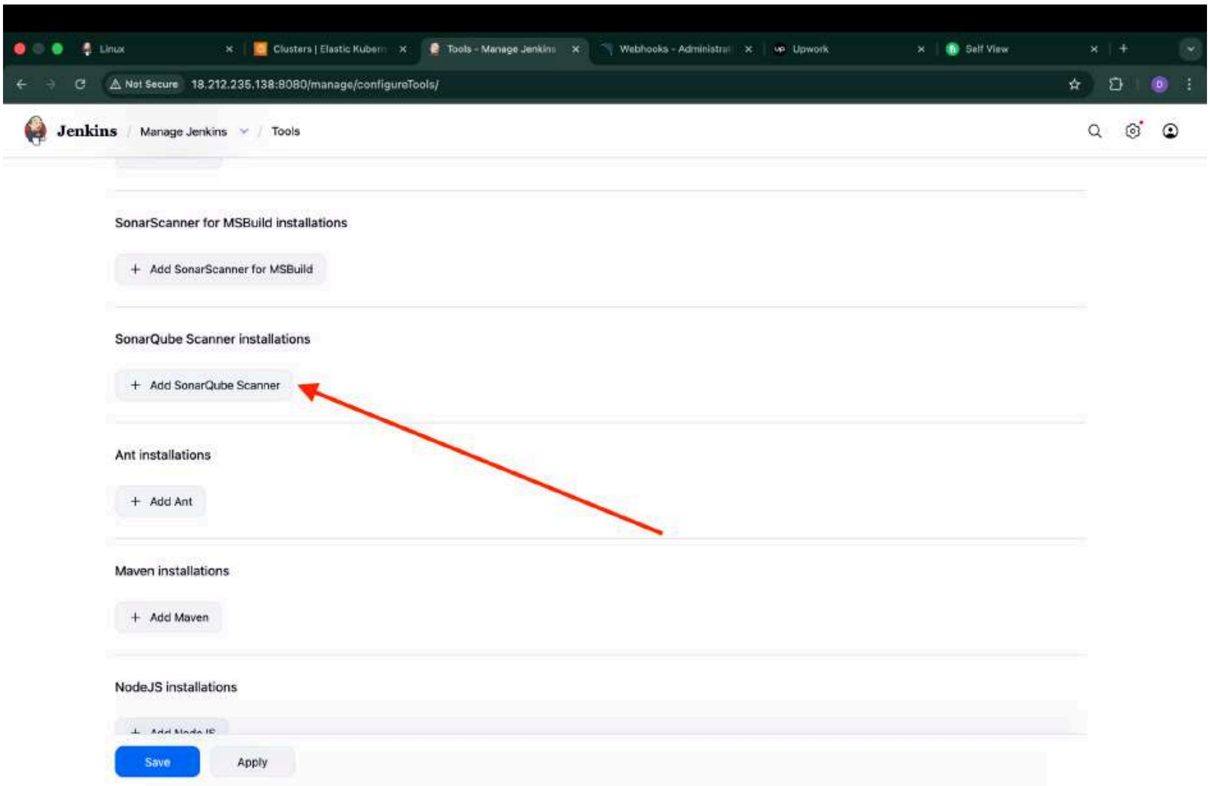
Linux
Clusters | Elastic Kuberr
Tools - Manage Jenkins
Webhooks - Administrat
Upwork
Self View
Not Secure
18.212.235.138:8080/manage/configureTools/
Jenkins
Manage Jenkins
Tools
Maven Configuration
Default settings provider
Use default maven settings
Default global settings provider
Use default maven global settings
JDK installations
+ Add JDK
Git installations
Git
Name
Default
Path to Git executable
Save
Apply

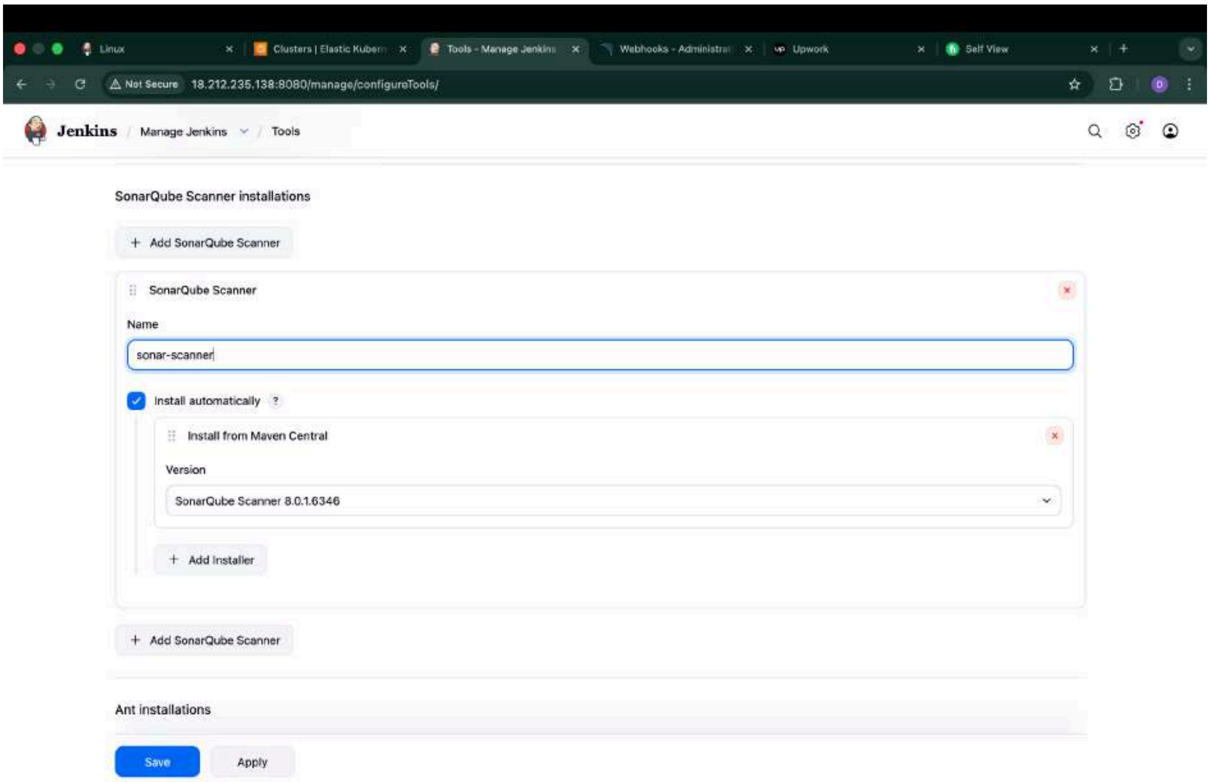


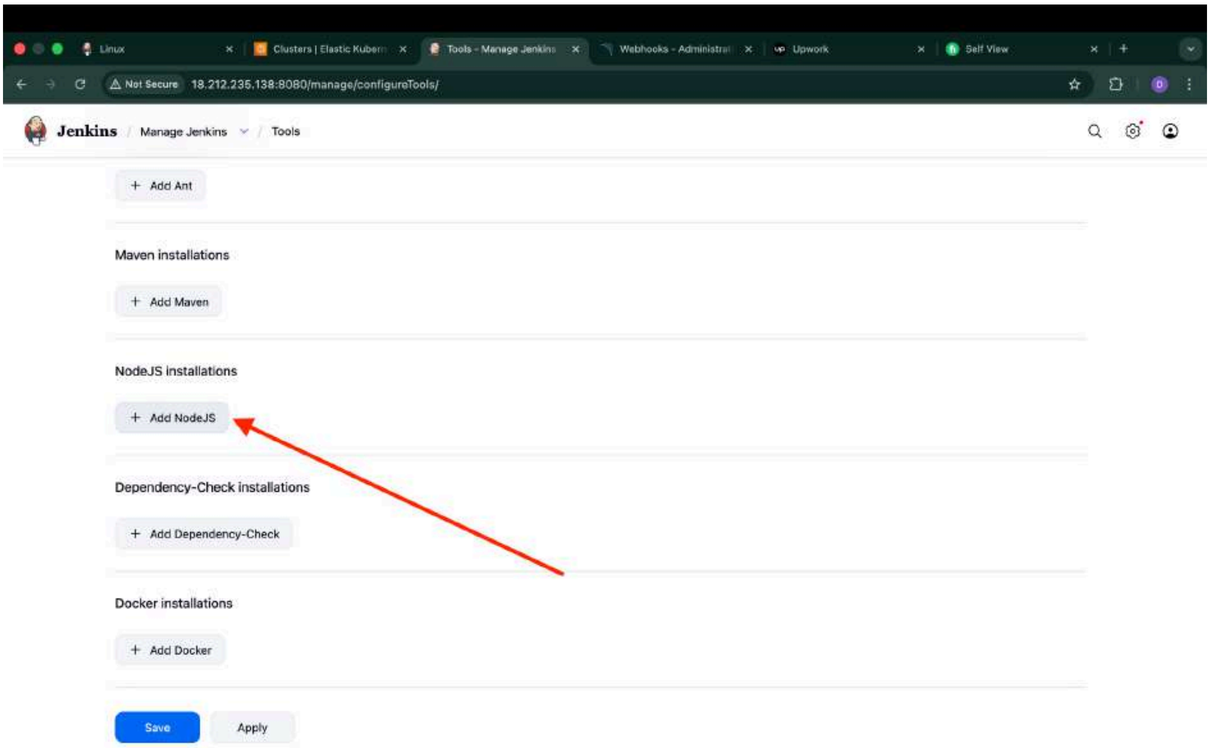
Linux
Clusters | Elastic Kubern
Tools - Manage Jenkins
Webhooks - Administrat
Upwork
Self View
Not Secure
18.212.235.138:8080/manage/configureTools/
Jenkins
Manage Jenkins
Tools
JDK installations
Add JDK
JDK
Name
jdk17
Install automatically
Add Installer
Filter
Extract *.zip/*.tar.gz
Install from adoptium.net
Run Batch Command
Run Shell Command
Git
Name
Default
Path to Git executable
Save
Apply

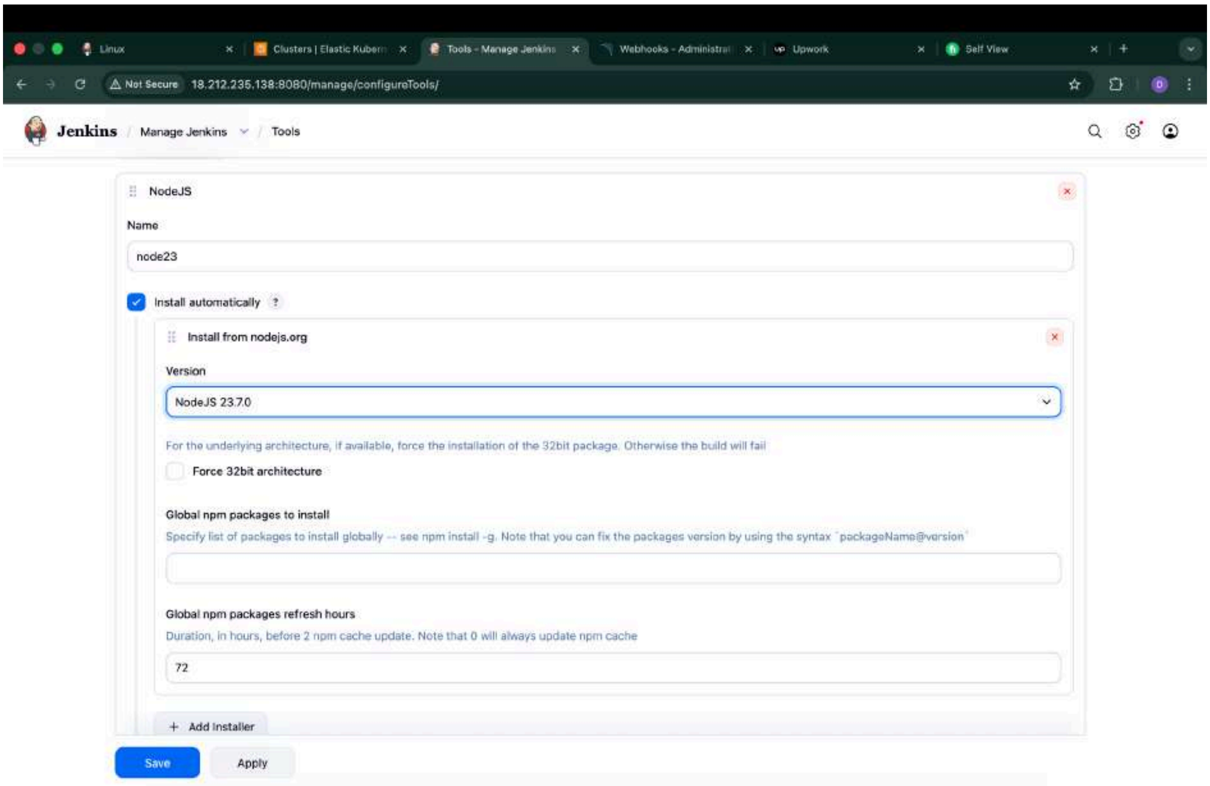



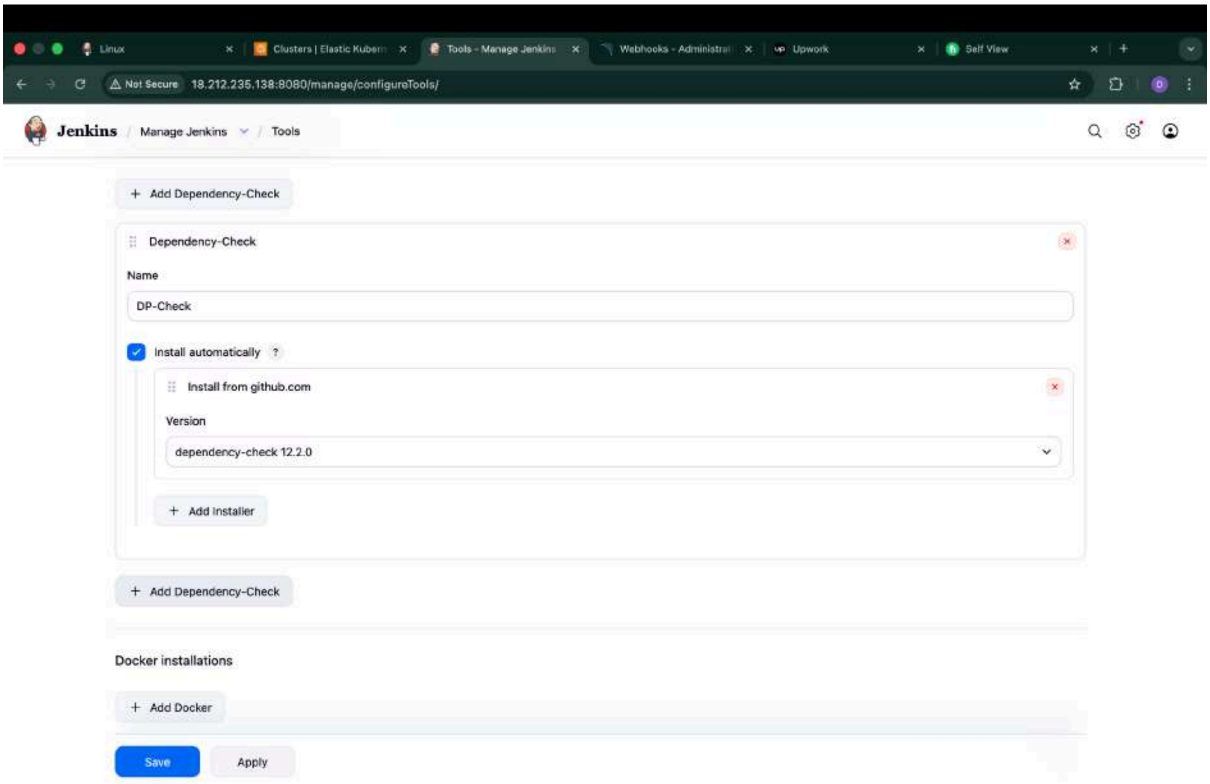
Linux
Clusters | Elastic Kubern
Tools - Manage Jenkins
Webhooks - Administrat
Upwork
Self View
Not Secure
18.212.235.138:8080/manage/configureTools/
Jenkins
Manage Jenkins
Tools
JDK installations
Add JDK
JDK
Name
jdk17
Install automatically
Install from adoptium.net
Version
jdk-17.0.8.1+1
Add Installer
Add JDK
Git installations
Save
Apply

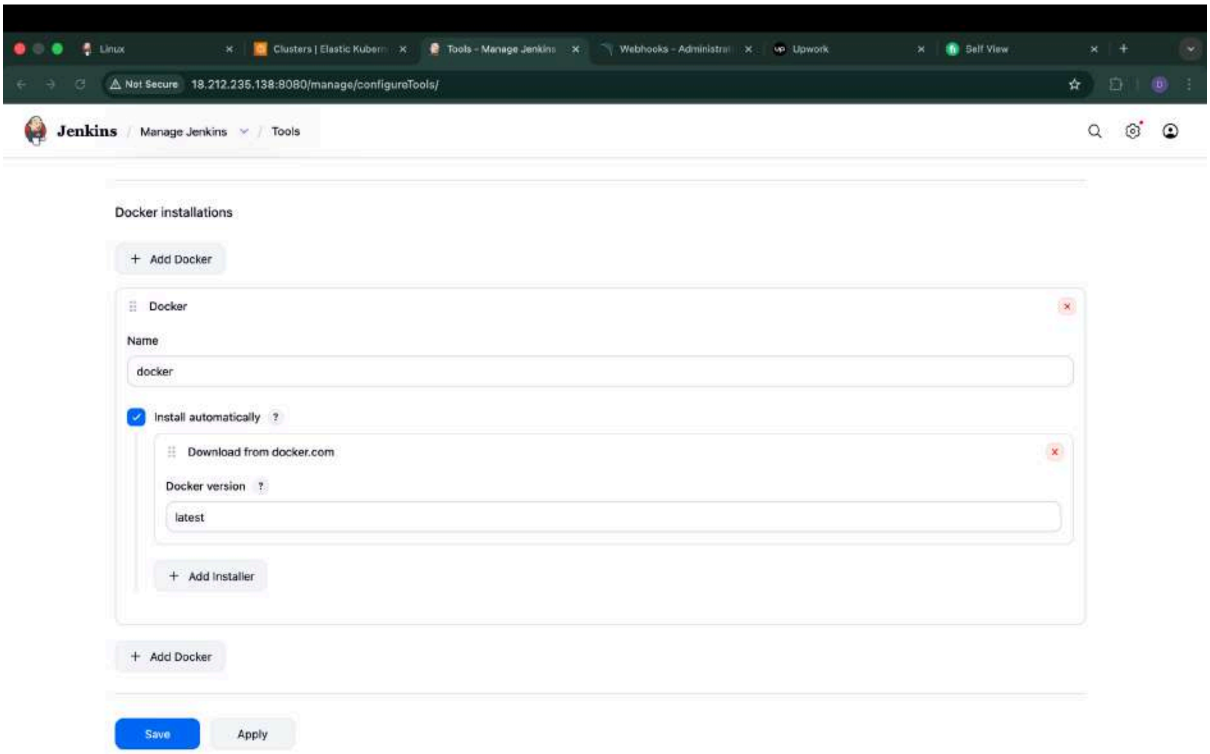

Jenkins
Manage Jenkins
Tools
SonarScanner for MSBuild installations
Add SonarScanner for MSBuild
SonarQube Scanner installations
Add SonarQube Scanner
Ant installations
Add Ant
Maven installations
Add Maven
NodeJS installations
Save
Apply

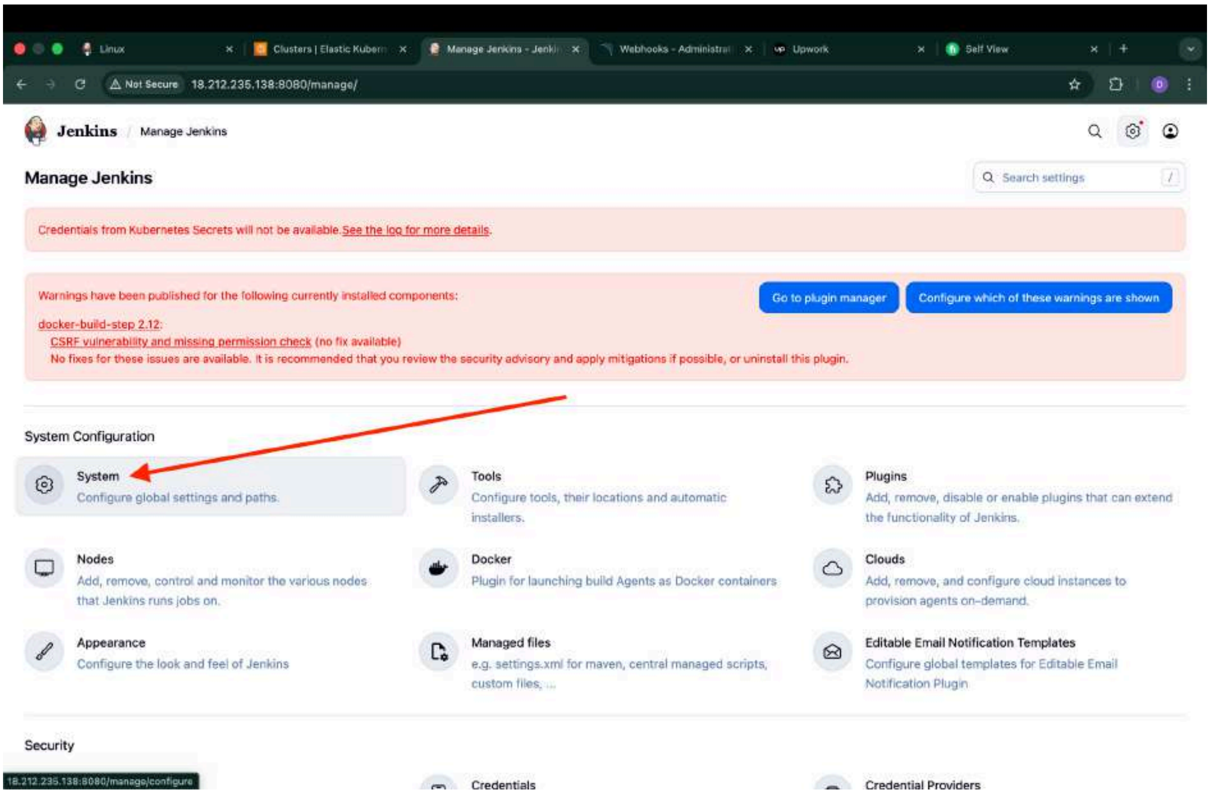

Jenkins
Manage Jenkins
Tools
SonarQube Scanner installations
Add SonarQube Scanner
SonarQube Scanner
Name
sonar-scanner
Install automatically
Install from Maven Central
Version
SonarQube Scanner 8.0.1.6346
Add Installer
Add SonarQube Scanner
Ant installations
Save
Apply

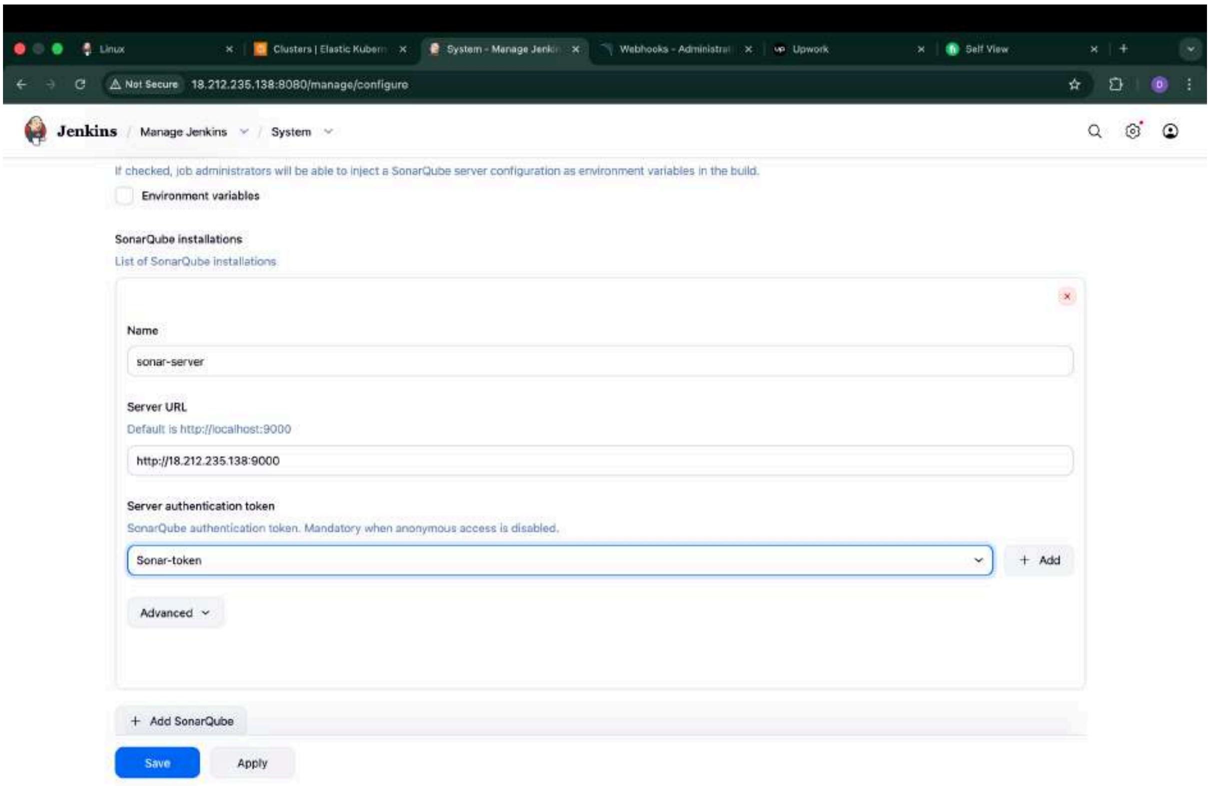

18.212.235.138:8080/manage/configureTools/
Jenkins
Manage Jenkins
Tools
Add Ant
Maven installations
Add Maven
NodeJS installations
Add NodeJS
Dependency-Check installations
Add Dependency-Check
Docker installations
Add Docker
Save
Apply


18.212.235.138:8080/manage/configureTools/
Jenkins
Manage Jenkins
Tools
NodeJS
Name
node23
Install automatically
Install from nodejs.org
Version
NodeJS 23.7.0
For the underlying architecture, if available, force the installation of the 32bit package. Otherwise the build will fail
Force 32bit architecture
Global npm packages to install
Specify list of packages to install globally -- see npm install -g. Note that you can fix the packages version by using the syntax `packageName@version`
Global npm packages refresh hours
Duration, in hours, before 2 npm cache update. Note that 0 will always update npm cache
72
Add Installer
Save
Apply


Linux
Clusters | Elastic Kubern
Tools - Manage Jenkins
Webhooks - Administrat
Upwork
Self View
Not Secure
18.212.235.138:8080/manage/configureTools/
Jenkins
Manage Jenkins
Tools
Add Dependency-Check
Dependency-Check
Name
DP-Check
Install automatically
Install from github.com
Version
dependency-check 12.2.0
Add Installer
Add Dependency-Check
Docker installations
Add Docker
Save
Apply


Linux
Clusters | Elastic Kubern
Tools - Manage Jenkins
Webhooks - Administrat
Upwork
Self View
Not Secure
18.212.235.138:8080/manage/configureTools/
Jenkins
Manage Jenkins
Tools
Docker installations
Add Docker
Docker
Name
docker
Install automatically
Download from docker.com
Docker version
latest
Add Installer
Add Docker
Save
Apply


18.212.235.138:8080/manage/
Jenkins / Manage Jenkins
Manage Jenkins
Credentials from Kubernetes Secrets will not be available. See the log for more details.
Warnings have been published for the following currently installed components:
docker-build-step 2.12:
CSRF vulnerability and missing permission check (no fix available)
No fixes for these issues are available. It is recommended that you review the security advisory and apply mitigations if possible, or uninstall this plugin.
Go to plugin manager
Configure which of these warnings are shown
System Configuration
System
Configure global settings and paths.
Tools
Configure tools, their locations and automatic installers.
Plugins
Add, remove, disable or enable plugins that can extend the functionality of Jenkins.
Nodes
Add, remove, control and monitor the various nodes that Jenkins runs jobs on.
Docker
Plugin for launching build Agents as Docker containers
Clouds
Add, remove, and configure cloud instances to provision agents on-demand.
Appearance
Configure the look and feel of Jenkins
Managed files
e.g. settings.xml for maven, central managed scripts, custom files, ...
Editable Email Notification Templates
Configure global templates for Editable Email Notification Plugin
Security
Credentials
Credential Providers

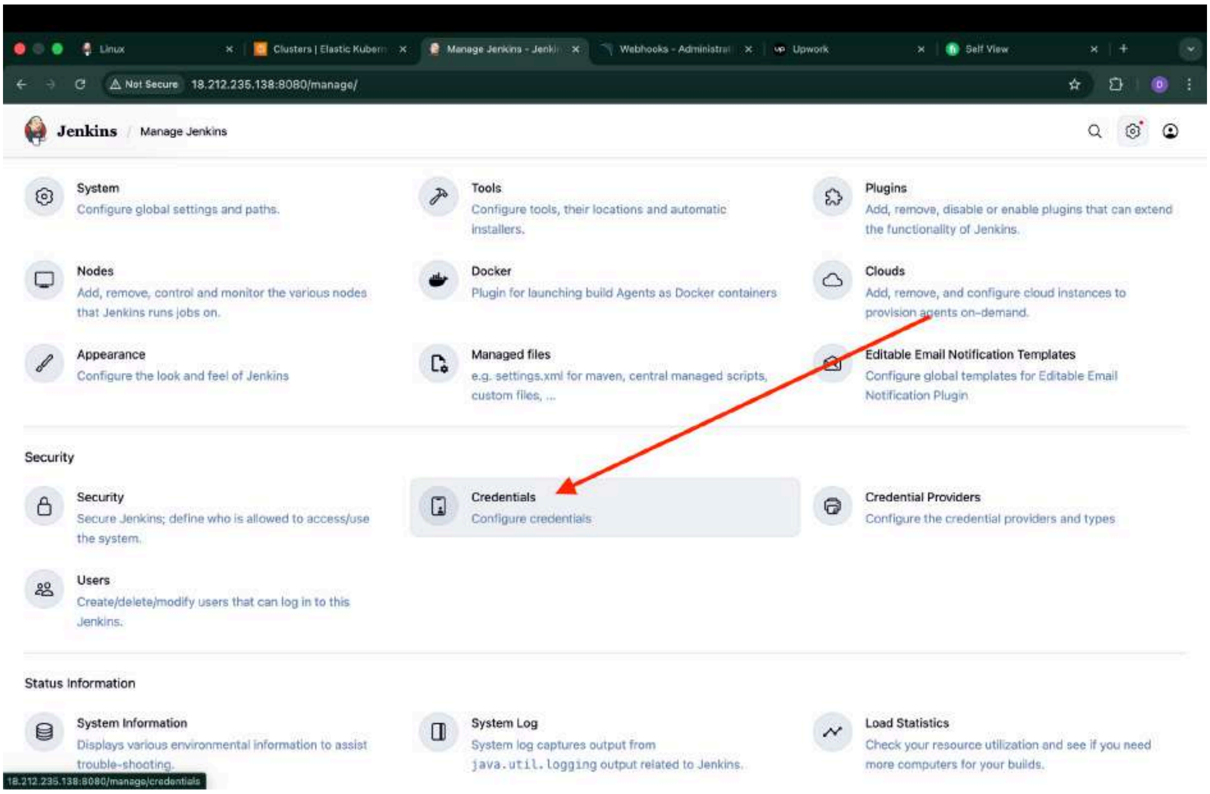

18.212.235.138:8080/manage/configure
Jenkins / Manage Jenkins / System
If checked, job administrators will be able to inject a SonarQube server configuration as environment variables in the build.
Environment variables
SonarQube installations
List of SonarQube installations
Name
sonar-server
Server URL
Default is http://localhost:9000
http://18.212.235.138:9000
Server authentication token
SonarQube authentication token. Mandatory when anonymous access is disabled.
Sonar-token
+ Add
Advanced
+ Add SonarQube
Save
Apply

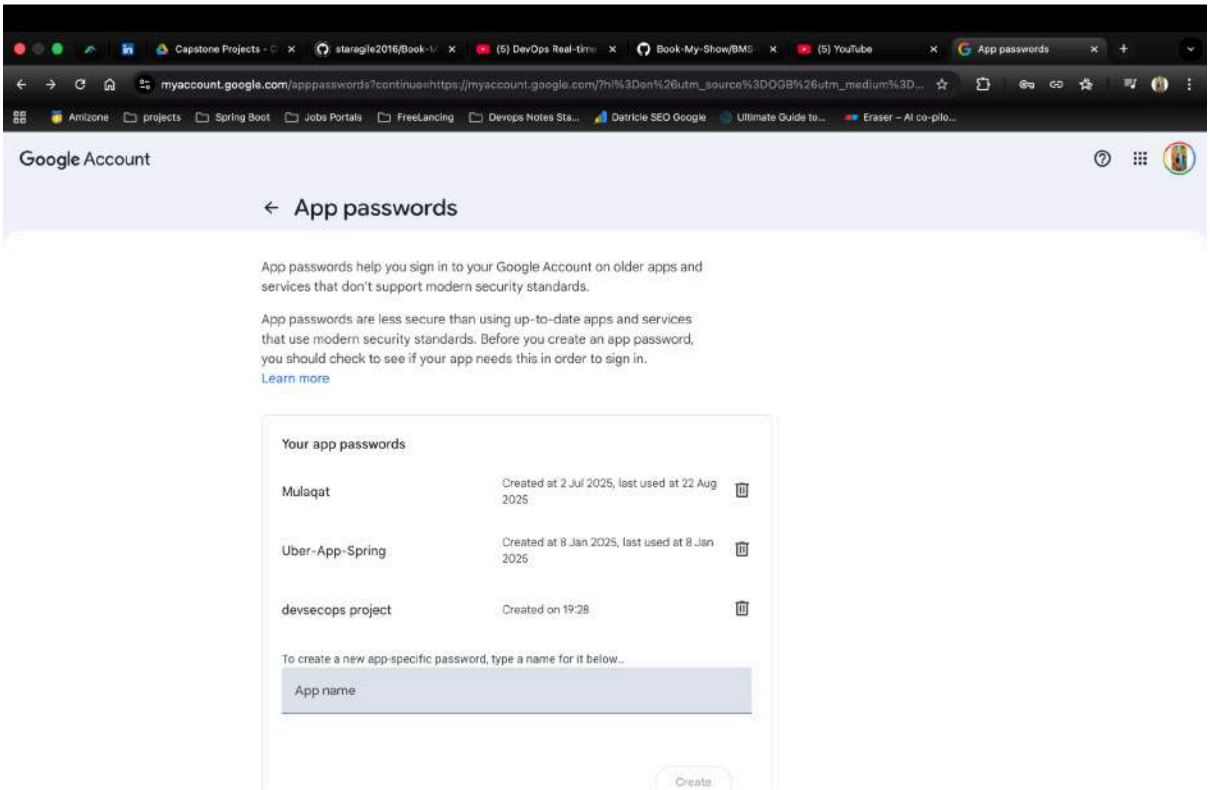
## Step 4: Email Integration

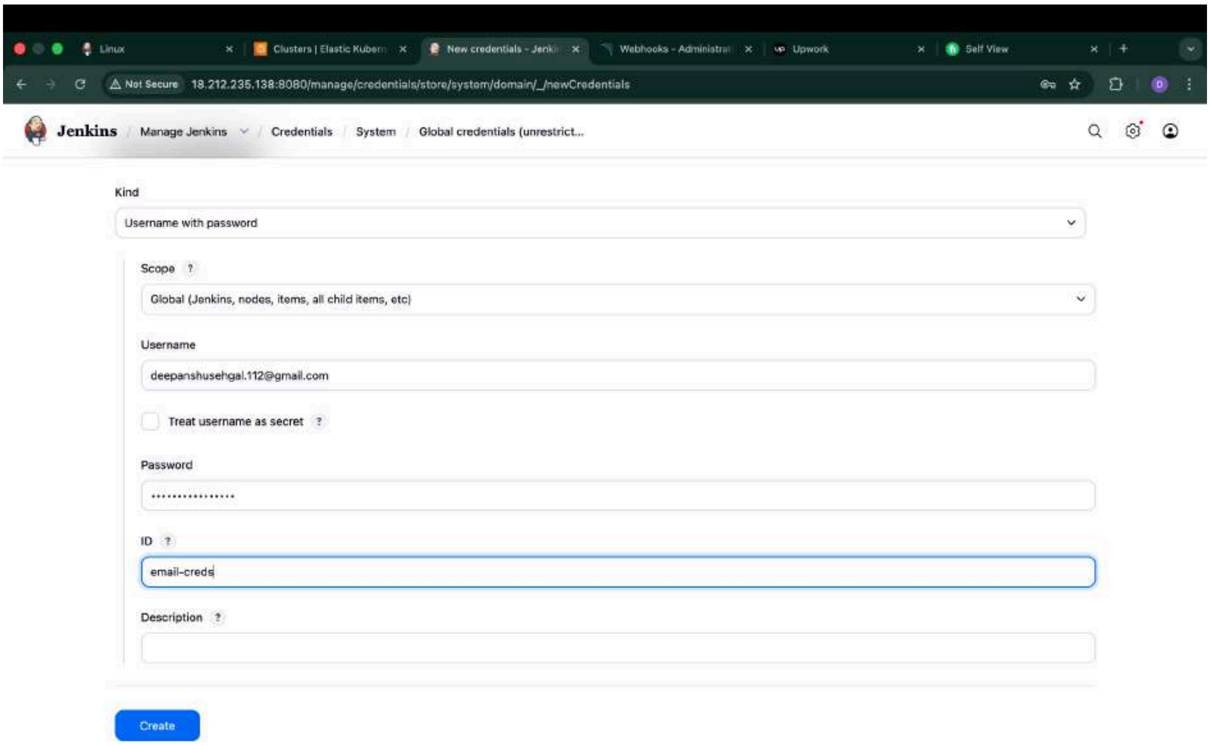
Use the Gmail app password.

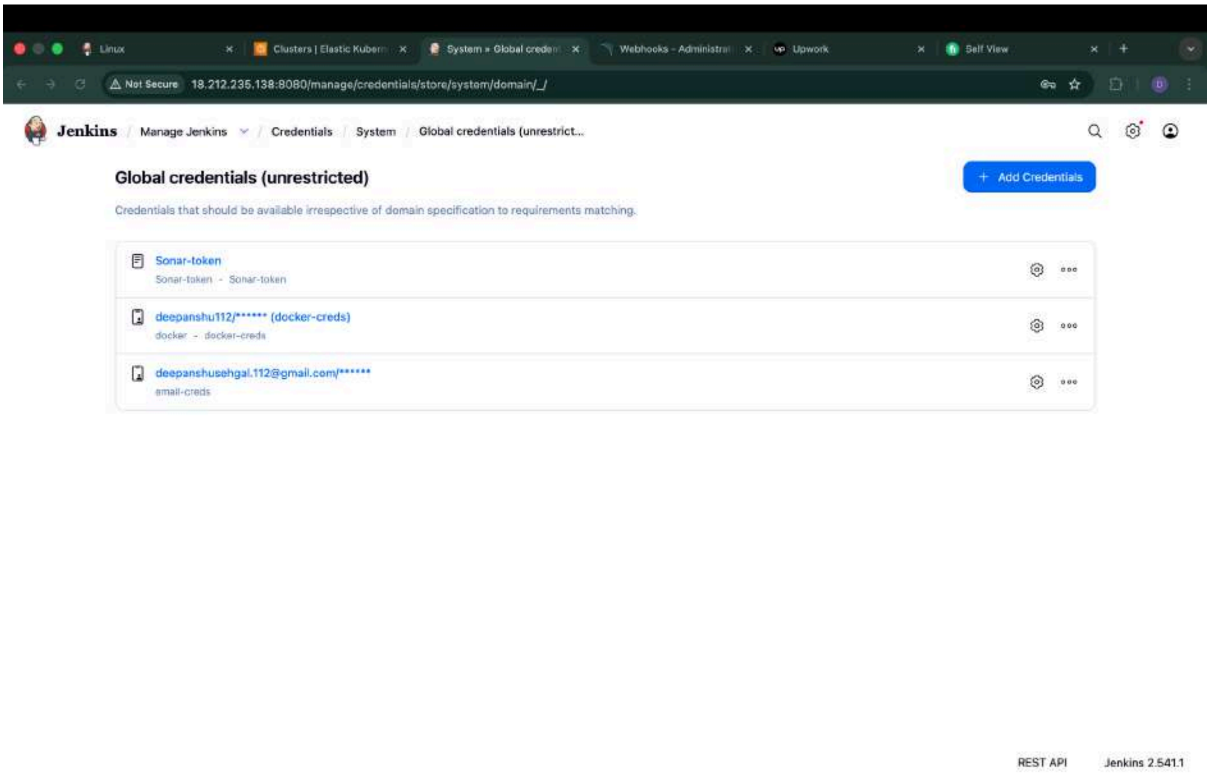
Configure in:

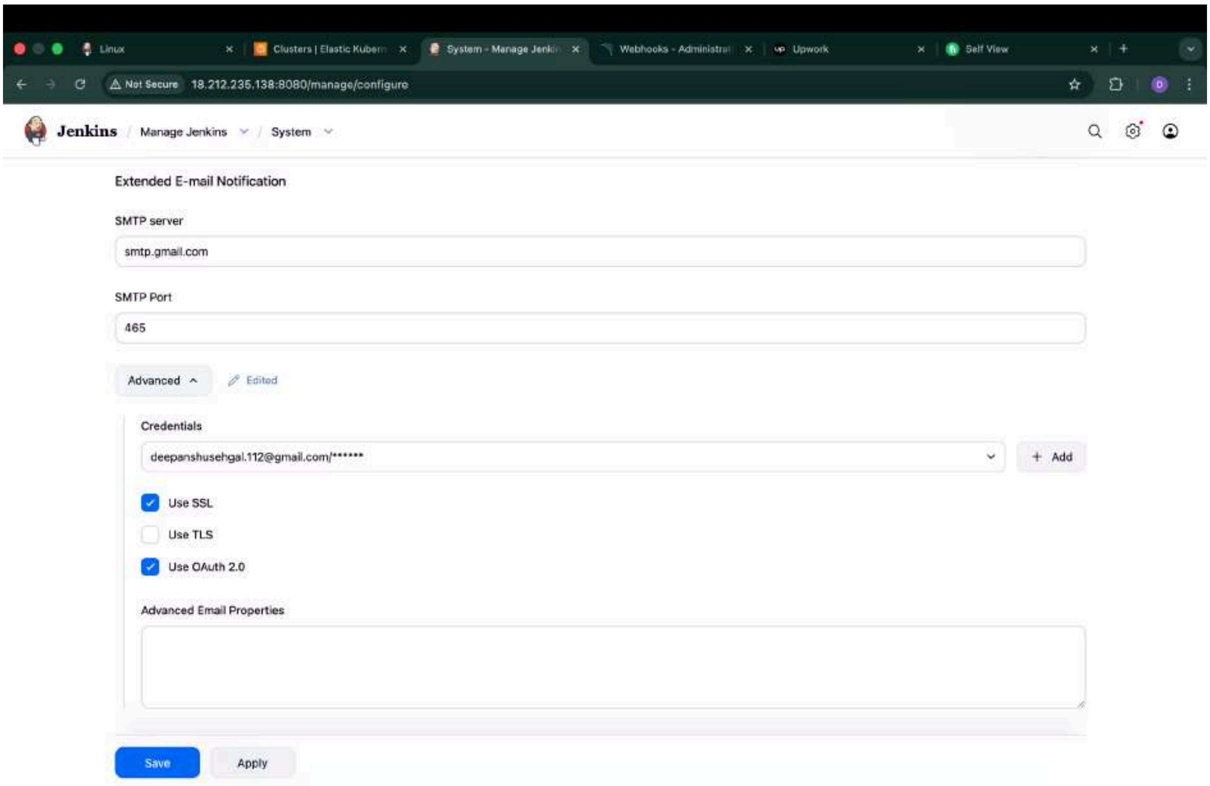
**Manage Jenkins → System**

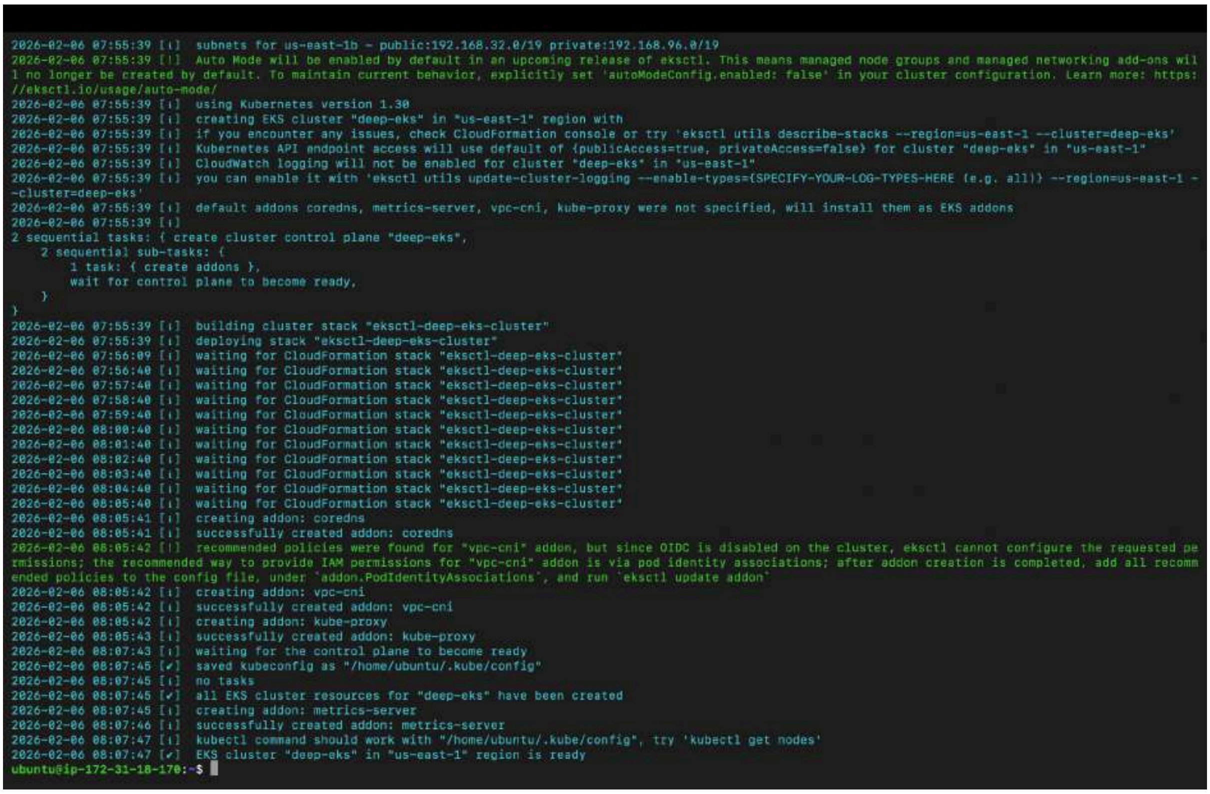


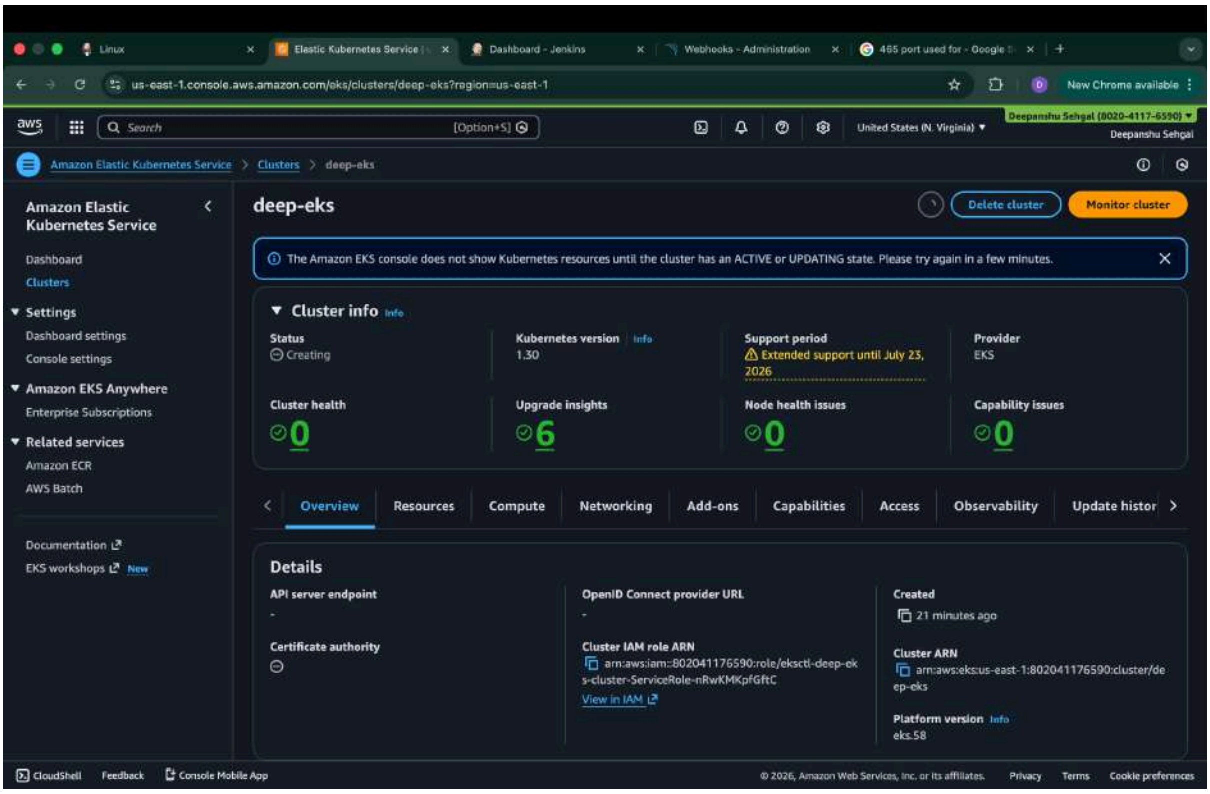

myaccount.google.com/apppasswords
Google Account
App passwords
App passwords help you sign in to your Google Account on older apps and services that don't support modern security standards.
App passwords are less secure than using up-to-date apps and services that use modern security standards. Before you create an app password, you should check to see if your app needs this in order to sign in.
Learn more
Your app passwords
Mulaqat
Created at 2 Jul 2025, last used at 22 Aug 2025
Uber-App-Spring
Created at 8 Jan 2025, last used at 8 Jan 2025
devsecops project
Created on 19:28
To create a new app-specific password, type a name for it below...
App name
Create

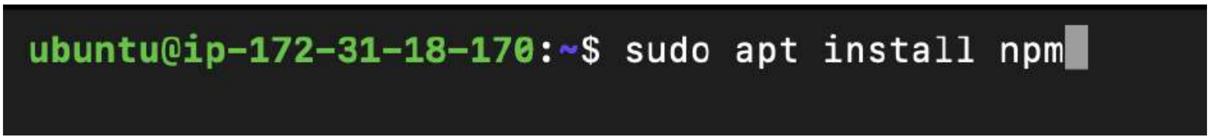

18.212.235.138:8080/manage/credentials/store/system/domain/_/newCredentials
Jenkins
Manage Jenkins
Credentials
System
Global credentials (unrestrict...
Kind
Username with password
Scope
Global (Jenkins, nodes, items, all child items, etc)
Username
deepanshusehgal.112@gmail.com
Treat username as secret
Password
ID
email-creds
Description
Create


Linux
Clusters | Elastic Kubern
System » Global creden
Webhooks - Administrat
Upwork
Self View
Not Secure
18.212.235.138:8080/manage/credentials/store/system/domain/_/
Jenkins
Manage Jenkins
Credentials
System
Global credentials (unrestrict...
Global credentials (unrestricted)
+ Add Credentials
Credentials that should be available irrespective of domain specification to requirements matching.
Sonar-token
Sonar-token - Sonar-token
deepanshu112/****** (docker-creds)
docker - docker-creds
deepanshusehgal.112@gmail.com/******
email-creds
REST API
Jenkins 2.541.1


Linux
Clusters | Elastic Kubern
System - Manage Jenkin
Webhooks - Administrat
Upwork
Self View
Not Secure
18.212.235.138:8080/manage/configure
Jenkins
Manage Jenkins
System
Extended E-mail Notification
SMTP server
smtp.gmail.com
SMTP Port
465
Advanced
Edited
Credentials
deepanshusehgal.112@gmail.com/******
+ Add
Use SSL
Use TLS
Use OAuth 2.0
Advanced Email Properties
Save
Apply

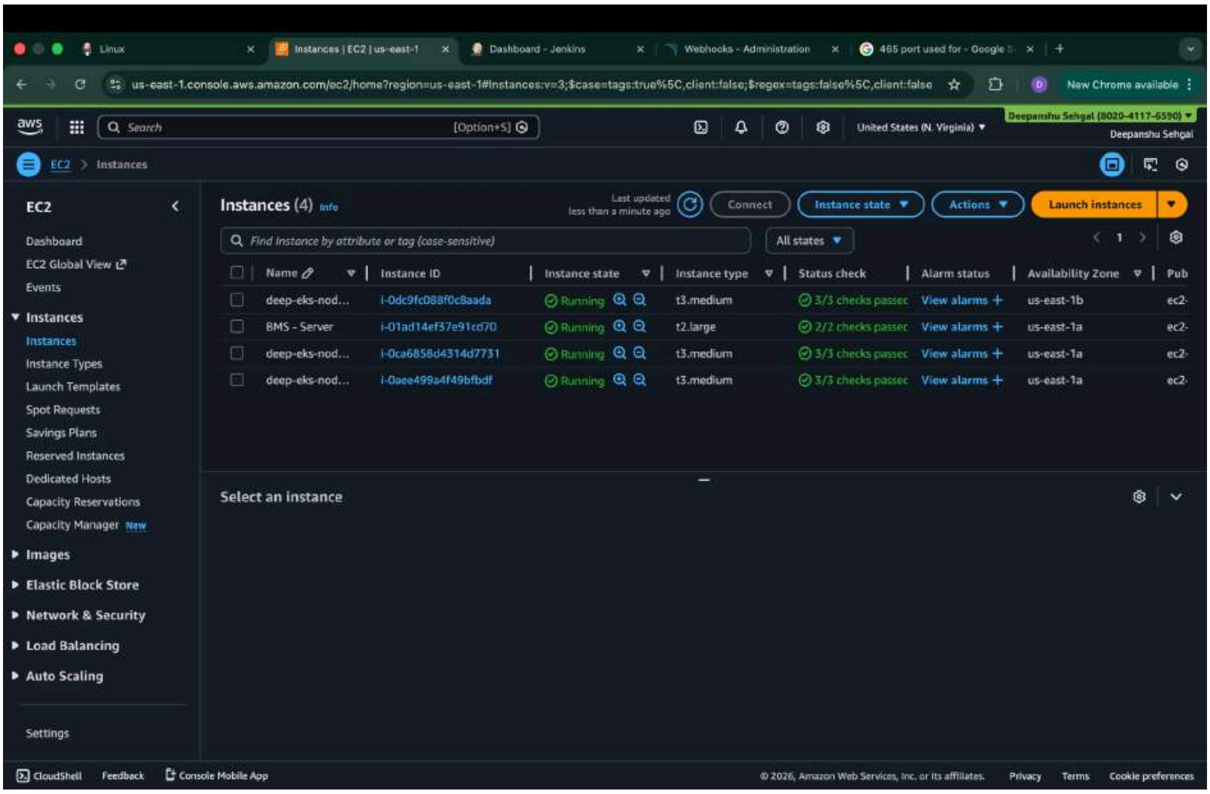

2026-02-06 07:55:39 [i] subnets for us-east-1b - public:192.168.32.0/19 private:192.168.96.0/19
2026-02-06 07:55:39 [!] Auto Mode will be enabled by default in an upcoming release of eksctl. This means managed node groups and managed networking add-ons will no longer be created by default. To maintain current behavior, explicitly set 'autoModeConfig.enabled: false' in your cluster configuration. Learn more: https://eksctl.io/usage/auto-mode/
2026-02-06 07:55:39 [i] using Kubernetes version 1.30
2026-02-06 07:55:39 [i] creating EKS cluster "deep-eks" in "us-east-1" region with
2026-02-06 07:55:39 [i] if you encounter any issues, check CloudFormation console or try 'eksctl utils describe-stacks --region=us-east-1 --cluster=deep-eks'
2026-02-06 07:55:39 [i] Kubernetes API endpoint access will use default of {publicAccess=true, privateAccess=false} for cluster "deep-eks" in "us-east-1"
2026-02-06 07:55:39 [i] CloudWatch logging will not be enabled for cluster "deep-eks" in "us-east-1"
2026-02-06 07:55:39 [i] you can enable it with 'eksctl utils update-cluster-logging --enable-types={SPECIFY-YOUR-LOG-TYPES-HERE (e.g. all)} --region=us-east-1 --cluster=deep-eks'
2026-02-06 07:55:39 [i] default addons coredns, metrics-server, vpc-cni, kube-proxy were not specified, will install them as EKS addons
2026-02-06 07:55:39 [i]
2 sequential tasks: { create cluster control plane "deep-eks",
2 sequential sub-tasks: {
1 task: { create addons },
wait for control plane to become ready,
}
}
2026-02-06 07:55:39 [i] building cluster stack "eksctl-deep-eks-cluster"
2026-02-06 07:55:39 [i] deploying stack "eksctl-deep-eks-cluster"
2026-02-06 07:56:09 [i] waiting for CloudFormation stack "eksctl-deep-eks-cluster"
2026-02-06 07:56:40 [i] waiting for CloudFormation stack "eksctl-deep-eks-cluster"
2026-02-06 07:57:40 [i] waiting for CloudFormation stack "eksctl-deep-eks-cluster"
2026-02-06 07:58:40 [i] waiting for CloudFormation stack "eksctl-deep-eks-cluster"
2026-02-06 07:59:40 [i] waiting for CloudFormation stack "eksctl-deep-eks-cluster"
2026-02-06 08:00:40 [i] waiting for CloudFormation stack "eksctl-deep-eks-cluster"
2026-02-06 08:01:40 [i] waiting for CloudFormation stack "eksctl-deep-eks-cluster"
2026-02-06 08:02:40 [i] waiting for CloudFormation stack "eksctl-deep-eks-cluster"
2026-02-06 08:03:40 [i] waiting for CloudFormation stack "eksctl-deep-eks-cluster"
2026-02-06 08:04:40 [i] waiting for CloudFormation stack "eksctl-deep-eks-cluster"
2026-02-06 08:05:40 [i] waiting for CloudFormation stack "eksctl-deep-eks-cluster"
2026-02-06 08:05:41 [i] creating addon: coredns
2026-02-06 08:05:41 [i] successfully created addon: coredns
2026-02-06 08:05:42 [!] recommended policies were found for "vpc-cni" addon, but since OIDC is disabled on the cluster, eksctl cannot configure the requested permissions; the recommended way to provide IAM permissions for "vpc-cni" addon is via pod identity associations; after addon creation is completed, add all recommended policies to the config file, under `addon.PodIdentityAssociations`, and run `eksctl update addon`
2026-02-06 08:05:42 [i] creating addon: vpc-cni
2026-02-06 08:05:42 [i] successfully created addon: vpc-cni
2026-02-06 08:05:42 [i] creating addon: kube-proxy
2026-02-06 08:05:43 [i] successfully created addon: kube-proxy
2026-02-06 08:07:43 [i] waiting for the control plane to become ready
2026-02-06 08:07:45 [✔] saved kubeconfig as "/home/ubuntu/.kube/config"
2026-02-06 08:07:45 [i] no tasks
2026-02-06 08:07:45 [✔] all EKS cluster resources for "deep-eks" have been created
2026-02-06 08:07:45 [i] creating addon: metrics-server
2026-02-06 08:07:46 [i] successfully created addon: metrics-server
2026-02-06 08:07:47 [i] kubectl command should work with "/home/ubuntu/.kube/config", try 'kubectl get nodes'
2026-02-06 08:07:47 [✔] EKS cluster "deep-eks" in "us-east-1" region is ready
ubuntu@ip-172-31-18-170:~$


Linux
Elastic Kubernetes Service
Dashboard - Jenkins
Webhooks - Administration
465 port used for - Google
us-east-1.console.aws.amazon.com/eks/clusters/deep-eks?region=us-east-1
New Chrome available
Search
[Option+S]
United States (N. Virginia)
Deepanshu Sehgal
Amazon Elastic Kubernetes Service
Clusters
deep-eks
Amazon Elastic Kubernetes Service
Dashboard
Clusters
Settings
Dashboard settings
Console settings
Amazon EKS Anywhere
Enterprise Subscriptions
Related services
Amazon ECR
AWS Batch
Documentation
EKS workshops
New
deep-eks
Delete cluster
Monitor cluster
The Amazon EKS console does not show Kubernetes resources until the cluster has an ACTIVE or UPDATING state. Please try again in a few minutes.
Cluster info
Info
Status
Creating
Kubernetes version
Info
1.30
Support period
Extended support until July 23, 2026
Provider
EKS
Cluster health
0
Upgrade insights
6
Node health issues
0
Capability issues
0
Overview
Resources
Compute
Networking
Add-ons
Capabilities
Access
Observability
Update histor
Details
API server endpoint
-
OpenID Connect provider URL
-
Created
21 minutes ago
Certificate authority
Cluster IAM role ARN
arn:aws:iam::802041176590:role/eksctl-deep-eks-cluster-ServiceRole-nRwKMKpfGftC
View in IAM
Cluster ARN
arn:aws:eks:us-east-1:802041176590:cluster/deep-eks
Platform version
Info
eks.58
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


ubuntu@ip-172-31-18-170:~$ sudo apt install npm

```
ubuntu@ip-172-31-18-170:~$ eksctl utils associate-iam-oidc-provider \
    --region us-east-1 \
    --cluster deep-eks \
    --approve
```

```
ubuntu@ip-172-31-18-170:~$ eksctl create nodegroup --cluster=deep-eks \
                       --region=us-east-1 \
                       --name=node2 \
                       --node-type=t3.medium \
                       --nodes=3 \
                       --nodes-min=2 \
                       --nodes-max=4 \
                       --node-volume-size=20 \
                       --ssh-access \
                       --ssh-public-key="jenkins machine" \
                       --managed \
                       --asg-access \
                       --external-dns-access \
                       --full-ecr-access \
                       --appmesh-access \
                       --alb-ingress-access
```

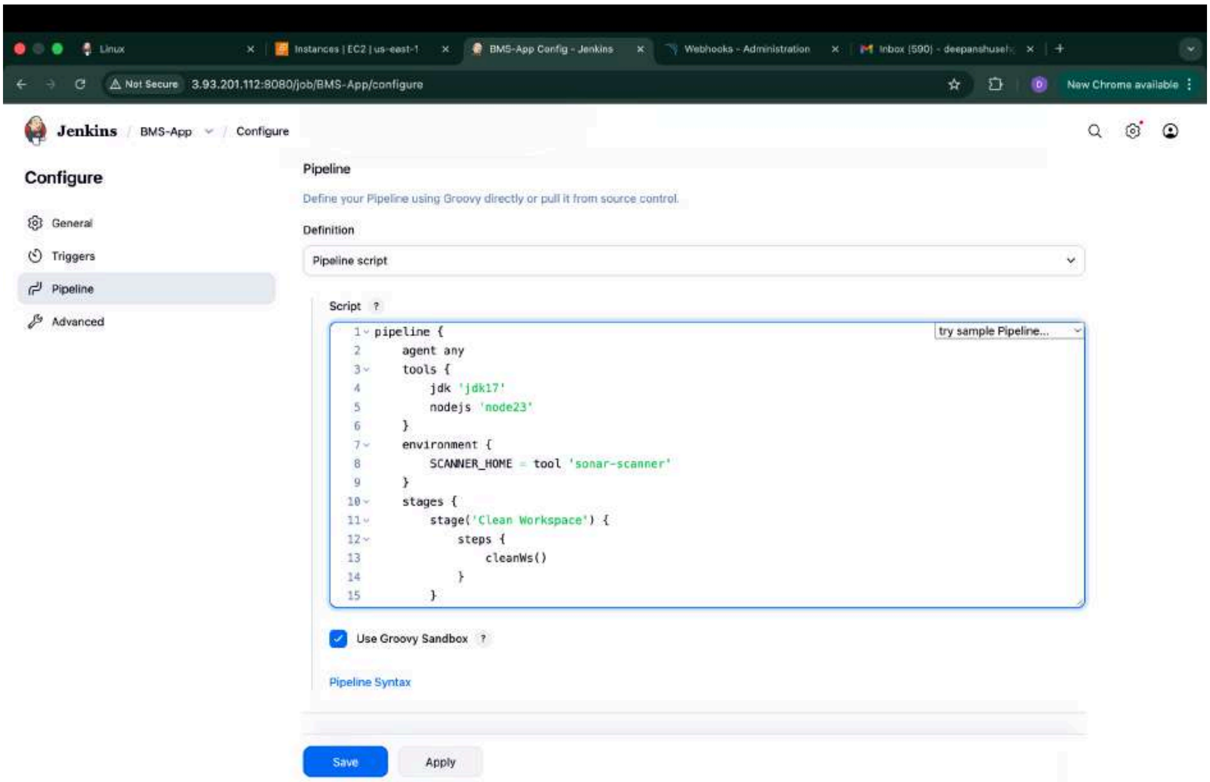


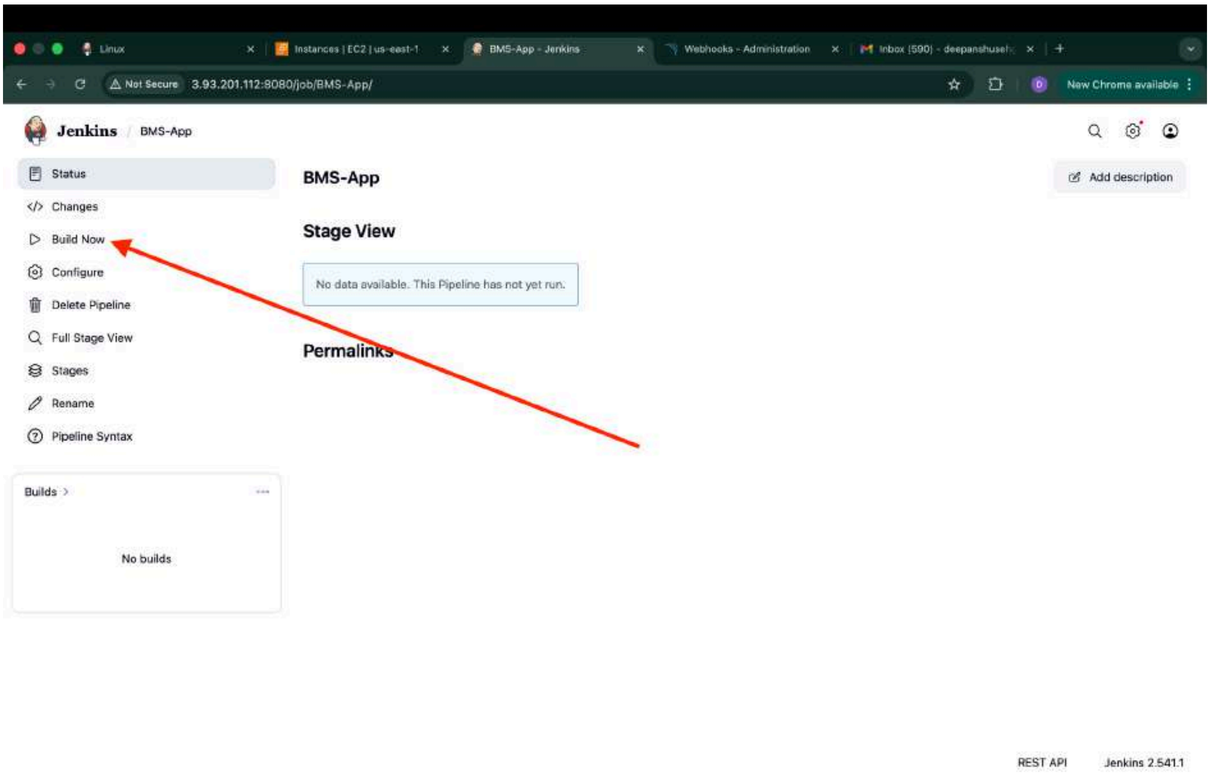
## Step 5: Pipeline Job

Pipeline includes:

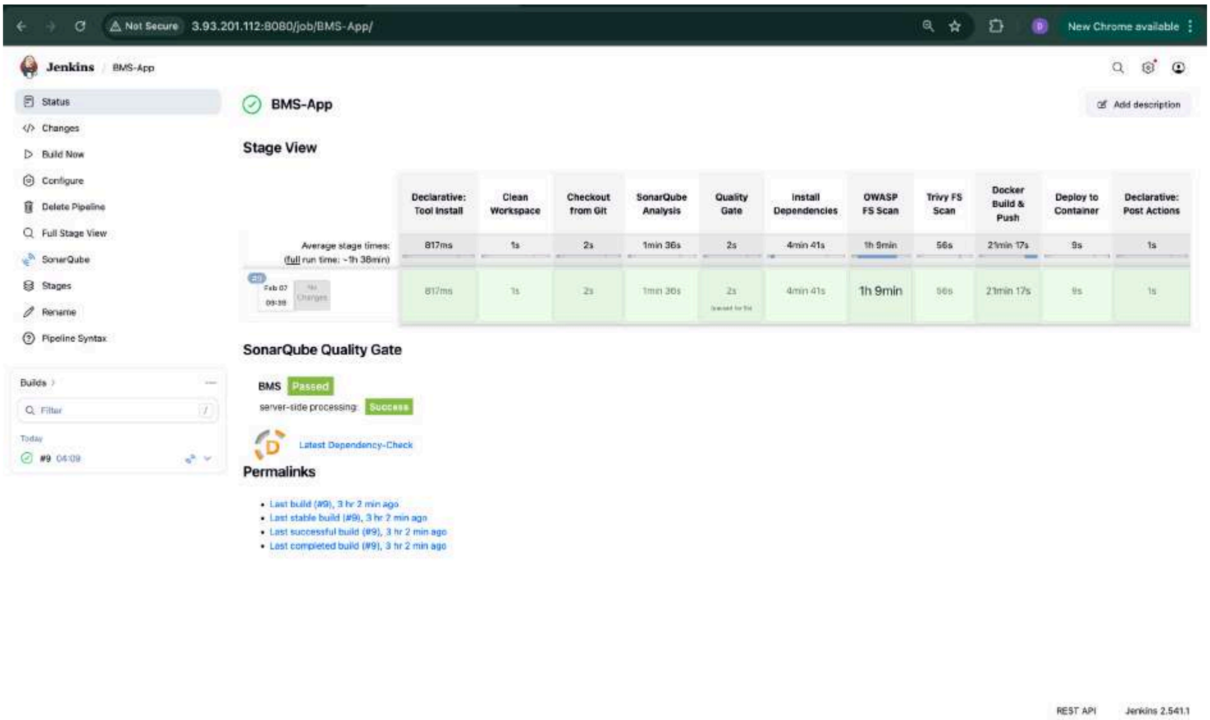
**Stages:**

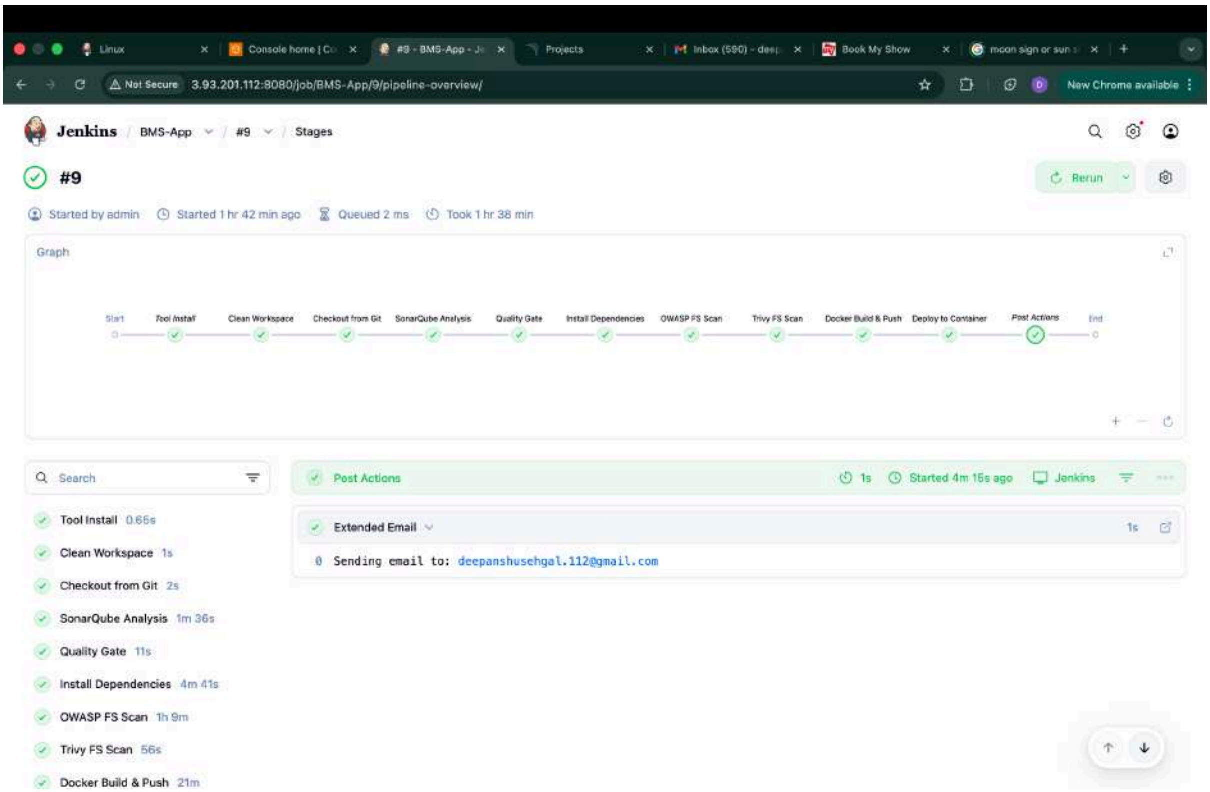
1. Clean Workspace
2. Git Checkout
3. SonarQube Analysis
4. Quality Gate
5. NPM Install
6. Trivy Scan
7. Docker Build & Push
8. Deploy Container

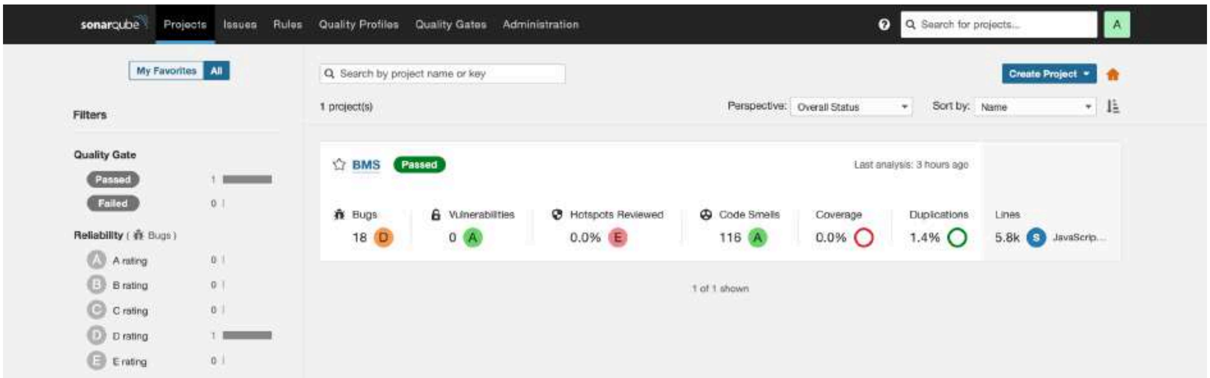


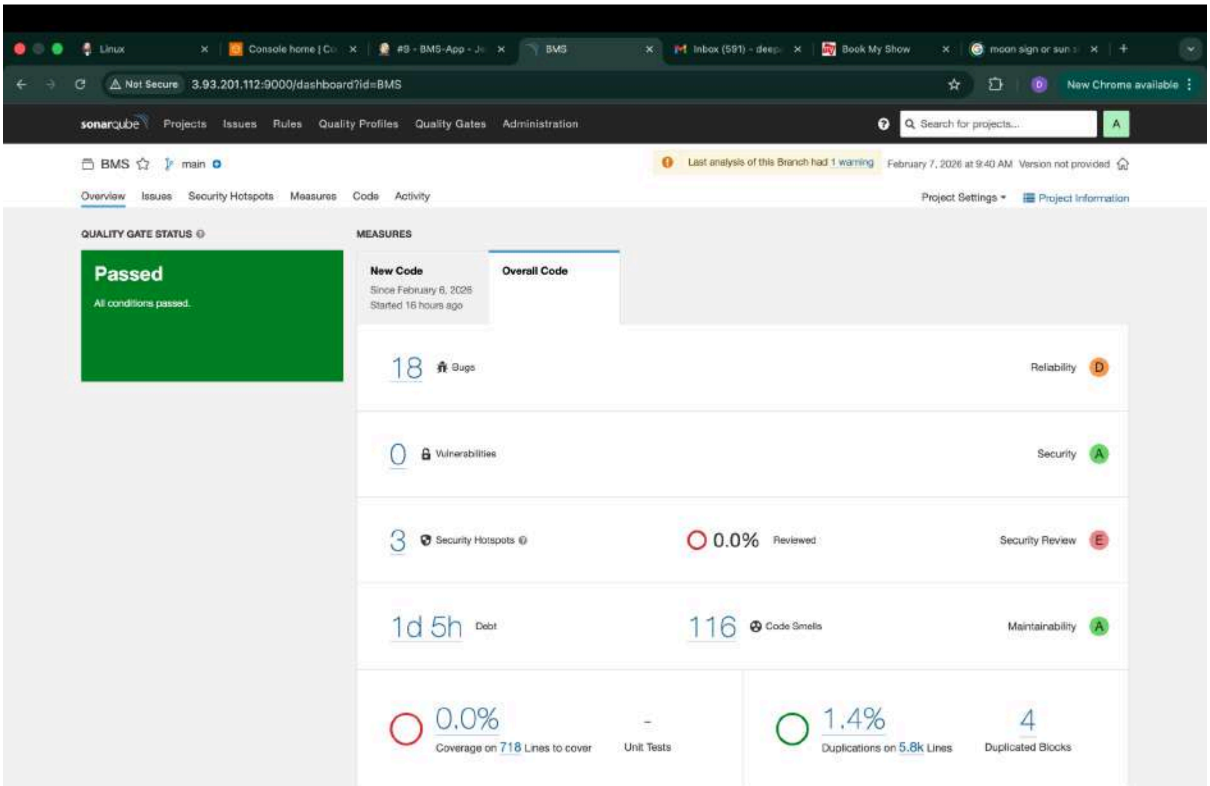


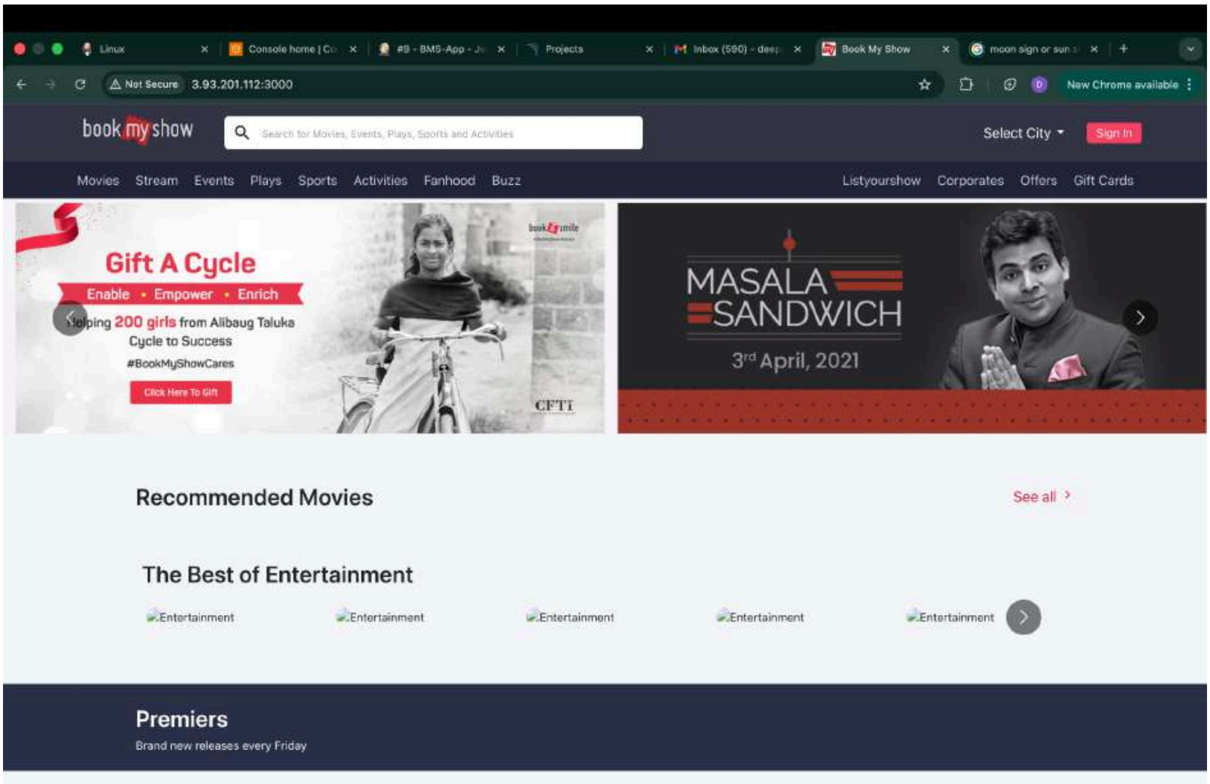
This pipeline has been deployed in a Docker container through the Jenkins pipeline, without using Kubernetes.




Not Secure 3.93.201.112:8080/job/BMS-App/9/pipeline-overview/
Jenkins
BMS-App
#9
Stages
#9
Rerun
Started by admin
Started 1 hr 42 min ago
Queued 2 ms
Took 1 hr 38 min
Graph
Start
Tool Install
Clean Workspace
Checkout from Git
SonarQube Analysis
Quality Gate
Install Dependencies
OWASP FS Scan
Trivy FS Scan
Docker Build & Push
Deploy to Container
Post Actions
End
Search
Tool Install 0.66s
Clean Workspace 1s
Checkout from Git 2s
SonarQube Analysis 1m 36s
Quality Gate 11s
Install Dependencies 4m 41s
OWASP FS Scan 1h 9m
Trivy FS Scan 56s
Docker Build & Push 21m
Post Actions
1s
Started 4m 15s ago
Jenkins
Extended Email
1s
Sending email to: deepanshusehgal.112@gmail.com


sonarqube
Projects
Issues
Rules
Quality Profiles
Quality Gates
Administration
Search for projects...
My Favorites
All
Search by project name or key
Create Project
1 project(s)
Perspective: Overall Status
Sort by: Name
Filters
Quality Gate
Passed 1
Failed 0
Reliability ( Bugs )
A rating 0
B rating 0
C rating 0
D rating 1
E rating 0
BMS
Passed
Last analysis: 3 hours ago
Bugs 18 D
Vulnerabilities 0 A
Hotspots Reviewed 0.0% E
Code Smells 116 A
Coverage 0.0%
Duplications 1.4%
Lines 5.8k S JavaScrip...
1 of 1 shown


Not Secure 3.93.201.112:9000/dashboard?id=BMS
sonarqube Projects Issues Rules Quality Profiles Quality Gates Administration
BMS main
Last analysis of this Branch had 1 warning February 7, 2026 at 9:40 AM Version not provided
Overview Issues Security Hotspots Measures Code Activity
Project Settings Project Information
QUALITY GATE STATUS
Passed
All conditions passed.
MEASURES
New Code
Since February 6, 2026
Started 16 hours ago
Overall Code
18 Bugs
Reliability D
0 Vulnerabilities
Security A
3 Security Hotspots
0.0% Reviewed
Security Review E
1d 5h Debt
116 Code Smells
Maintainability A
0.0%
Coverage on 718 Lines to cover
Unit Tests
1.4%
Duplications on 5.8k Lines
4
Duplicated Blocks


Not Secure 3.93.201.112:3000
book my show
Select City Sign In
Movies Stream Events Plays Sports Activities Fanhood Buzz
Listyourshow Corporates Offers Gift Cards
Gift A Cycle
Enable • Empower • Enrich
Helping 200 girls from Alibaug Taluka
Cycle to Success
#BookMyShowCares
Click Here To Gift
MASALA
SANDWICH
3rd April, 2021
Recommended Movies
See all
The Best of Entertainment
Entertainment
Entertainment
Entertainment
Entertainment
Entertainment
Premiers
Brand new releases every Friday

# Kubernetes Deployment & Monitoring

### K8s Deployment

## Switch to jenkins user:

To know Jenkins is running on which user:

```
ubuntu@ip-172-31-18-170:~$ ps aux | grep jenkins
jenkins    41933  2.3  8.8 4776392 715964 ?      Ssl  06:37   1:34 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/j
enkins/war --httpPort=8080
ubuntu     43170  0.0  0.0   7076  2168 pts/1    S+   07:44   0:00 grep --color=auto jenkins
ubuntu@ip-172-31-18-170:~$
```

Since Jenkins runs under the Jenkins user, we need to switch to the Jenkins user.

Configure the Aws in Jenkins user also

```
ubuntu@ip-172-31-18-170:~$ sudo -su jenkins
jenkins@ip-172-31-18-170:/home/ubuntu$ aws configure
 WS Access Key ID [None]: p
jenkins@ip-172-31-18-170:/home/ubuntu$ aws configure
AWS Access Key ID [None]: AKIA3VPLBEIHJJXIB5HT
AWS Secret Access Key [None]: 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
Default region name [None]:
Default output format [None]:
jenkins@ip-172-31-18-170:/home/ubuntu$
```

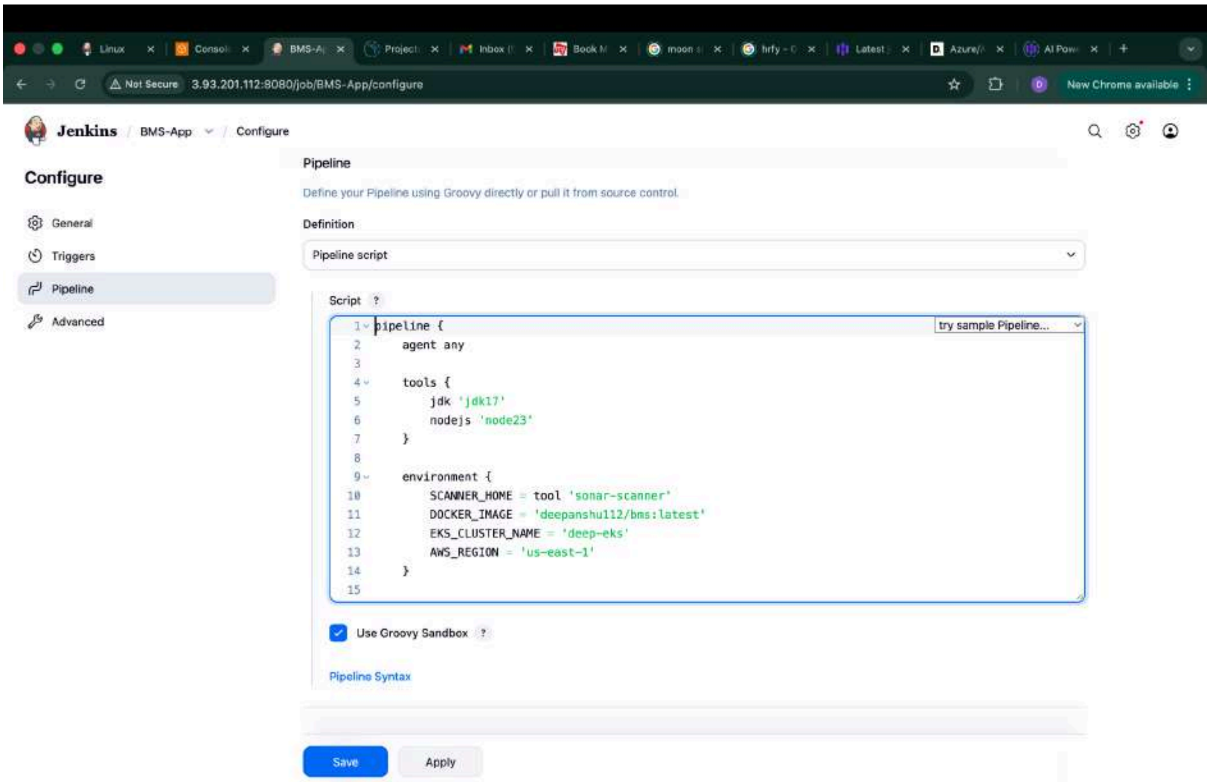
Verify the credentials

```
jenkins@ip-172-31-18-170:/home/ubuntu$ aws sts get-caller-identity
{
    "UserId": "AIDA3VPLBEIHCADNPPH2L",
    "Account": "802041176590",
    "Arn": "arn:aws:iam::802041176590:user/eks-deep-user"
}
jenkins@ip-172-31-18-170:/home/ubuntu$
```

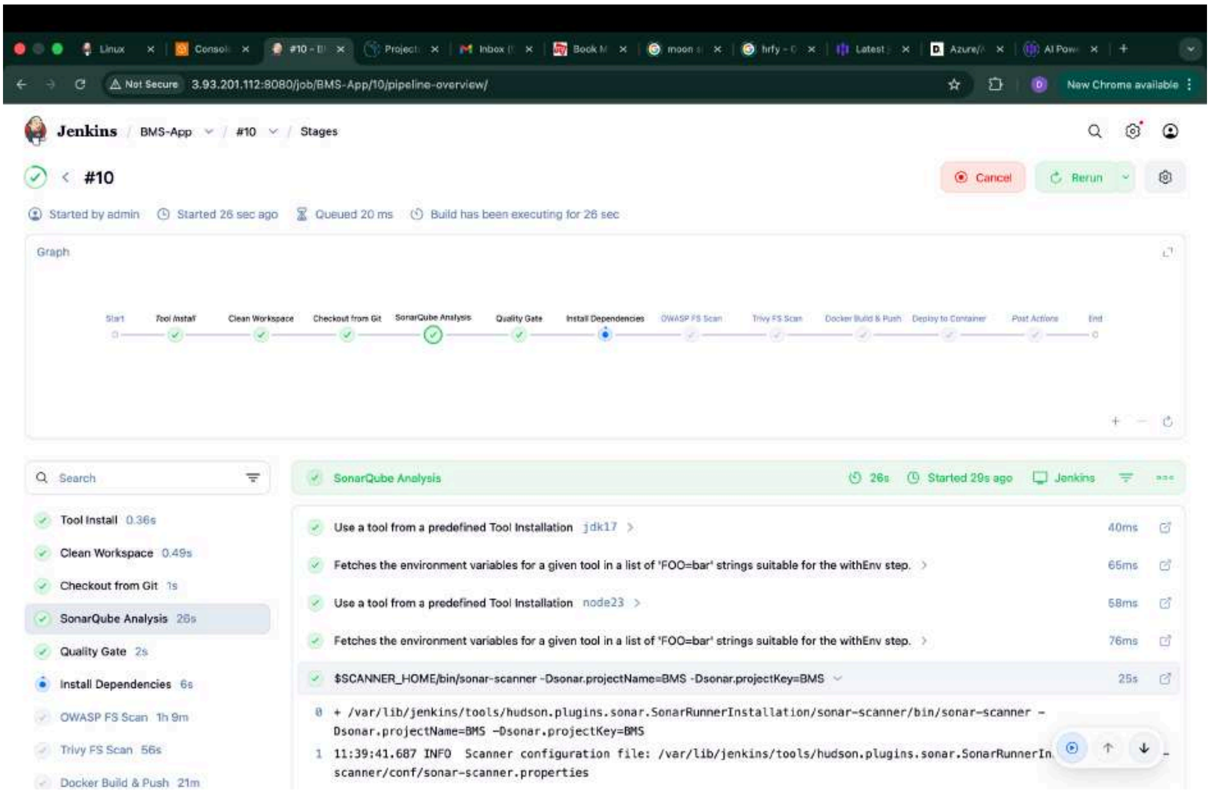
Comeout of the Jenkins user to Restart Jenkins

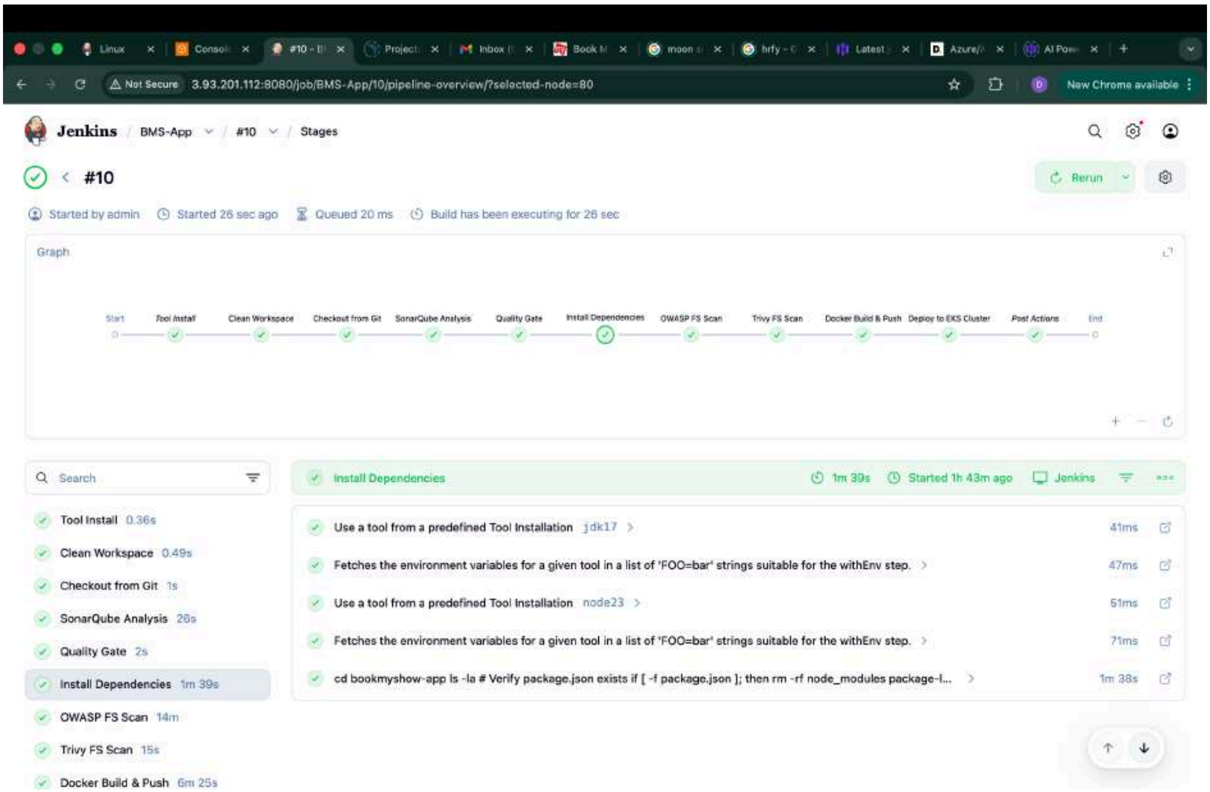
```
ubuntu@ip-172-31-18-170:~$ sudo -su jenkins
jenkins@ip-172-31-18-170:/home/ubuntu$ aws eks update-kubeconfig --name deep-eks --region us-east-1
Added new context arn:aws:eks:us-east-1:802041176590:cluster/deep-eks to /var/lib/jenkins/.kube/config
jenkins@ip-172-31-18-170:/home/ubuntu$
```

Pipeline deploys using:




Now trigger the K8S pipeline

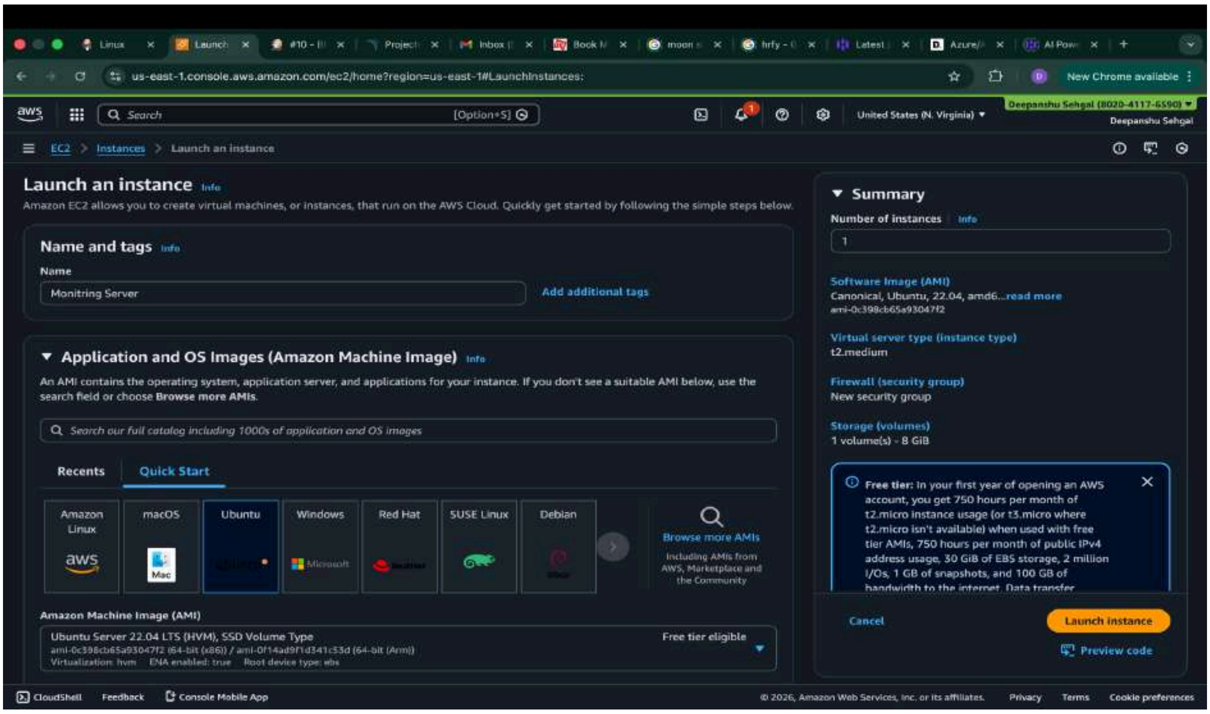


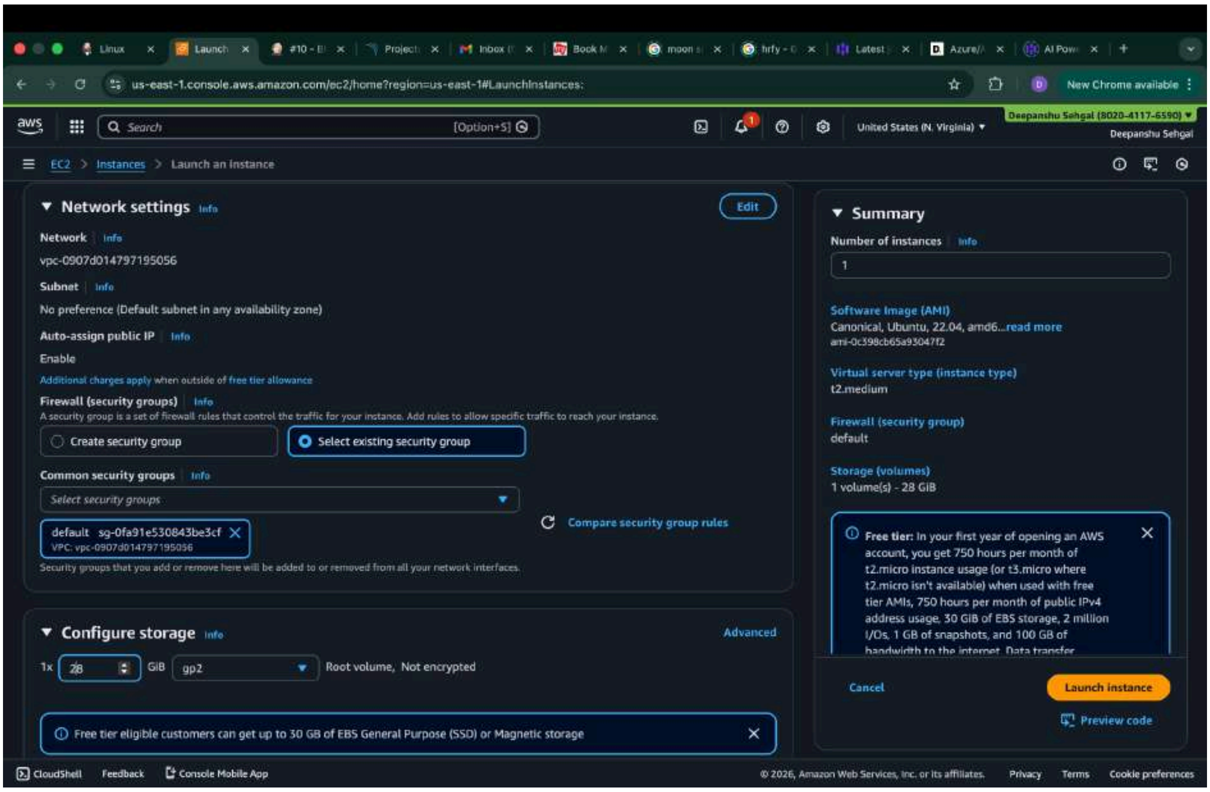

Not Secure 3.93.201.112:8080/job/BMS-App/10/pipeline-overview/?selected-node=80
New Chrome available
Jenkins
BMS-App
#10
Stages
#10
Rerun
Started by admin
Started 26 sec ago
Queued 20 ms
Build has been executing for 26 sec
Graph
Start
Tool Install
Clean Workspace
Checkout from Git
SonarQube Analysis
Quality Gate
Install Dependencies
OWASP FS Scan
Trivy FS Scan
Docker Build & Push
Deploy to EKS Cluster
Post Actions
End
Search
Tool Install 0.36s
Clean Workspace 0.49s
Checkout from Git 1s
SonarQube Analysis 26s
Quality Gate 2s
Install Dependencies 1m 39s
OWASP FS Scan 14m
Trivy FS Scan 15s
Docker Build & Push 6m 25s
Install Dependencies
1m 39s
Started 1h 43m ago
Jenkins
Use a tool from a predefined Tool Installation jdk17
41ms
Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.
47ms
Use a tool from a predefined Tool Installation node23
51ms
Fetches the environment variables for a given tool in a list of 'FOO=bar' strings suitable for the withEnv step.
71ms
cd bookmyshow-app ls -la # Verify package.json exists if [ -f package.json ]; then rm -rf node_modules package-l...
1m 38s

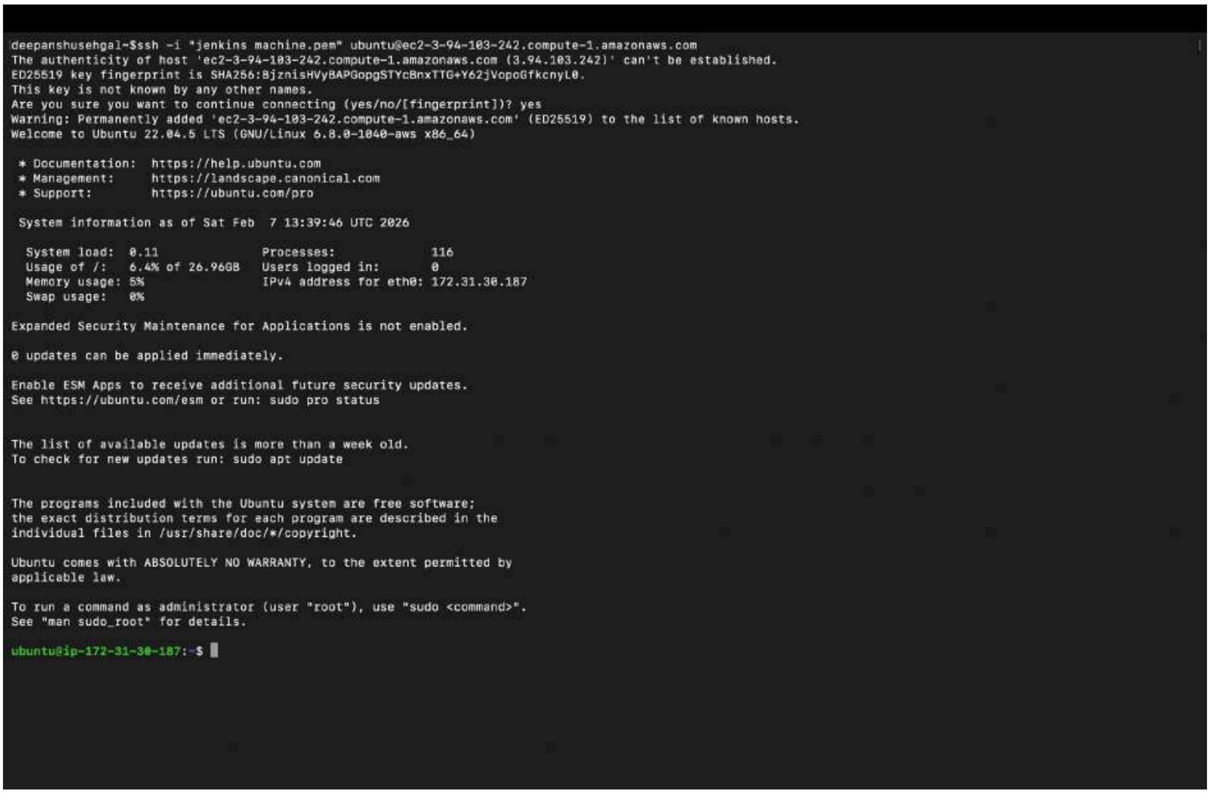
## Step 6: Monitoring Setup

Launch a new VM:

- Ubuntu 22.04
- t2.medium
- Name: Monitoring Server







Connect to the Monitoring Server VM (Execute in Monitoring Server VM)
Create a dedicated Linux user sometimes called a 'system' account for Prometheus

sudo apt update

```
ubuntu@ip-172-31-30-187:~$ sudo useradd \
    --system \
    --no-create-home \
    --shell /bin/false prometheus
```

With the above command, we have created a 'Prometheus' user

```
ubuntu@ip-172-31-30-187:~$ sudo wget https://github.com/prometheus/prometheus/releases/download/v2.47.1/prometheus-2.47.1.linux-amd64.tar.gz
--2026-02-07 13:42:56--  https://github.com/prometheus/prometheus/releases/download/v2.47.1/prometheus-2.47.1.linux-amd64.tar.gz
Resolving github.com (github.com)... 140.82.113.3
Connecting to github.com (github.com)|140.82.113.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/6838921/2f9b7b37-63a0-428b-adb5-0294482fd743?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-02-07T14%3A34%3A39Z&rscd=attachment%3B+filename%3Dprometheus-2.47.1.linux-amd64.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-02-07T13%3A34%3A28Z&ske=2026-02-07T14%3A34%3A39Z&sks=b&skv=2018-11-09&sig=KbZjBIup5DI6ty8X5u1VnnPDGD2Ai52RF6YqL3AlpqE%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1hc3NldHMuZ2l0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5IjoiaZV5MSIsImV4cCI6MTc3MDQ3MzU3NiwibmJmIjoxNzcwNDcxNzc2LCJwYXRoIjoicmVsZWFzZWFzc2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.VSPyZq5ZrAVK5vU8ytHlg8vXz4BVAQUgImHBpxaSqIk&response-content-disposition=attachment%3B%20filename%3Dprometheus-2.47.1.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream [following]
--2026-02-07 13:42:56--  https://release-assets.githubusercontent.com/github-production-release-asset/6838921/2f9b7b37-63a0-428b-adb5-0294482fd743?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-02-07T14%3A34%3A39Z&rscd=attachment%3B+filename%3Dprometheus-2.47.1.linux-amd64.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-02-07T13%3A34%3A28Z&ske=2026-02-07T14%3A34%3A39Z&sks=b&skv=2018-11-09&sig=KbZjBIup5DI6ty8X5u1VnnPDGD2Ai52RF6YqL3AlpqE%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1hc3NldHMuZ2l0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc3MDQ3MzU3NiwibmJmIjoxNzcwNDcxNzc2LCJwYXRoIjoicmVsZWFzZWFzc2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.VSPyZq5ZrAVK5vU8ytHlg8vXz4BVAQUgImHBpxaSqIk&response-content-disposition=attachment%3B%20filename%3Dprometheus-2.47.1.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.109.133, 185.199.110.133, 185.199.111.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.109.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 95713066 (91M) [application/octet-stream]
Saving to: 'prometheus-2.47.1.linux-amd64.tar.gz'

prometheus-2.47.1.linux-amd64.tar.gz     100%[===================================================================>]  91.28M   116MB/s    in 0.8s

2026-02-07 13:42:57 (116 MB/s) - 'prometheus-2.47.1.linux-amd64.tar.gz' saved [95713066/95713066]

ubuntu@ip-172-31-30-187:~$
```

```
ubuntu@ip-172-31-30-187:~$ tar -xvf prometheus-2.47.1.linux-amd64.tar.gz
```

```
ubuntu@ip-172-31-30-187:~$ tar -xvf prometheus-2.47.1.linux-amd64.tar.gz
prometheus-2.47.1.linux-amd64/
prometheus-2.47.1.linux-amd64/LICENSE
prometheus-2.47.1.linux-amd64/NOTICE
prometheus-2.47.1.linux-amd64/prometheus.yml
prometheus-2.47.1.linux-amd64/consoles/
prometheus-2.47.1.linux-amd64/consoles/prometheus.html
prometheus-2.47.1.linux-amd64/consoles/prometheus-overview.html
prometheus-2.47.1.linux-amd64/consoles/node-cpu.html
prometheus-2.47.1.linux-amd64/consoles/index.html.example
prometheus-2.47.1.linux-amd64/consoles/node.html
prometheus-2.47.1.linux-amd64/consoles/node-disk.html
prometheus-2.47.1.linux-amd64/consoles/node-overview.html
prometheus-2.47.1.linux-amd64/promtool
prometheus-2.47.1.linux-amd64/console_libraries/
prometheus-2.47.1.linux-amd64/console_libraries/prom.lib
prometheus-2.47.1.linux-amd64/console_libraries/menu.lib
prometheus-2.47.1.linux-amd64/prometheus
ubuntu@ip-172-31-30-187:~$ sudo mkdir -p /data /etc/prometheus
ubuntu@ip-172-31-30-187:~$ cd prometheus-2.47.1.linux-amd64/
ubuntu@ip-172-31-30-187:~/prometheus-2.47.1.linux-amd64$
```

Move the Prometheus binary and a promtool to the /usr/local/bin/. promtool is used to check configuration files and Prometheus rules.

```
ubuntu@ip-172-31-30-187:~/prometheus-2.47.1.linux-amd64$ sudo mv prometheus promtool /usr/local/bin/
```

Move console libraries to the Prometheus configuration directory

```
ubuntu@ip-172-31-30-187:~/prometheus-2.47.1.linux-amd64$ sudo mv consoles/ console_libraries/ /etc/prometheus/
ubuntu@ip-172-31-30-187:~/prometheus-2.47.1.linux-amd64$
```

Move the example of the main Prometheus configuration file

```
ubuntu@ip-172-31-30-187:~/prometheus-2.47.1.linux-amd64$ sudo mv prometheus.yml /etc/prometheus/prometheus.yml
ubuntu@ip-172-31-30-187:~/prometheus-2.47.1.linux-amd64$
```

Set the correct ownership for the /etc/prometheus/ and data directory

```
ubuntu@ip-172-31-30-187:~/prometheus-2.47.1.linux-amd64$ sudo chown -R prometheus:prometheus /etc/prometheus/ /data/
ubuntu@ip-172-31-30-187:~/prometheus-2.47.1.linux-amd64$
```

Delete the archive and a Prometheus tar.gz file

```
ubuntu@ip-172-31-30-187:~/prometheus-2.47.1.linux-amd64$ cd
ubuntu@ip-172-31-30-187:~$ rm -rf prometheus-2.47.1.linux-amd64.tar.gz
ubuntu@ip-172-31-30-187:~$
```

```
ubuntu@ip-172-31-30-187:~$ prometheus --version
prometheus, version 2.47.1 (branch: HEAD, revision: c4d1a8beff37cc004f1dc4ab9d2e73193f51aaeb)
  build user:       root@4829330363be
  build date:       20231004-10:31:16
  go version:       go1.21.1
  platform:         linux/amd64
  tags:             netgo,builtinassets,stringlabels
ubuntu@ip-172-31-30-187:~$
```

We're going to use Systemd, which is a system and service manager for Linux operating systems. For that, we need to create a Systemd unit configuration file.

```
ubuntu@ip-172-31-30-187:~$ sudo vi /etc/systemd/system/prometheus.service
```

```
[Unit]
Description=Prometheus
Wants=network-online.target
After=network-online.target
StartLimitIntervalSec=500
StartLimitBurst=5
[Service]
User=prometheus
Group=prometheus
Type=simple
Restart=on-failure
RestartSec=5s
ExecStart=/usr/local/bin/prometheus \
  --config.file=/etc/prometheus/prometheus.yml \
  --storage.tsdb.path=/data \
  --web.console.templates=/etc/prometheus/consoles \
  --web.console.libraries=/etc/prometheus/console_libraries \
  --web.listen-address=0.0.0.0:9090 \
  --web.enable-lifecycle
[Install]
WantedBy=multi-user.target
~
~
~
~
~
~
~
```

To automatically start the Prometheus after reboot run the below command

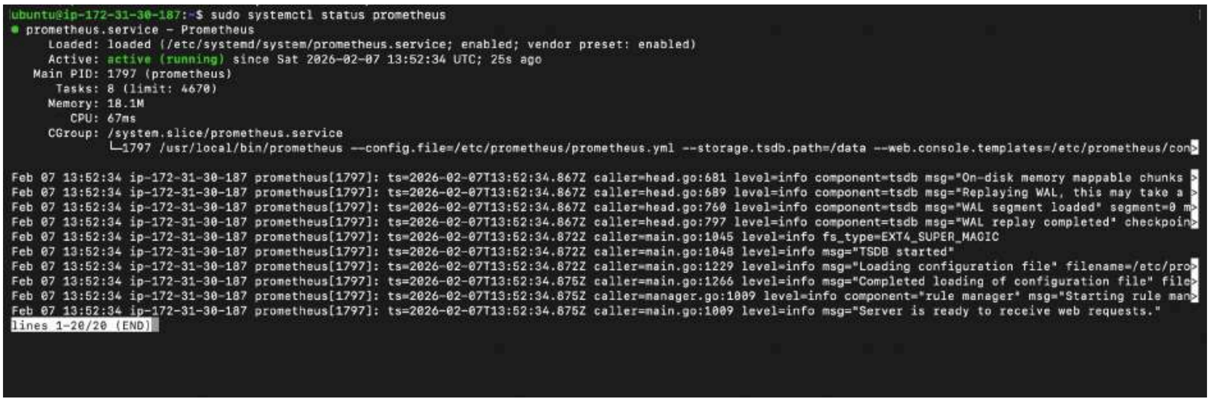
```
ubuntu@ip-172-31-30-187:~$ sudo systemctl enable prometheus
Created symlink /etc/systemd/system/multi-user.target.wants/prometheus.service → /etc/systemd/system/prometheus.service.
ubuntu@ip-172-31-30-187:~$
```

Start the Prometheus

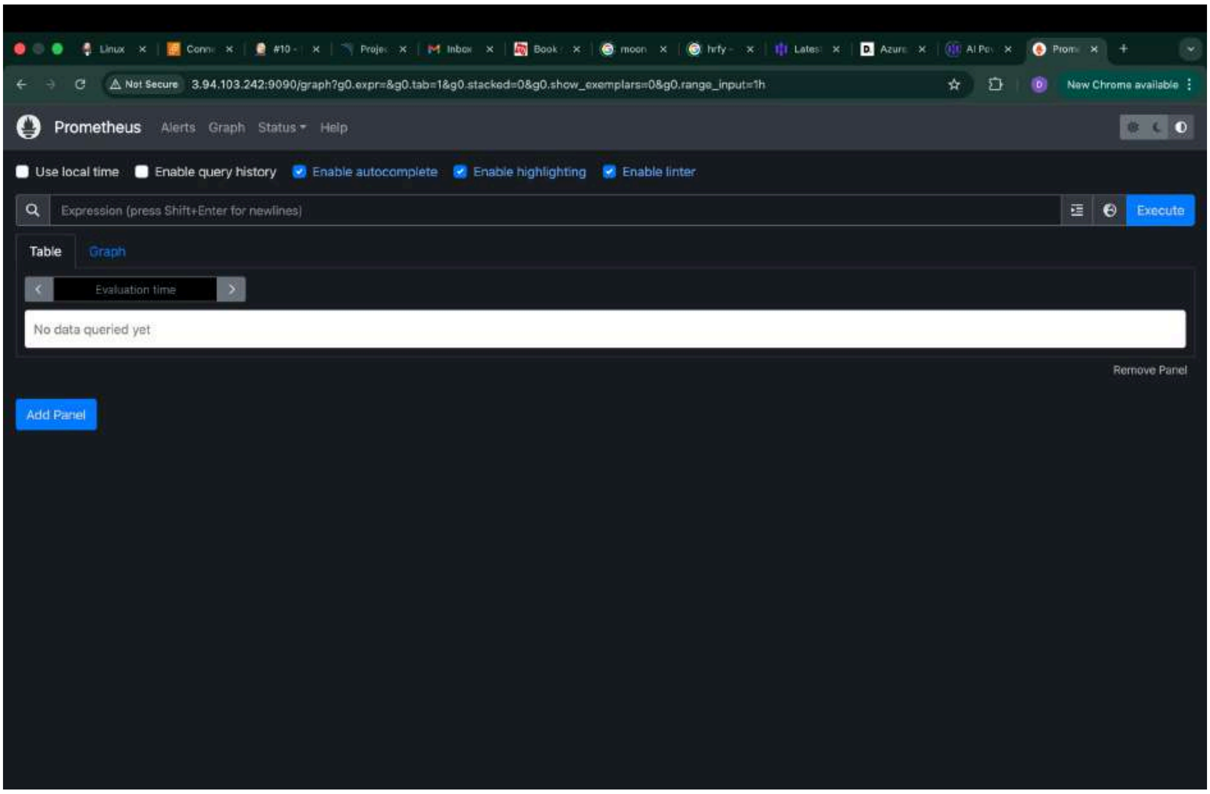
sudo systemctl start prometheus

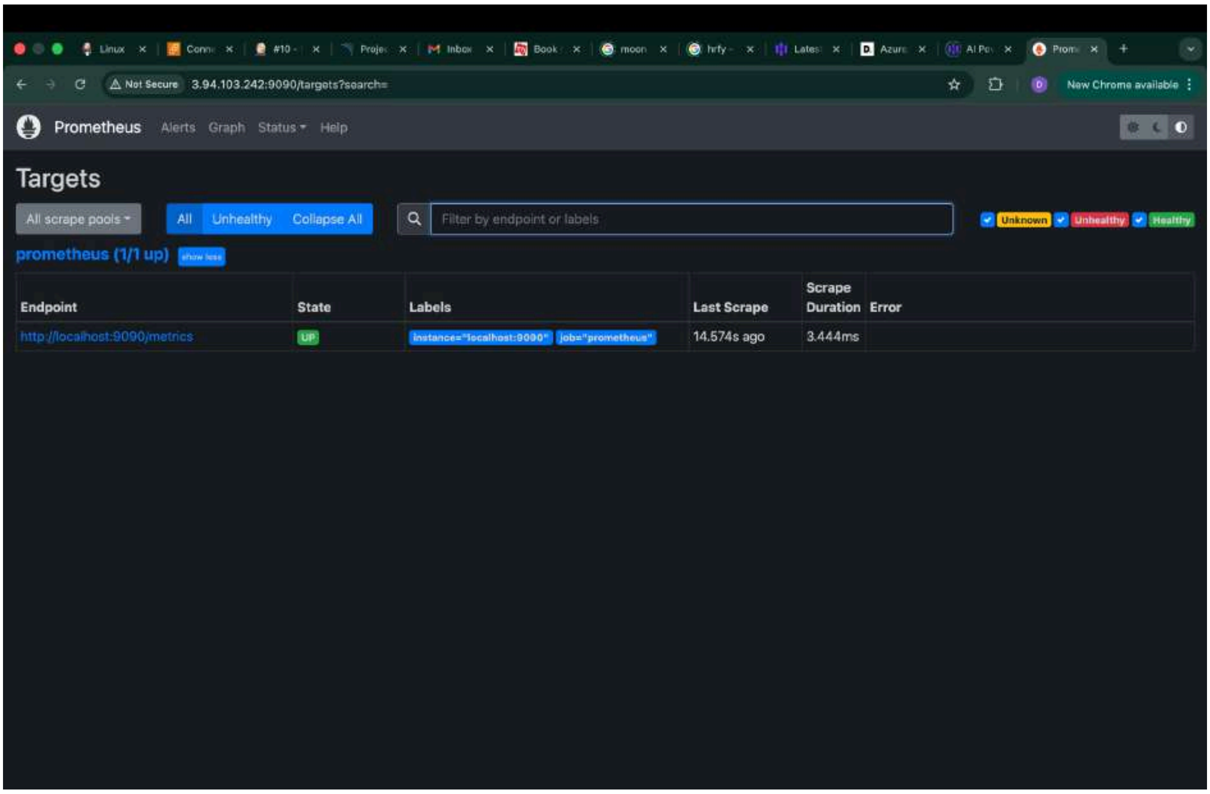
```
ubuntu@ip-172-31-30-187:~$ sudo systemctl start prometheus
ubuntu@ip-172-31-30-187:~$
```

Check the status of Prometheus




Open Port No. 9090 for Monitoring Server VM and Access Prometheus






Create a system user for Node Exporter and download Node Exporter:

```
ubuntu@ip-172-31-30-187:~$ sudo useradd --system --no-create-home --shell /bin/false node_exporter
ubuntu@ip-172-31-30-187:~$ wget https://github.com/prometheus/node_exporter/releases/download/v1.6.1/node_exporter-1.6.1.linux-amd64.tar.gz
--2026-02-07 14:00:15--  https://github.com/prometheus/node_exporter/releases/download/v1.6.1/node_exporter-1.6.1.linux-amd64.tar.gz
Resolving github.com (github.com)... 140.82.112.3
Connecting to github.com (github.com)|140.82.112.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/9524057/5509b569-5c34-471e-8598-c05c0733bb7f?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-02-07T14%3A53%3A15Z&rscd=attachment%3B+filename%3Dnode_exporter-1.6.1.linux-amd64.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-02-07T13%3A52%3A57Z&ske=2026-02-07T14%3A53%3A15Z&sks=b&skv=2018-11-09&sig=ip5Y%2BcGLQoNQPMaleyz%2B12BxIJPIOX%2F5ONFprv%2FxsOI%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1hc3NldHMuZ2l0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc3MDQ3MzExNSwibmJmIjoxNzcwNDcyODE1LCJwYXRoIjoicmVsZWFzZWFzc2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.PXie1lGdgolfP_IcvbYv9Q8f3ve62mXR6Q1ELA_hnvs&response-content-disposition=attachment%3B%20filename%3Dnode_exporter-1.6.1.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream [following]
--2026-02-07 14:00:15--  https://release-assets.githubusercontent.com/github-production-release-asset/9524057/5509b569-5c34-471e-8598-c05c0733bb7f?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-02-07T14%3A53%3A15Z&rscd=attachment%3B+filename%3Dnode_exporter-1.6.1.linux-amd64.tar.gz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-02-07T13%3A52%3A57Z&ske=2026-02-07T14%3A53%3A15Z&sks=b&skv=2018-11-09&sig=ip5Y%2BcGLQoNQPMaleyz%2B12BxIJPIOX%2F5ONFprv%2FxsOI%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwiYXVkIjoicmVsZWFzZS1hc3NldHMuZ2l0aHVidXNlcmNvbnRlbnQuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc3MDQ3MzExNSwibmJmIjoxNzcwNDcyODE1LCJwYXRoIjoicmVsZWFzZWFzc2V0cHJvZHVjdGlvbi5ibG9iLmNvcmUud2luZG93cy5uZXQifQ.PXie1lGdgolfP_IcvbYv9Q8f3ve62mXR6Q1ELA_hnvs&response-content-disposition=attachment%3B%20filename%3Dnode_exporter-1.6.1.linux-amd64.tar.gz&response-content-type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 10368103 (9.9M) [application/octet-stream]
Saving to: 'node_exporter-1.6.1.linux-amd64.tar.gz'

node_exporter-1.6.1.linux-amd64.tar.gz   100%[===================================================================================>]   9.89M  --.-KB/s    in 0.09s

2026-02-07 14:00:15 (116 MB/s) - 'node_exporter-1.6.1.linux-amd64.tar.gz' saved [10368103/10368103]

ubuntu@ip-172-31-30-187:~$
```

Extract Node Exporter files, move the binary, and clean up:

```
ubuntu@ip-172-31-30-187:~$ ls
node_exporter-1.6.1.linux-amd64.tar.gz  prometheus-2.47.1.linux-amd64
ubuntu@ip-172-31-30-187:~$ tar -xvf node_exporter-1.6.1.linux-amd64.tar.gz
node_exporter-1.6.1.linux-amd64/
node_exporter-1.6.1.linux-amd64/NOTICE
node_exporter-1.6.1.linux-amd64/node_exporter
node_exporter-1.6.1.linux-amd64/LICENSE
ubuntu@ip-172-31-30-187:~$ sudo mv node_exporter-1.6.1.linux-amd64/node_exporter /usr/local/bin/
ubuntu@ip-172-31-30-187:~$ rm -rf node_exporter*
ubuntu@ip-172-31-30-187:~$ ls
prometheus-2.47.1.linux-amd64
ubuntu@ip-172-31-30-187:~$
```

```
ubuntu@ip-172-31-30-187:~$ node_exporter --version
node_exporter, version 1.6.1 (branch: HEAD, revision: 4a1b77600c1873a8233f3ffb55afcedbb63b8d84)
  build user:       root@586879db11e5
  build date:       20230717-12:10:52
  go version:       go1.20.6
  platform:         linux/amd64
  tags:             netgo osusergo static_build
ubuntu@ip-172-31-30-187:~$
```

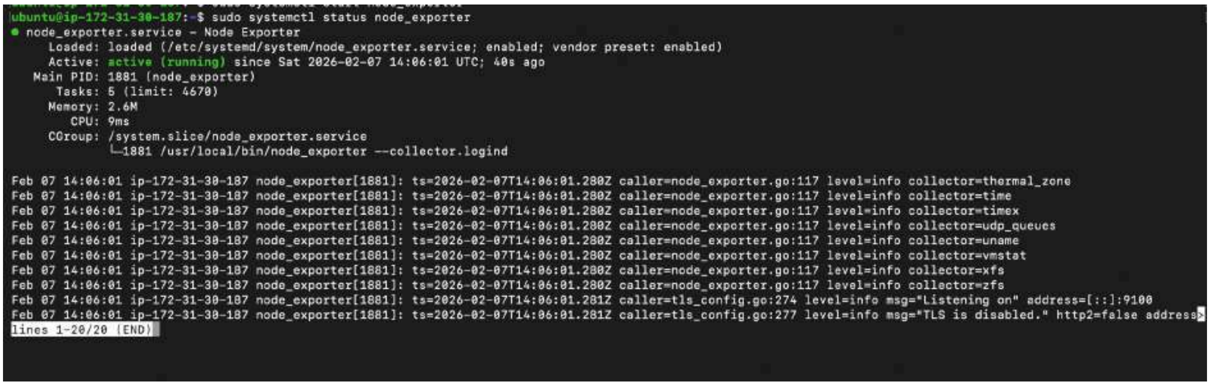
Create a systemd unit configuration file for Node Exporter:

```
[Unit]
Description=Node Exporter
Wants=network-online.target
After=network-online.target

StartLimitIntervalSec=500
StartLimitBurst=5

[Service]
User=node_exporter
Group=node_exporter
Type=simple
Restart=on-failure
RestartSec=5s
ExecStart=/usr/local/bin/node_exporter --collector.logind

[Install]
WantedBy=multi-user.target
~
~
~
~
~
~
~
```

Enable and start Node Exporter:

```
ubuntu@ip-172-31-30-187:~$ sudo systemctl enable node_exporter
ubuntu@ip-172-31-30-187:~$ sudo systemctl start node_exporter
ubuntu@ip-172-31-30-187:~$
```

Verify the Node Exporter's status:




## 6.3. Configure Prometheus Plugin Integration

As of now we created Prometheus service, but we need to add a job in order to fetch the details by node exporter. So for that we need to create 2 jobs, one with 'node exporter' and the other with 'jenkins' as shown below;

Integrate Jenkins with Prometheus to monitor the CI/CD pipeline.

**Prometheus Configuration:**

To configure Prometheus to scrape metrics from Node Exporter and Jenkins, you need to modify the prometheus.yml file.

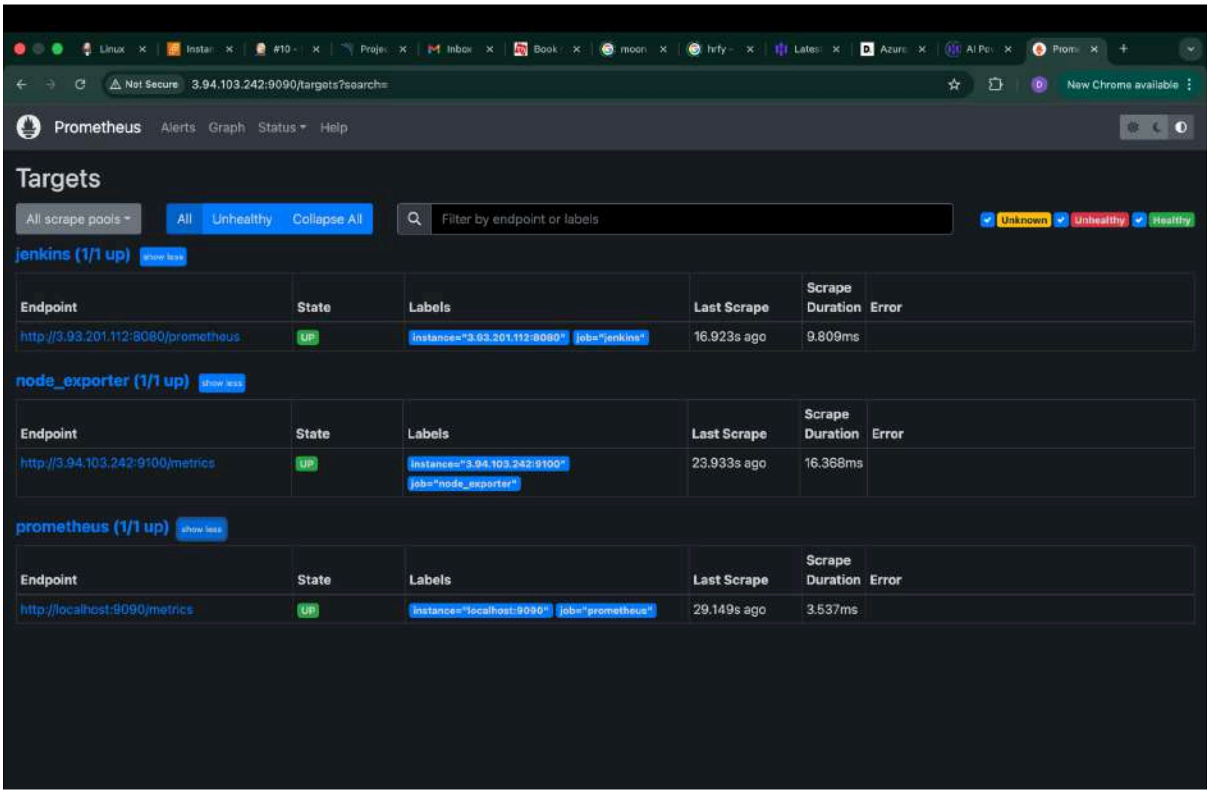
```yaml
# my global config
global:
  scrape_interval: 15s # Set the scrape interval to every 15 seconds. Default is every 1 minute.
  evaluation_interval: 15s # Evaluate rules every 15 seconds. The default is every 1 minute.
  # scrape_timeout is set to the global default (10s).

# Alertmanager configuration
alerting:
  alertmanagers:
    - static_configs:
        - targets:
          # - alertmanager:9093

# Load rules once and periodically evaluate them according to the global 'evaluation_interval'.
rule_files:
  # - "first_rules.yml"
  # - "second_rules.yml"

# A scrape configuration containing exactly one endpoint to scrape:
# Here it's Prometheus itself.
scrape_configs:
  # The job name is added as a label `job=<job_name>` to any timeseries scraped from this config.
  - job_name: "prometheus"

    # metrics_path defaults to '/metrics'
    # scheme defaults to 'http'.

    static_configs:
      - targets: ["localhost:9090"]

  - job_name: 'node_exporter'
    static_configs:
      - targets: ['3.94.103.242:9100']

  - job_name: 'jenkins'
    metrics_path: '/prometheus'
    static_configs:
      - targets: ['3.93.201.112:8080']

~
~
~
~
~
~
~
~
~
"prometheus.yml" 38L, 1140B                                          31,2          All
```

Check the validity of the configuration file:

```
ubuntu@ip-172-31-30-187:/etc/prometheus$ promtool check config /etc/prometheus/prometheus.yml
Checking /etc/prometheus/prometheus.yml
 SUCCESS: /etc/prometheus/prometheus.yml is valid prometheus config file syntax

ubuntu@ip-172-31-30-187:/etc/prometheus$
```

Reload the Prometheus configuration without restarting:

```
ubuntu@ip-172-31-30-187:/etc/prometheus$ curl -X POST http://localhost:9090/-/reload
ubuntu@ip-172-31-30-187:/etc/prometheus$
```



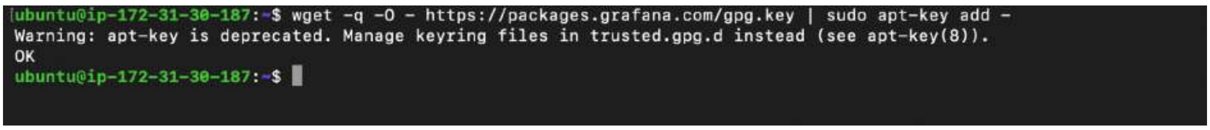

## 6.4. Install Grafana (Execute in Monitoring Server VM)

Install Grafana on Monitoring Server;

### Step 1: Install Dependencies:

First, ensure that all necessary dependencies are installed:

### Step 2: Add the GPG Key:



## Step 3: Add Grafana Repository:

Add the repository for Grafana stable releases:

```
ubuntu@ip-172-31-30-187:~$ echo "deb https://packages.grafana.com/oss/deb stable main" | sudo tee -a /etc/apt/sources.list.d/grafana.list
deb https://packages.grafana.com/oss/deb stable main
ubuntu@ip-172-31-30-187:~$
```

## Step 4: Update and Install Grafana:

Update the package list and install Grafana:

```
ubuntu@ip-172-31-30-187:~$ echo "deb https://packages.grafana.com/oss/deb stable main" | sudo tee -a /etc/apt/sources.list.d/grafana.list
deb https://packages.grafana.com/oss/deb stable main
ubuntu@ip-172-31-30-187:~$ sudo apt-get update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://security.ubuntu.com/ubuntu jammy-security InRelease
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:5 https://packages.grafana.com/oss/deb stable InRelease [7660 B]
Get:6 https://packages.grafana.com/oss/deb stable/main amd64 Packages [462 kB]
Fetched 725 kB in 1s (941 kB/s)
Reading package lists... Done
W: https://packages.grafana.com/oss/deb/dists/stable/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt/trusted.gpg), see the DEPRECATION section i
n apt-key(8) for details.
ubuntu@ip-172-31-30-187:~$ sudo apt-get -y install grafana
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  musl
The following NEW packages will be installed:
  grafana musl
0 upgraded, 2 newly installed, 0 to remove and 66 not upgraded.
Need to get 194 MB of archives.
After this operation, 711 MB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 musl amd64 1.2.2-4 [407 kB]
Get:2 https://packages.grafana.com/oss/deb stable/main amd64 grafana amd64 12.3.2 [194 MB]
Fetched 194 MB in 4s (50.0 MB/s)
Selecting previously unselected package musl:amd64.
(Reading database ... 65997 files and directories currently installed.)
Preparing to unpack .../musl_1.2.2-4_amd64.deb ...
Unpacking musl:amd64 (1.2.2-4) ...
Selecting previously unselected package grafana.
Preparing to unpack .../grafana_12.3.2_amd64.deb ...
Unpacking grafana (12.3.2) ...
Setting up musl:amd64 (1.2.2-4) ...
Setting up grafana (12.3.2) ...
Adding system user `grafana' (UID 115) ...
Adding new user `grafana' (UID 115) with group `grafana' ...
Not creating home directory `/usr/share/grafana'.
### NOT starting on installation, please execute the following statements to configure grafana to start automatically using systemd
 sudo /bin/systemctl daemon-reload
 sudo /bin/systemctl enable grafana-server
### You can start grafana-server by executing
 sudo /bin/systemctl start grafana-server
Processing triggers for man-db (2.10.2-1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.
```

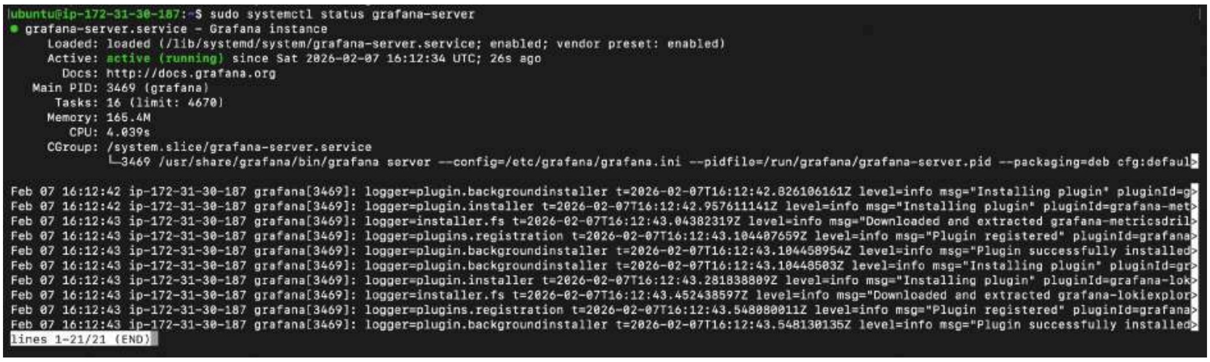
## Step 5: Enable and Start Grafana Service:

To automatically start Grafana after a reboot, enable the service:

```
ubuntu@ip-172-31-30-187:~$ sudo systemctl enable grafana-server
Synchronizing state of grafana-server.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable grafana-server
Created symlink /etc/systemd/system/multi-user.target.wants/grafana-server.service → /lib/systemd/system/grafana-server.service.
ubuntu@ip-172-31-30-187:~$ sudo systemctl start grafana-server
ubuntu@ip-172-31-30-187:~$
```

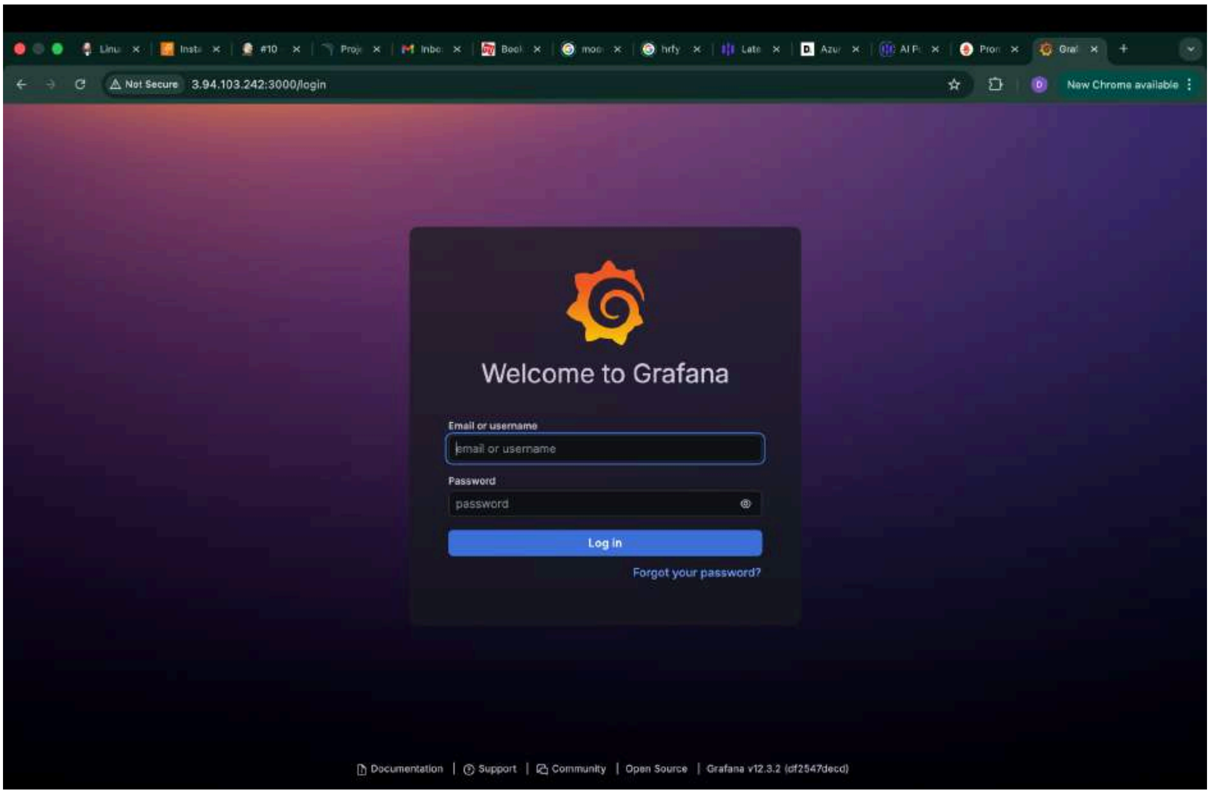
## Step 6: Check Grafana Status:

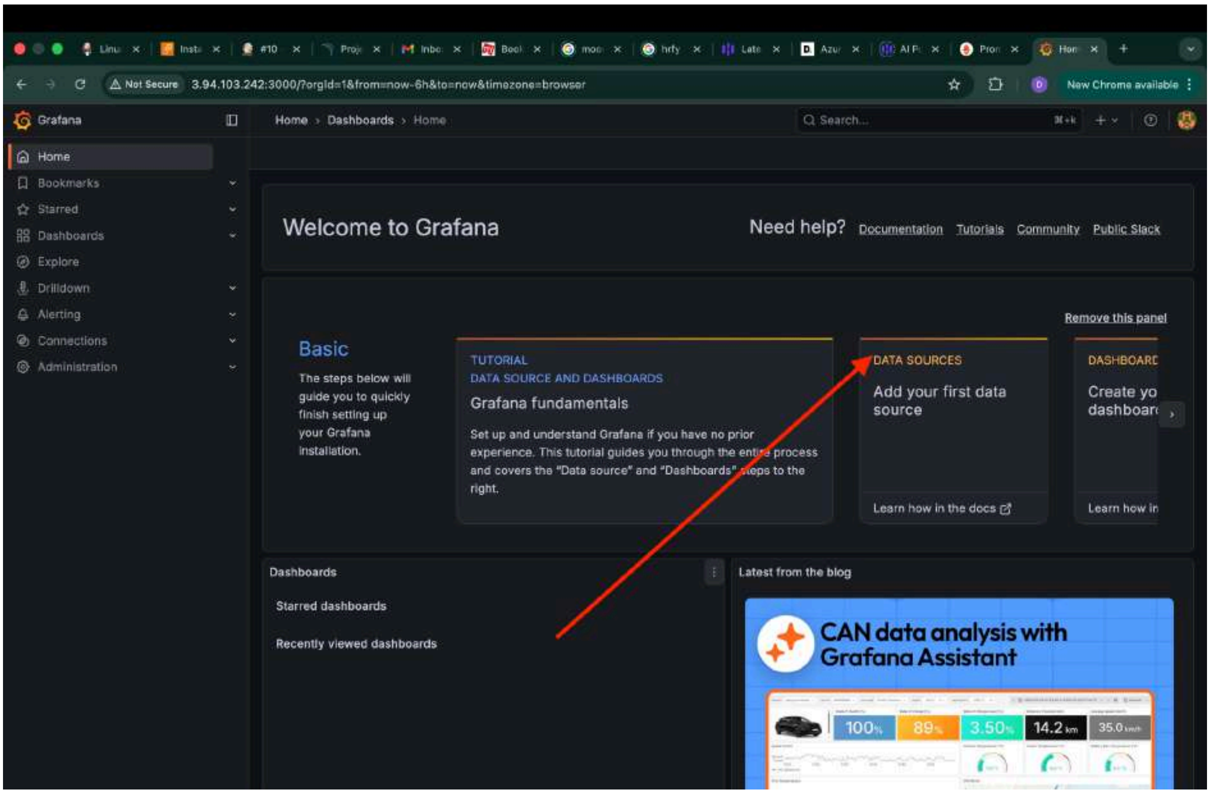
Verify the status of the Grafana service to ensure it's running correctly:

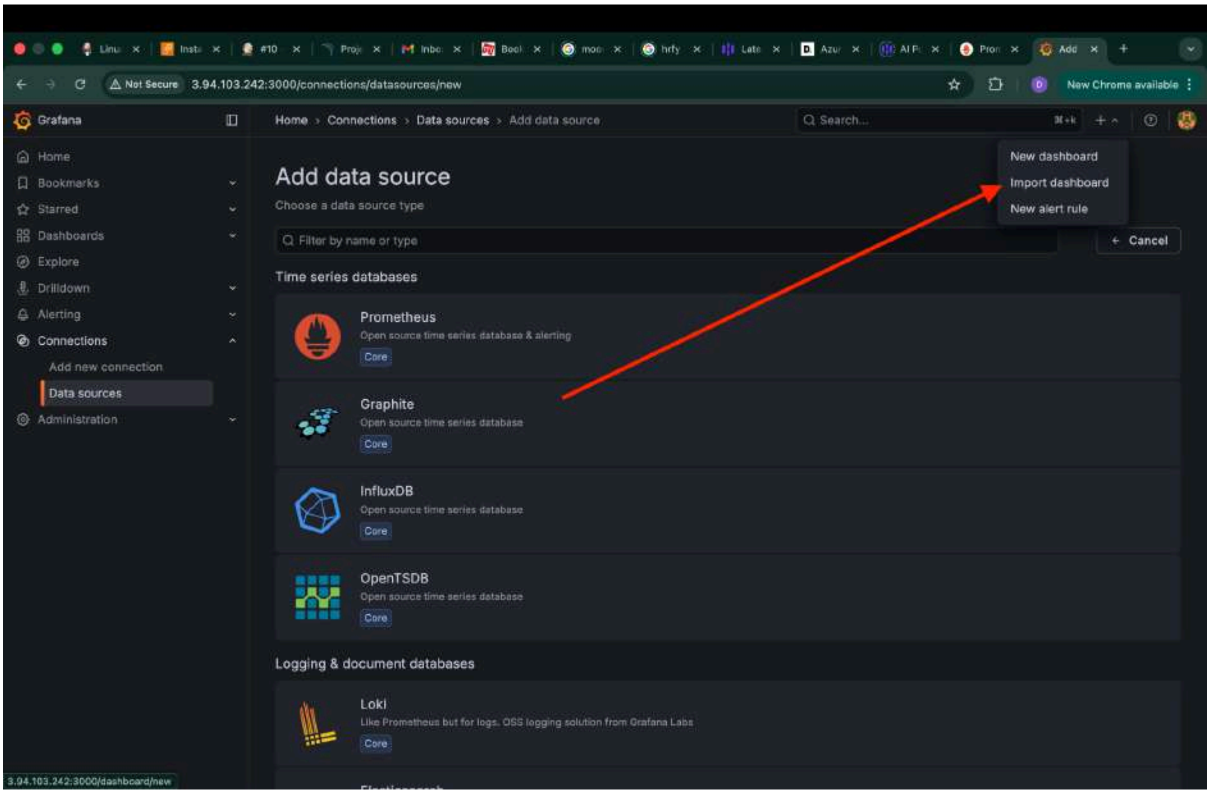


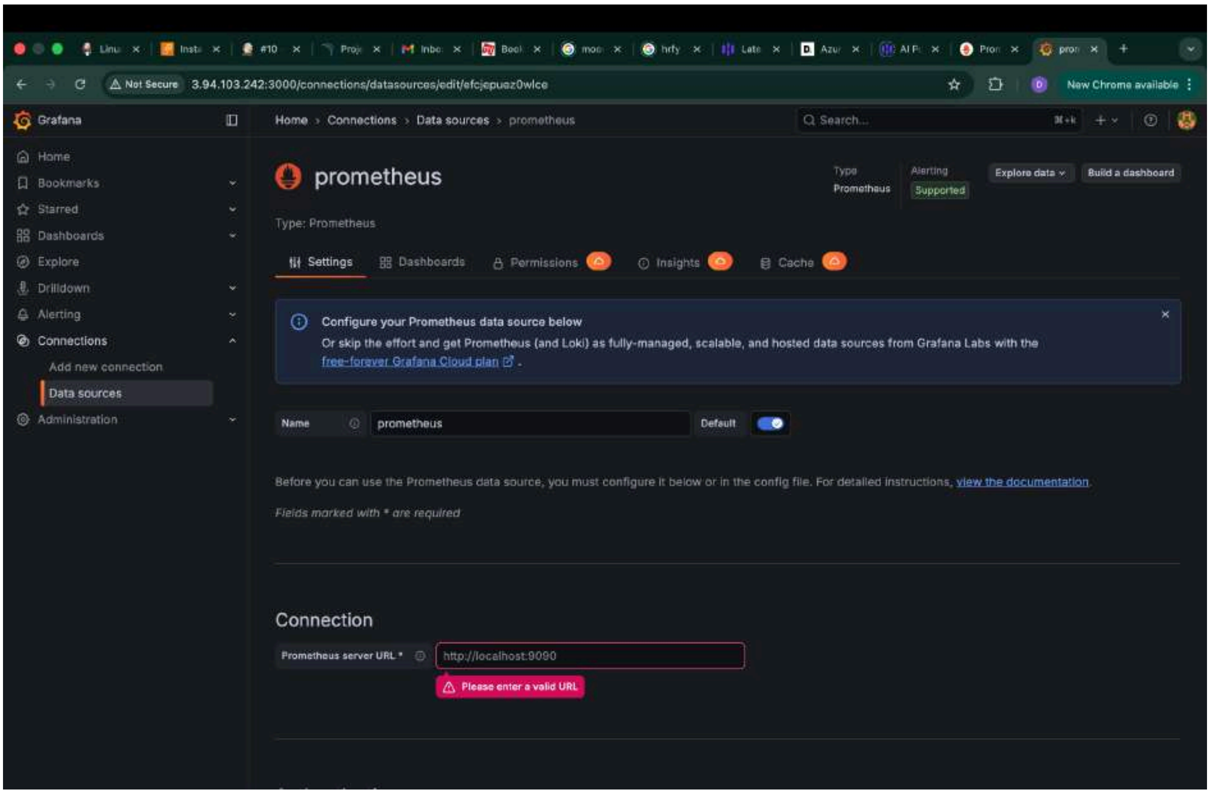

## Step 7: Access Grafana Web Interface:

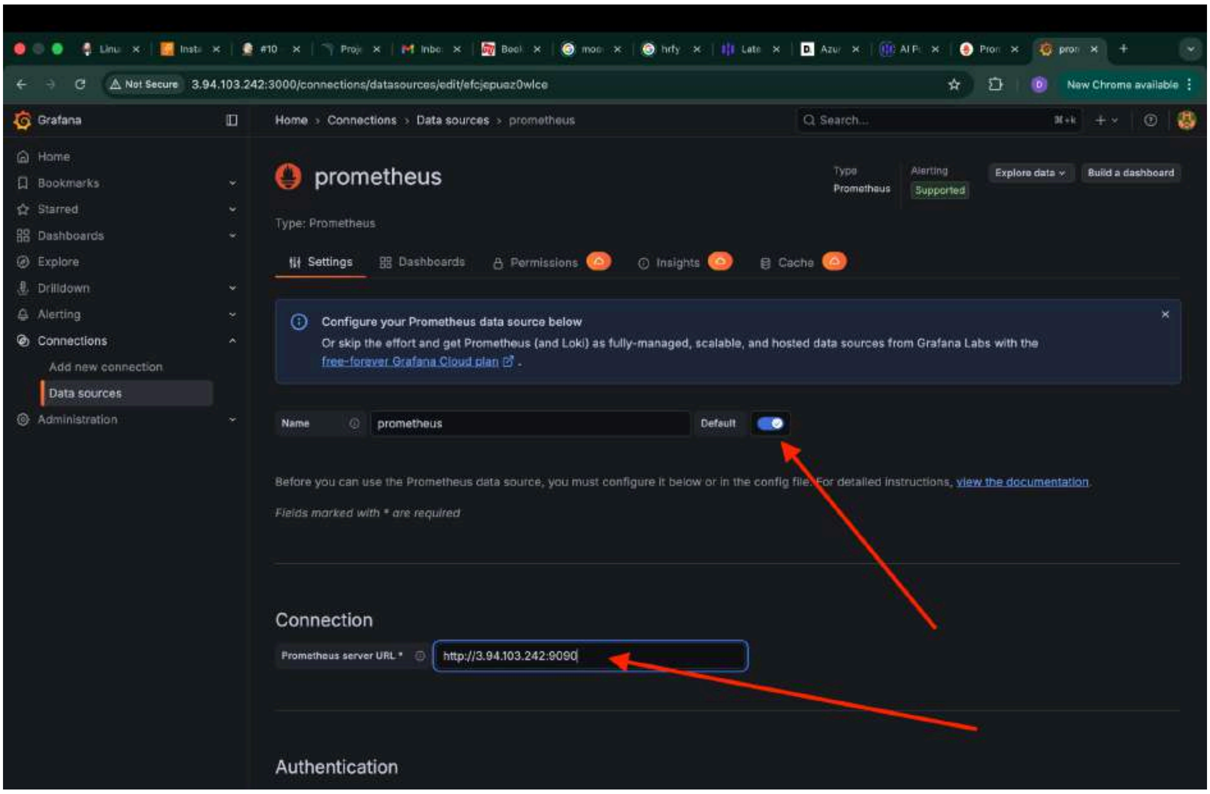
The default port for Grafana is 3000

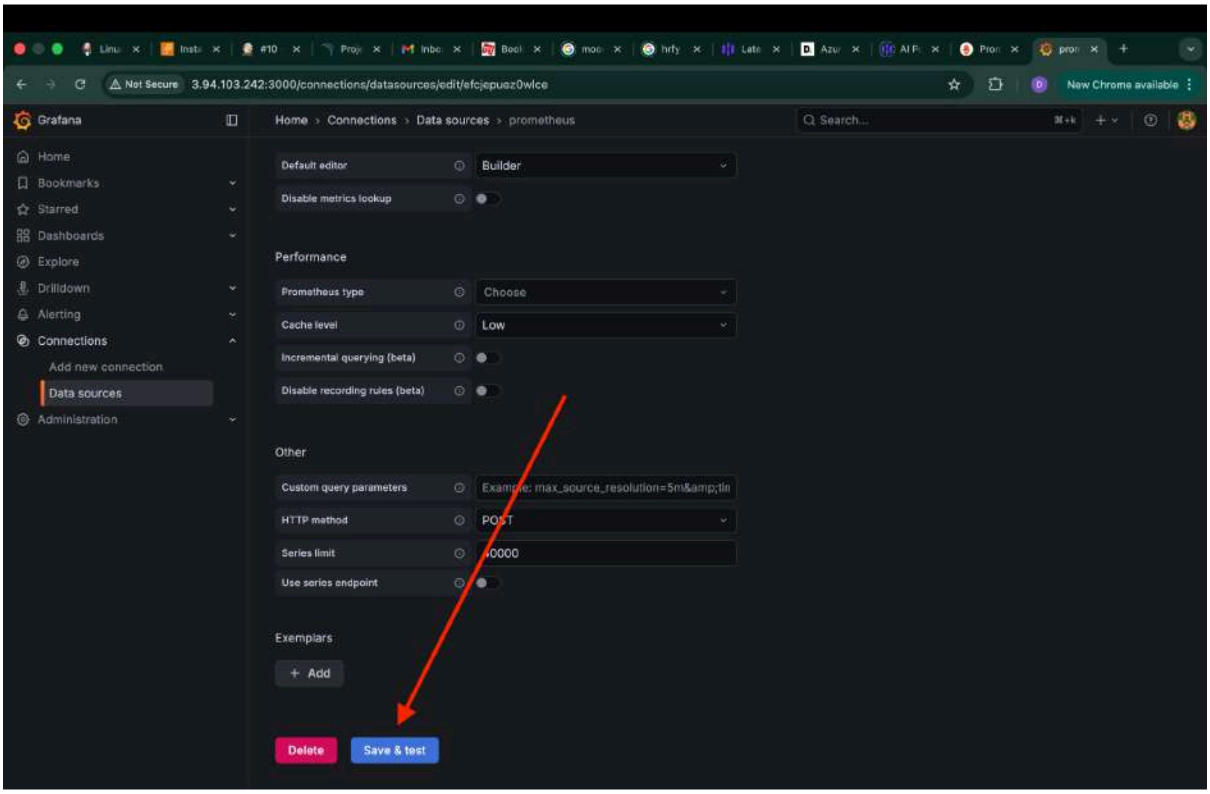


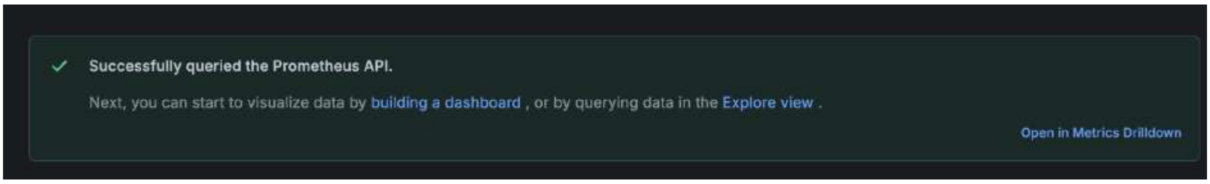

3.94.103.242:3000/?orgId=1&from=now-6h&to=now&timezone=browser
Grafana
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Administration
Home › Dashboards › Home
Welcome to Grafana
Need help? Documentation Tutorials Community Public Slack
Remove this panel
Basic
The steps below will guide you to quickly finish setting up your Grafana installation.
TUTORIAL
DATA SOURCE AND DASHBOARDS
Grafana fundamentals
Set up and understand Grafana if you have no prior experience. This tutorial guides you through the entire process and covers the "Data source" and "Dashboards" steps to the right.
DATA SOURCES
Add your first data source
Learn how in the docs
Dashboards
Starred dashboards
Recently viewed dashboards
Latest from the blog
CAN data analysis with Grafana Assistant

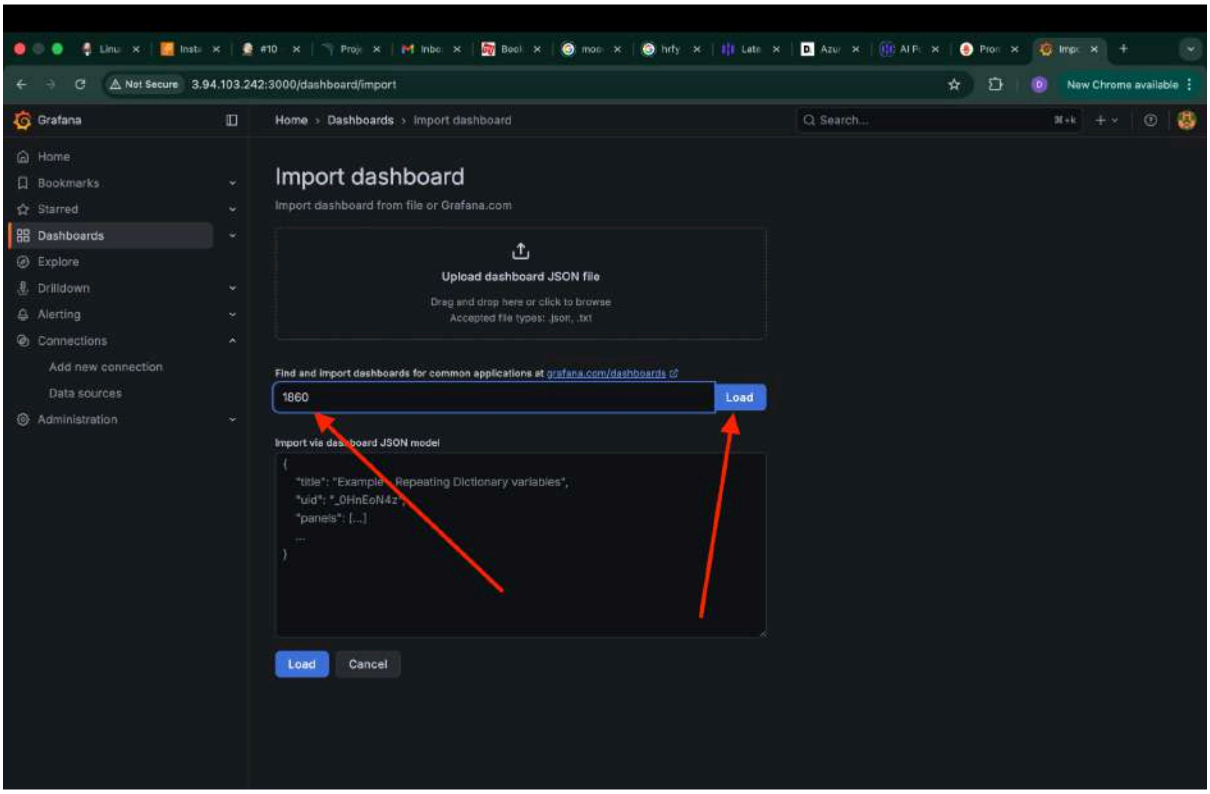

3.94.103.242:3000/connections/datasources/new
Grafana
Home › Connections › Data sources › Add data source
New dashboard
Import dashboard
New alert rule
Add data source
Choose a data source type
Filter by name or type
Cancel
Time series databases
Prometheus
Open source time series database & alerting
Core
Graphite
Open source time series database
Core
InfluxDB
Open source time series database
Core
OpenTSDB
Open source time series database
Core
Logging & document databases
Loki
Like Prometheus but for logs. OSS logging solution from Grafana Labs
Core
Connections
Add new connection
Data sources
3.94.103.242:3000/dashboard/new

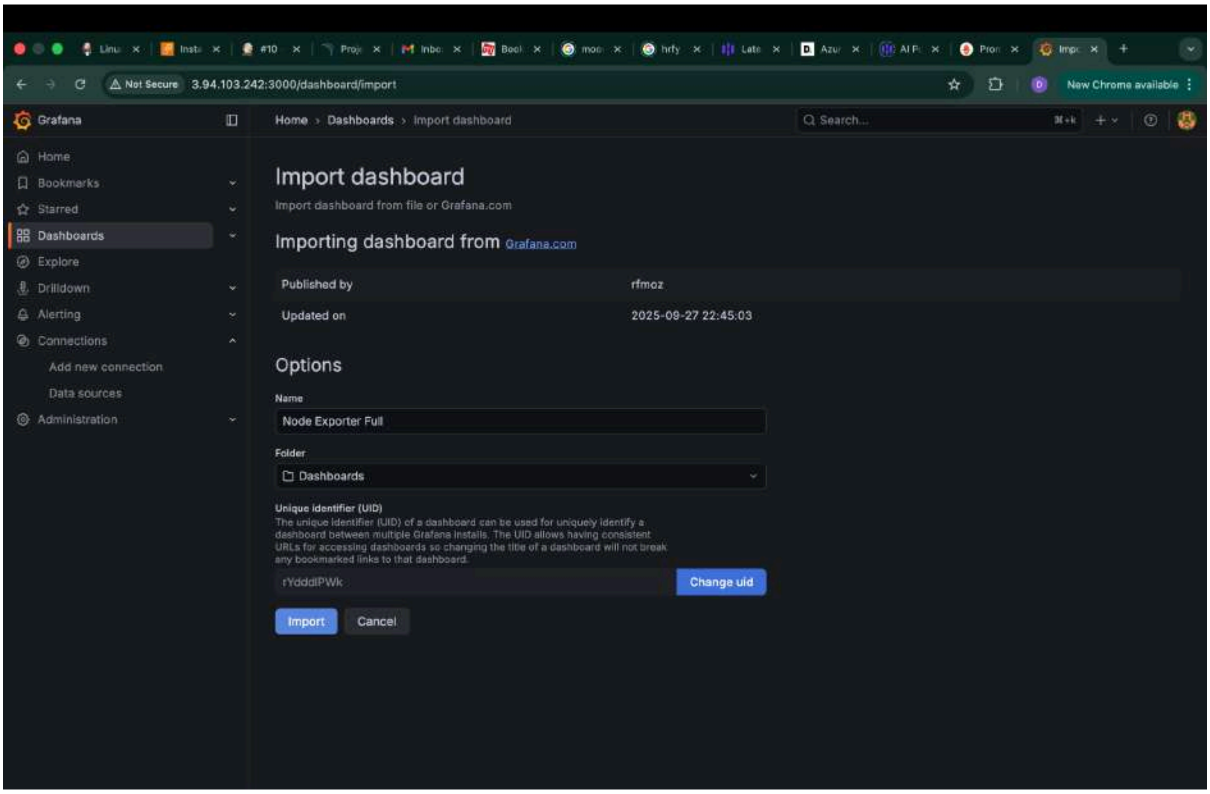

3.94.103.242:3000/connections/datasources/edit/efcjepuez0wlce
Grafana
Home › Connections › Data sources › prometheus
prometheus
Type: Prometheus
Settings
Dashboards
Permissions
Insights
Cache
Configure your Prometheus data source below
Or skip the effort and get Prometheus (and Loki) as fully-managed, scalable, and hosted data sources from Grafana Labs with the free-forever Grafana Cloud plan .
Name
prometheus
Default
Before you can use the Prometheus data source, you must configure it below or in the config file. For detailed instructions, view the documentation.
Fields marked with * are required
Connection
Prometheus server URL *
http://localhost:9090
Please enter a valid URL

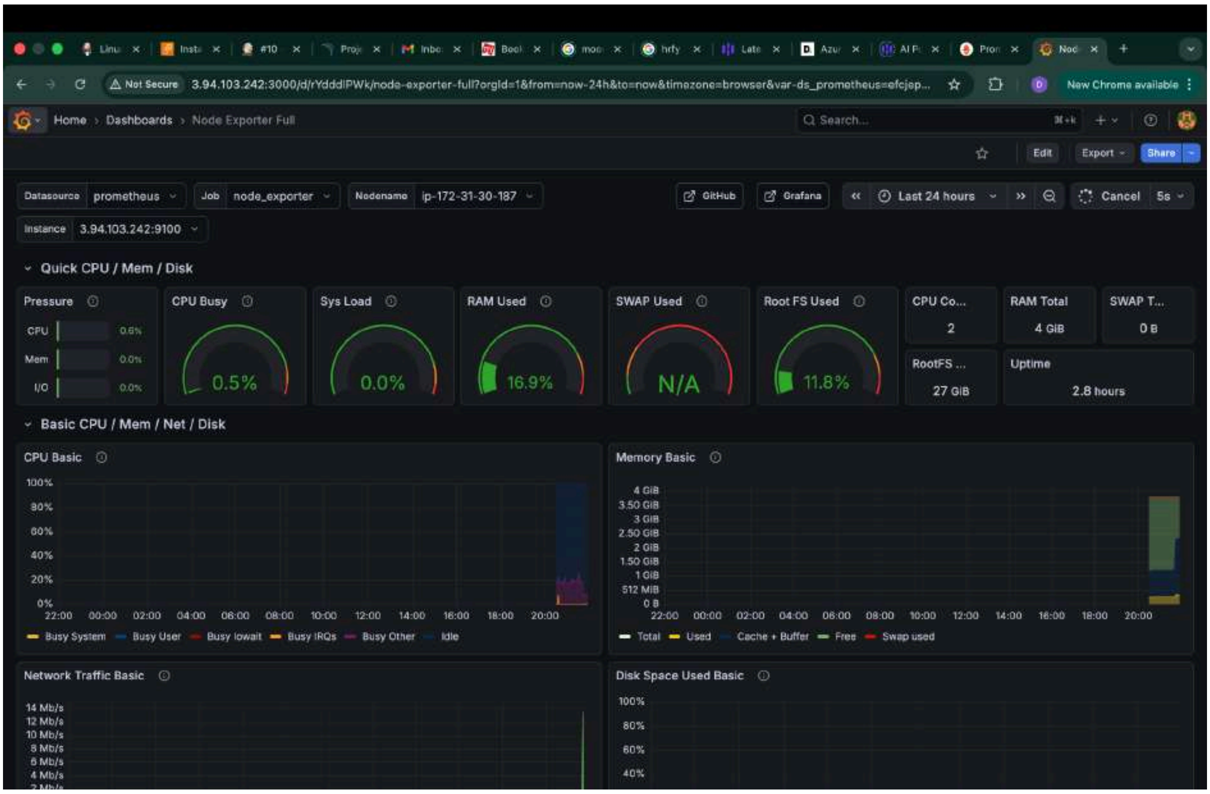

3.94.103.242:3000/connections/datasources/edit/efcjepuez0wlce
Grafana
Home › Connections › Data sources › prometheus
prometheus
Type: Prometheus
Settings
Dashboards
Permissions
Insights
Cache
Configure your Prometheus data source below
Or skip the effort and get Prometheus (and Loki) as fully-managed, scalable, and hosted data sources from Grafana Labs with the free-forever Grafana Cloud plan .
Name
prometheus
Default
Before you can use the Prometheus data source, you must configure it below or in the config file. For detailed instructions, view the documentation.
Fields marked with * are required
Connection
Prometheus server URL *
http://3.94.103.242:9090
Authentication

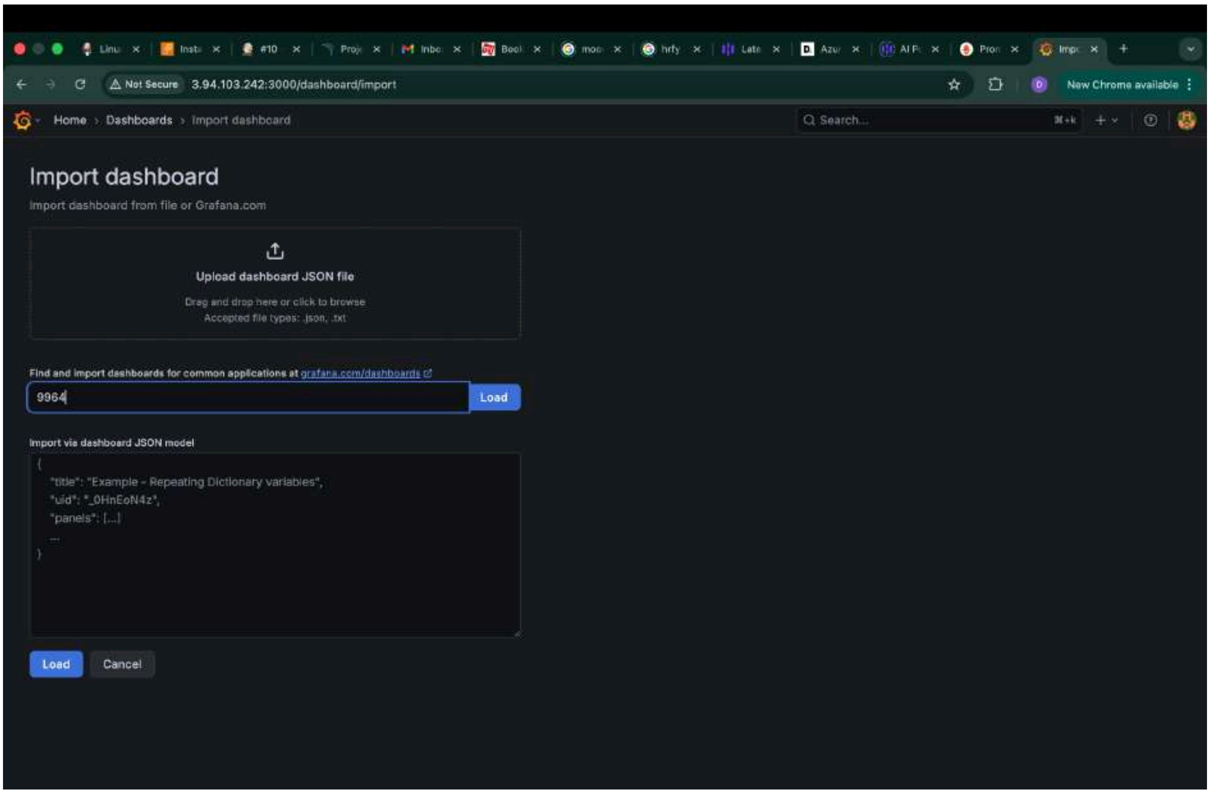

3.94.103.242:3000/connections/datasources/edit/efcjepuez0wlce
Grafana
Home › Connections › Data sources › prometheus
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Default editor
Builder
Disable metrics lookup
Performance
Prometheus type
Choose
Cache level
Low
Incremental querying (beta)
Disable recording rules (beta)
Other
Custom query parameters
HTTP method
POST
Series limit
Use series endpoint
Exemplars
+ Add
Delete
Save & test

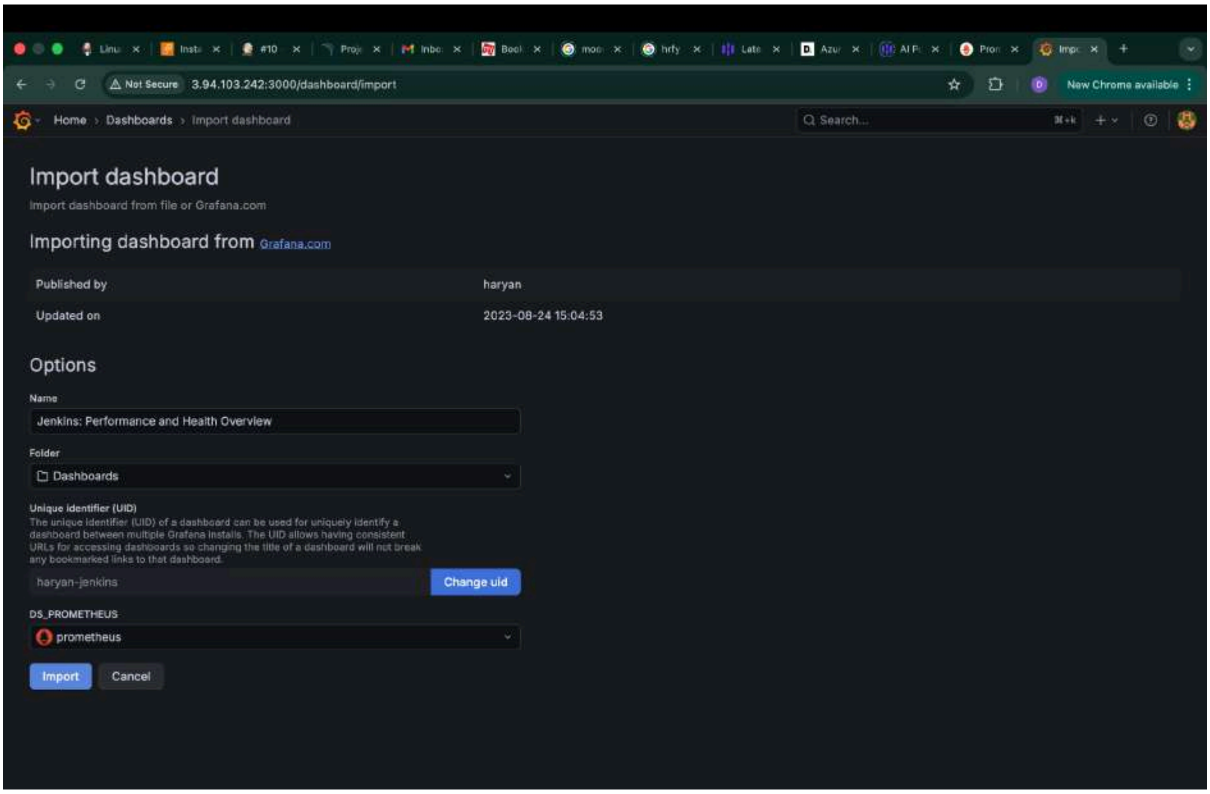

Successfully queried the Prometheus API.
Next, you can start to visualize data by building a dashboard , or by querying data in the Explore view .
Open in Metrics Drilldown

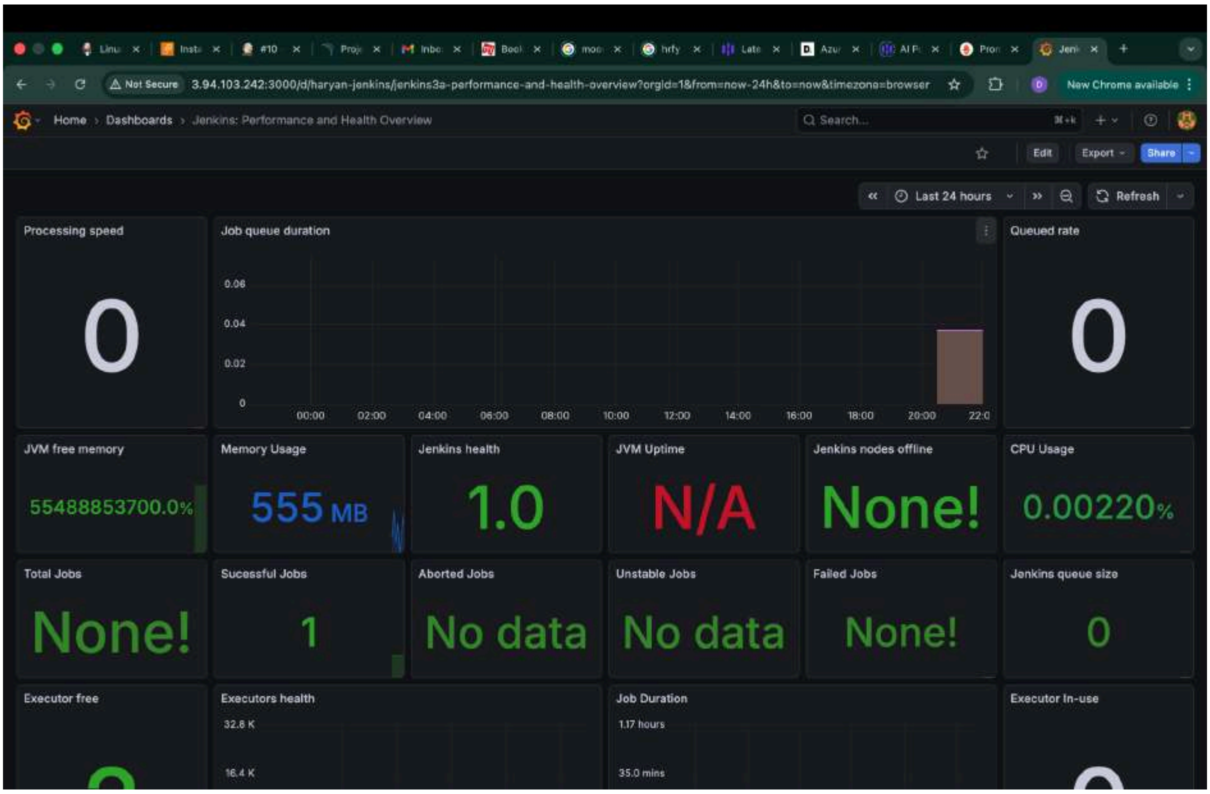
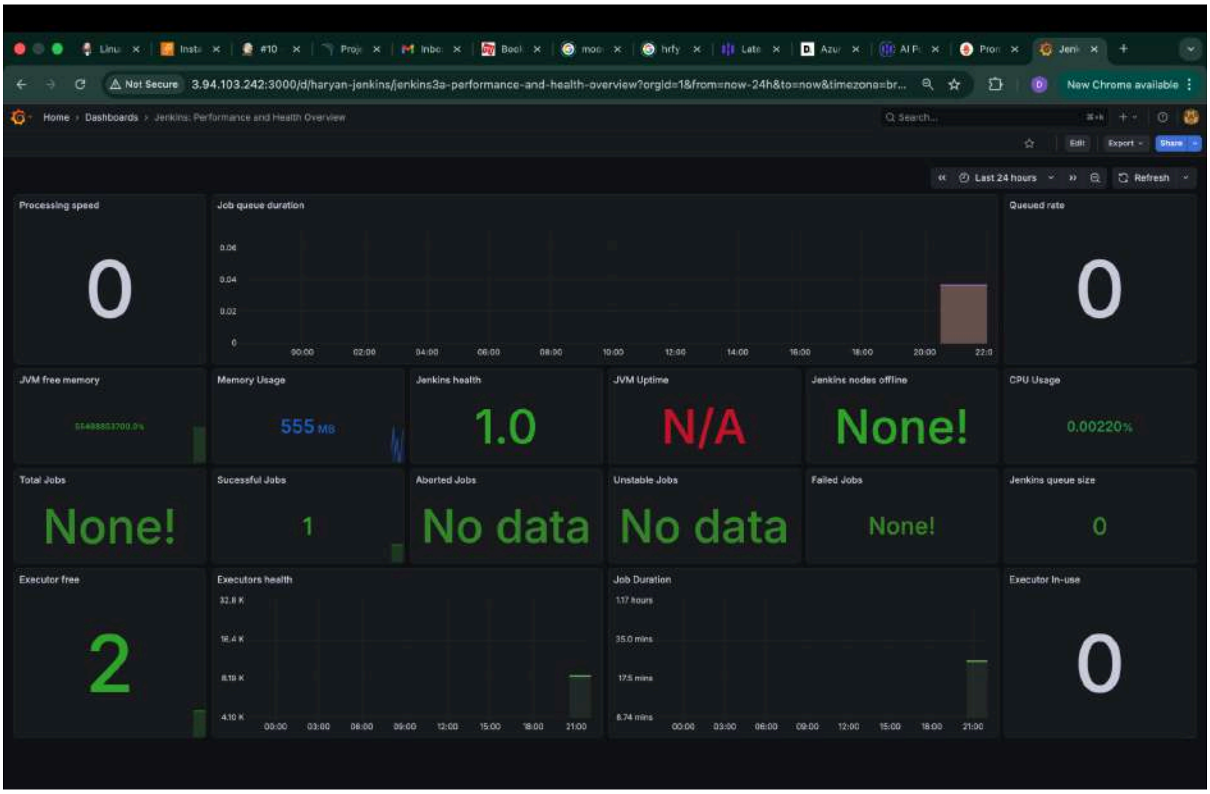

3.94.103.242:3000/dashboard/import
Grafana
Home › Dashboards › Import dashboard
Import dashboard
Import dashboard from file or Grafana.com
Upload dashboard JSON file
Drag and drop here or click to browse
Accepted file types: .json, .txt
Find and import dashboards for common applications at grafana.com/dashboards
1860
Load
Import via dashboard JSON model
Load
Cancel

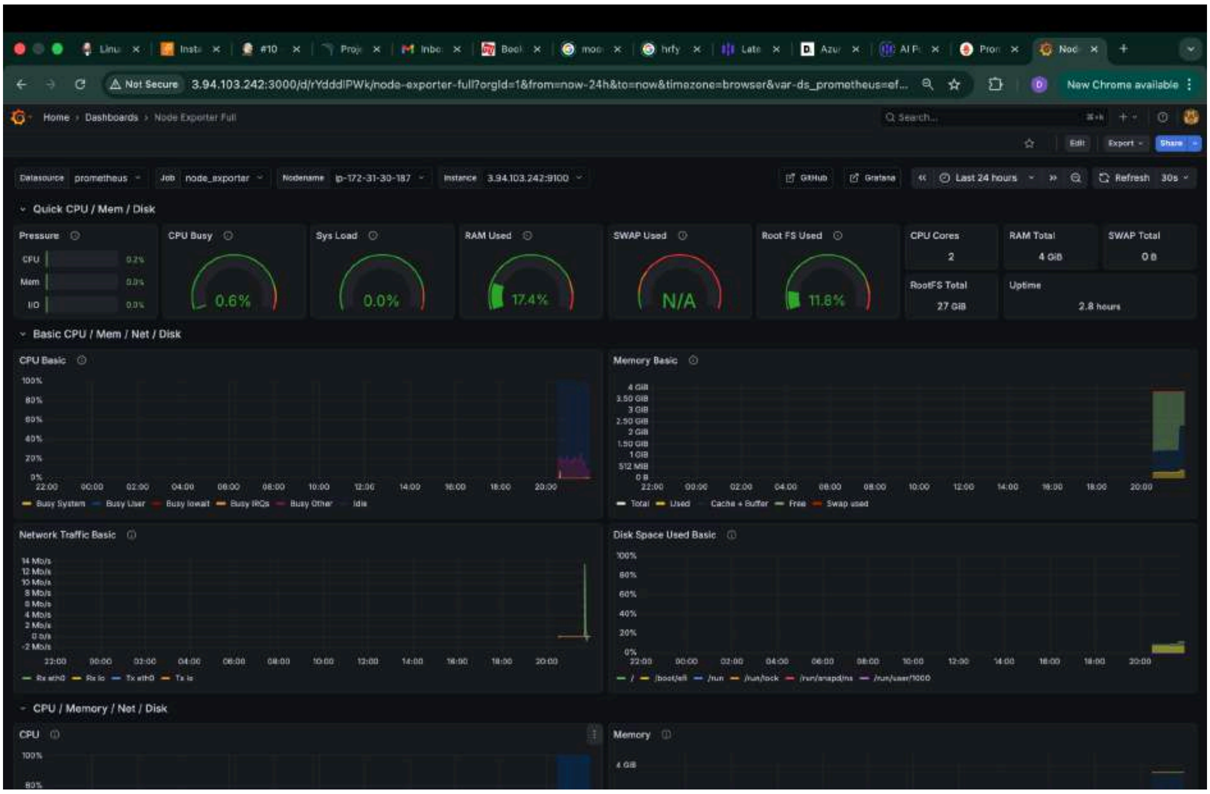

3.94.103.242:3000/dashboard/import
Grafana
Home
Bookmarks
Starred
Dashboards
Explore
Drilldown
Alerting
Connections
Add new connection
Data sources
Administration
Home › Dashboards › Import dashboard
Import dashboard
Import dashboard from file or Grafana.com
Importing dashboard from Grafana.com
Published by rfmoz
Updated on 2025-09-27 22:45:03
Options
Name
Node Exporter Full
Folder
Dashboards
Unique identifier (UID)
rYdddlPWk
Change uid
Import
Cancel


3.94.103.242:3000/d/rYdddlPWk/node-exporter-full?orgId=1&from=now-24h&to=now&timezone=browser&var-ds_prometheus=efcjep...
Home › Dashboards › Node Exporter Full
Edit
Export
Share
Datasource prometheus
Job node_exporter
Nodename ip-172-31-30-187
Instance 3.94.103.242:9100
GitHub
Grafana
Last 24 hours
Cancel
5s
Quick CPU / Mem / Disk
Pressure
CPU Busy
0.5%
Sys Load
0.0%
RAM Used
16.9%
SWAP Used
N/A
Root FS Used
11.8%
CPU Co...
2
RAM Total
4 GiB
SWAP T...
0 B
RootFS ...
27 GiB
Uptime
2.8 hours
Basic CPU / Mem / Net / Disk
CPU Basic
Busy System
Busy User
Busy Iowait
Busy IRQs
Busy Other
Idle
Memory Basic
Total
Used
Cache + Buffer
Free
Swap used
Network Traffic Basic
Disk Space Used Basic

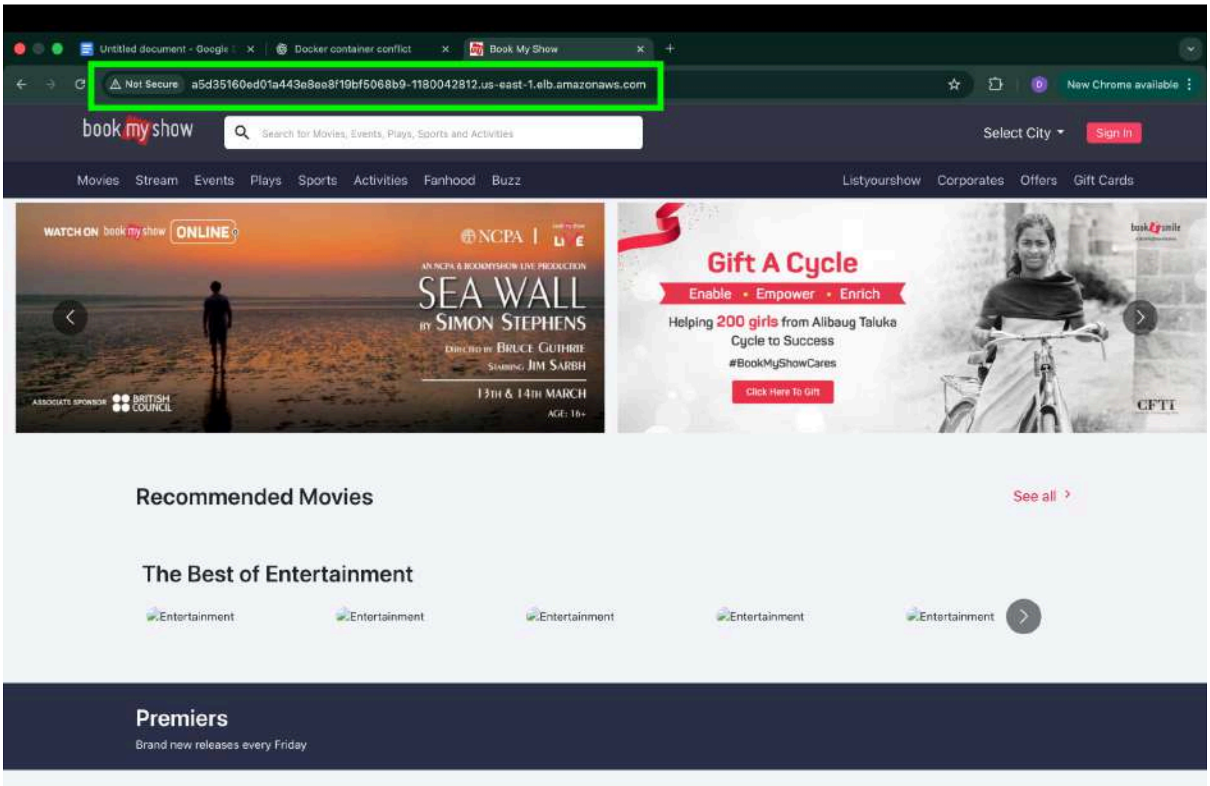

Not Secure 3.94.103.242:3000/dashboard/import
New Chrome available
Home › Dashboards › Import dashboard
Search...
Import dashboard
Import dashboard from file or Grafana.com
Upload dashboard JSON file
Drag and drop here or click to browse
Accepted file types: .json, .txt
Find and import dashboards for common applications at grafana.com/dashboards
9964
Load
Import via dashboard JSON model
{
"title": "Example - Repeating Dictionary variables",
"uid": "_0HnEoN4z",
"panels": [...]
...
}
Load
Cancel


Not Secure 3.94.103.242:3000/dashboard/import
New Chrome available
Home › Dashboards › Import dashboard
Search...
Import dashboard
Import dashboard from file or Grafana.com
Importing dashboard from Grafana.com
Published by
haryan
Updated on
2023-08-24 15:04:53
Options
Name
Jenkins: Performance and Health Overview
Folder
Dashboards
Unique identifier (UID)
The unique identifier (UID) of a dashboard can be used for uniquely identify a dashboard between multiple Grafana installs. The UID allows having consistent URLs for accessing dashboards so changing the title of a dashboard will not break any bookmarked links to that dashboard.
haryan-jenkins
Change uid
DS_PROMETHEUS
prometheus
Import
Cancel


Not Secure
3.94.103.242:3000/d/haryan-jenkins/jenkins3a-performance-and-health-overview?orgId=1&from=now-24h&to=now&timezone=browser
New Chrome available
Home › Dashboards › Jenkins: Performance and Health Overview
Search...
Edit
Export
Share
Last 24 hours
Refresh
Processing speed
0
Job queue duration
Queued rate
0
JVM free memory
55488853700.0%
Memory Usage
555 MB
Jenkins health
1.0
JVM Uptime
N/A
Jenkins nodes offline
None!
CPU Usage
0.00220%
Total Jobs
None!
Sucessful Jobs
1
Aborted Jobs
No data
Unstable Jobs
No data
Failed Jobs
None!
Jenkins queue size
0
Executor free
Executors health
Job Duration
Executor In-use

Not Secure
New Chrome available
Home › Dashboards › Jenkins: Performance and Health Overview
Processing speed
0
Job queue duration
Queued rate
0
JVM free memory
Memory Usage
555 MB
Jenkins health
1.0
JVM Uptime
N/A
Jenkins nodes offline
None!
CPU Usage
0.00220%
Total Jobs
None!
Sucessful Jobs
1
Aborted Jobs
No data
Unstable Jobs
No data
Failed Jobs
None!
Jenkins queue size
0
Executor free
2
Executors health
Job Duration
Executor In-use
0




```
ubuntu@ip-172-31-18-170:~$ kubectl get nodes
NAME                           STATUS   ROLES    AGE   VERSION
ip-192-168-28-171.ec2.internal   Ready    <none>   31h   v1.30.14-eks-ecaa3a6
ip-192-168-3-46.ec2.internal     Ready    <none>   31h   v1.30.14-eks-ecaa3a6
ip-192-168-37-56.ec2.internal    Ready    <none>   31h   v1.30.14-eks-ecaa3a6
ubuntu@ip-172-31-18-170:~$ kubectl get svc
NAME          TYPE           CLUSTER-IP      EXTERNAL-IP                                                                PORT(S)        AGE
bms-service   LoadBalancer   10.100.230.18   a5d35160ed01a443e8ee8f19bf5068b9-1180042812.us-east-1.elb.amazonaws.com   80:32680/TCP   4h34m
kubernetes    ClusterIP      10.100.0.1      <none>                                                                     443/TCP        32h
ubuntu@ip-172-31-18-170:~$
```


Untitled document - Google
Docker container conflict
Book My Show
Not Secure
a5d35160ed01a443e8ee8f19bf5068b9-1180042812.us-east-1.elb.amazonaws.com
New Chrome available
book my show
Search for Movies, Events, Plays, Sports and Activities
Select City
Sign In
Movies
Stream
Events
Plays
Sports
Activities
Fanhood
Buzz
Listyourshow
Corporates
Offers
Gift Cards
WATCH ON book my show ONLINE
NCPA
SEA WALL
BY SIMON STEPHENS
DIRECTED BY BRUCE GUTHRIE
STARRING JIM SARBH
13TH & 14TH MARCH
AGE: 16+
ASSOCIATE SPONSOR BRITISH COUNCIL
Gift A Cycle
Enable • Empower • Enrich
Helping 200 girls from Alibaug Taluka
Cycle to Success
#BookMyShowCares
Click Here To Gift
CFTI
Recommended Movies
See all
The Best of Entertainment
Entertainment
Entertainment
Entertainment
Entertainment
Entertainment
Premiers
Brand new releases every Friday